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Artificial Intelligence and Machine Learning for Official Statistics

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Workpackage 9

USE CASE: IMPUTATION FOCUS – STATISTICALLY VALID AND EFFICIENT EDITING AND IMPUTATION IN OFFICIAL STATISTICS AI/ML – WITH A SPECIAL FOCUS ON IMPUTATION

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Executive summary

This deliverable, **D9.2 – Methodological aspects from use cases in Machine Learning techniques for post-collection imputation**, is the second output of Workpackage 9 within the **AIML4OS** (One-Stop-Shop Artificial Intelligence and Machine Learning for Official Statistics) project. The primary objective of this report is to evaluate the effectiveness, stability, and operational feasibility of Machine Learning (ML) algorithms for **post-collection imputation**, the process of treating units and values missing after the data collection phase has concluded.

As National Statistical Institutes (NSIs) face increasing data complexity and declining response rates, traditional imputation methods such as Hot-deck or simple linear regressions often fail to capture the intricate, non-linear relationships present in modern administrative and survey data. This document provides a cross-country and cross-applications methodological assessment based on six diverse use cases from **Poland, Luxembourg, Spain, Portugal, Slovenia and Austria**.

This document is divided into six chapters. Chapter 1 includes the introduction where a brief background on the techniques used is presented. Chapter 2 has the descriptions of the six projects related to post-collection imputation in WP9. Chapter 3 contains all the information and suggestions about the preprocessing of the data including cleaning and feature engineering, in this particular case there is also a data preparation related to the generation of synthetic population to be able to evaluate results from different methods. Chapter 4 consists of the issues related to model selection: algorithms and hyperparameters to select the best model, train and test steps and model evaluation at micro and macro levels. In Chapter 5 the results of each project are included. Finally in Chapter 6 some conclusions can be found regarding methodological outputs, the production focused strategy and some final remarks. At the end of the document, Appendix A contains the information about the specific regressors used in each project.

List of Acronyms

Acronym	Definition
AI	Artificial Intelligence
AIML4OS	Artificial Intelligence and Machine Learning for Official Statistics
API	Application Programming Interface
AUC	Area Under the ROC Curve
CART	Classification and Regression Trees
CN	Combined Nomenclature (EU foreign trade product classification)
CoD	Coefficient of Dispersion
CPI	Consumer Price Index
CV	Cross-Validation / Coefficient of Variation
DT	Decision Tree
EAN	European Article Number
EPA	Encuesta de Población Activa (Spanish Labour Force Survey)
ESS	European Statistical System
F1	Harmonic mean of Precision and Recall
GBR	Gradient Boosting Regression / Regressor
GDPR	General Data Protection Regulation
GEKS	Gini–Eltetö–Köves–Szulc (multilateral price index formula)
GIGO	Garbage In, Garbage Out
GK	Geary–Khamis (multilateral price index formula)
GLM	Generalized Linear Model
GSBPM	Generic Statistical Business Process Model
GUS	Główny Urząd Statystyczny (Statistics Poland)
HICP	Harmonised Index of Consumer Prices
HPO	Hyperparameter Optimization
IAEC	Inquérito Anual às Empresas de Construção (Annual Construction Survey, PT)
IES	Informação Empresarial Simplificada (Simplified Business Information, PT)
INE	Instituto Nacional de Estadística (Spain / Portugal)
INSS	Instituto Nacional de la Seguridad Social (Spain)
IQR	Interquartile Range
kNN	k-Nearest Neighbors
LFS	Labour Force Survey
LightGBM	Light Gradient Boosting Machine
LRMSE	Log Root Mean Squared Error

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Acronym	Definition
MAE	Mean Absolute Error
MAD	Mean Absolute Deviation
MAPE	Mean Absolute Prediction Error
MAR	Missing at Random
MCAR	Missing Completely at Random
MICE	Multiple Imputation by Chained Equations
ML	Machine Learning
MMPER	Max-Min Percentage Error Range
MNAR	Missing Not at Random
MPE	Mean Prediction Error
MSE	Mean Squared Error
NACE	Statistical Classification of Economic Activities in the European Community
NND	Nearest Neighbor Distance
NSI	National Statistical Institute
NUTS	Nomenclature of Territorial Units for Statistics
OPI	Situación Ocupacional / Occupational Status (Target variable in ES-LFS)
pMSE	Propensity Score Mean Squared Error
PMM	Predictive Mean Matching
PPS	Probability-Proportional-to-Size
RF	Random Forest
RMSE	Root Mean Squared Error
SCIE	Sistema de Contas Integradas das Empresas (Integrated Business Accounts, PT)
SEPE	Servicio Público de Empleo Estatal (Spanish Public Employment Service)
SHD	Sequential Hot Deck
STATEC	Institut national de la statistique et des études économiques (Luxembourg)
STL	Seasonal-Trend decomposition using Loess
SURS	Statistični urad Republike Slovenije (Statistical Office of Slovenia)
TPD	Time Product Dummy
UNECE	United Nations Economic Commission for Europe
VIM	Visualization and Imputation of Missing Values (R package)
WP	Workpackage
XGB / XGBoost	Extreme Gradient Boosting

1

General Introduction

1.1. Background and Motivation

The production of official statistics is increasingly challenged by the growing complexity of data sources and the rising rates of unit and item non-response. Traditional statistical imputation procedures, while robust, often struggle to capture non-linear relationships, high-dimensional interactions, and strong seasonal patterns inherent in modern datasets.

The **AIML4OS (One-Stop-Shop Artificial Intelligence and Machine Learning for Official Statistics)** project, specifically within Workpackage 9, addresses these challenges by exploring the transition from conventional methods to Machine Learning (ML) techniques for **post-collection imputation**. Post-collection imputation focuses on reconstructing missing values after the data collection period has ended, ensuring that the final microdataset is complete and statistically valid for the estimation of population aggregates.

1.2. Technical Framework for Post-Collection Imputation

Post-collection imputation is a critical stage in the statistical production chain, occurring after the data collection phase to address both unit and item non-response. The primary goal is to create a "clean" microdataset that preserves the underlying joint distribution of the variables, thereby avoiding bias in the final published aggregates.

The primary objective of WP9 in the AIML4OS project, as detailed in the introduction chapter of AIML4OS WP9. European Commission [2026], is to enhance the quality, reliability, and timeliness of official statistics by integrating Machine Learning (ML) into the standard production chain. Within the context of international standards, specifically the **Generic Statistical Business Process Model (GSBPM)**, imputation is identified as a core component of the "Process" phase. The project addresses the persistent challenge of non-response by leveraging ML to reconstruct missing values with a level of precision that traditional methods often fail to achieve. By aligning these advanced techniques with the GSBPM, the project ensures that imputation processes are not isolated experiments but are fully integrated into the harmonized and validated workflows of National Statistical Institutes (NSIs).

1.2.1. Traditional Imputation Paradigms

Historically, National Statistical Institutes (NSIs) have relied on a well-established taxonomy of traditional methods, which can be classified according to two main methodological dimensions [Haziza, 2009]:

- **Donor vs. Predicted Value Imputation:** This classification focuses on the source of the replacement value. In *donor imputation methods*, the missing value of a non-respondent (the recipient) is replaced by the observed value of a respondent (the donor) who shares similar characteristics. In contrast, *predicted value imputation methods* utilize a statistical model (such as a regression) to calculate an estimated value based on the auxiliary information available for the unit.
- **Deterministic vs. Random (Stochastic) Methods:** This distinction refers to the stability of the output. *Deterministic methods* are those that yield a fixed, unique imputed value given the sample if the process is repeated (e.g., mean imputation or kNN with $k = 1$). On the other hand, *random methods* incorporate a stochastic component—such as adding a random residual to a prediction or selecting a donor from a pool via probability—meaning they do not necessarily yield the same imputed value in repeated applications.

The choice between these paradigms is fundamental, as random methods are generally preferred when the objective is to preserve the variance and the original distribution of the data, whereas deterministic methods may be more efficient for point estimation but often lead to "variance compression" [Haziza, 2009, de Waal et al., 2011].

Among the most widely utilized traditional techniques is **Hot-deck imputation**. This donor-based method replaces missing values with observed values from "similar" responding units. Variants such as *Random Hot-deck* and *Sequential Hot-deck* have been the standard for decades due to their ability to preserve the observed variance and avoid the "mean-reversion" effect common in simple mean imputation [Eurostat, 2022]. However, Hot-deck often struggles in high-dimensional spaces where finding a suitable donor becomes computationally expensive or statistically impossible.

Another cornerstone of traditional imputation is **Regression-based imputation**, typically utilizing Generalized Linear Models (GLMs). In this framework, a functional relationship is assumed between the target variable and a set of auxiliary predictors. While effective for capturing global trends, these models often fail to account for complex interactions or non-linearities unless they are manually specified by the statistician [Bajari et al., 2025].

Beyond the binary classifications mentioned above, modern statistical practice often employs *mixed approaches* that bridge the gap between model-based and donor-based paradigms. A prominent example is **Predictive Mean Matching (PMM)**. In this framework, a predictive model is first used to calculate intermediate values for both respondents and non-respondents. Subsequently, the missing value is replaced by the observed value of a donor whose *predicted* value is closest to the *predicted* value of the recipient. This hybrid strategy leverages the strengths of both worlds: it ensures that imputed values are always "real" and observed (avoiding the risk of implausible values from pure regression) while using the sophisticated predictive power of a model to guide the matching process. By doing so, PMM maintains the distributional integrity of donor methods while reducing the bias associated with simple distance-based matching in high-dimensional spaces [see e.g. Haziza, 2009, de Waal et al., 2011].

1.2.2. The Role of Nearest Neighbor Methods: From NND Hot-deck to kNN

The **k-Nearest Neighbors (kNN)** algorithm serves as a fundamental bridge between traditional donor-based paradigms and modern Machine Learning. In statistical production, this connection is best exemplified by the **Nearest Neighbor Distance (NND) hot-deck** imputation, a widely used technique that corresponds mathematically to a **1NN** approach ($k = 1$). By

identifying the single most similar respondent to a recipient based on a distance metric, NND hot-deck provides a practical starting point for the transition toward more complex algorithmic frameworks.

By generalizing this logic to $k > 1$, kNN extends the traditional donor approach into the world of Machine Learning. Using distance metrics—such as Euclidean for continuous data or Gower distance for mixed types—to identify the k most similar units, the algorithm provides a localized imputation that does not require strong parametric assumptions. As demonstrated in several use cases within this project, kNN remains a robust benchmark for post-collection imputation. It is particularly effective when the relationship between variables is locally stable but globally complex, allowing for a more stable estimation than 1NN by averaging (in deterministic versions) or sampling (in random versions) from a larger pool of similar donors [Steurer et al., 2021].

1.2.3. Transition to Machine Learning

Despite the reliability of traditional methods, the increasing availability of administrative data and the need for higher precision have pushed the boundaries of these techniques. Modern **Machine Learning** approaches, such as Random Forest and Gradient Boosting, extend the logic of kNN and regression by automatically discovering interactions and structural breaks in the data [Zeng and Rao, 2025]. This deliverable evaluates how these advanced algorithms can be integrated into the existing statistical frameworks to improve the accuracy of post-collection imputation while maintaining the theoretical rigor required by official statistics [May and Lamboray, 2026].

The integration of Machine Learning (ML) into the production of official statistics represents a paradigm shift in how National Statistical Institutes (NSIs) handle data gaps. As highlighted by Beck et al. [2021], the transition from traditional parametric models to ML is driven by the need to process increasingly complex datasets where traditional assumptions of linearity and homoscedasticity often fail. Their review emphasizes that ML techniques are particularly well-suited for post-collection imputation, as they can autonomously identify intricate patterns in high-dimensional data without the rigid constraints of classical statistical models.

Furthermore, recent assessments of the state of the art in official statistics underscore that the focus has shifted from purely experimental applications to the systematic implementation of ML within the Generic Statistical Business Process Model (GSBPM). According to the latest reviews in the field, the maturity of ML algorithms now allows for a more robust treatment of non-response, provided that these methods are accompanied by rigorous quality frameworks that ensure transparency and reproducibility [see e.g. Ranalli, 2025, UNECE, 2021]. This global trend provides the methodological foundation for the AIML4OS project, which seeks to operationalize these advanced techniques across diverse statistical domains.

1.2.4. Specific Methodological Challenges in Post-Collection Imputation

Unlike early imputation [AIML4OS WP9. European Commission, 2026], post-collection imputation often lacks observed ground truth for validation, as the missing values are never recovered during the production cycle. To address this limitation, a robust evaluation framework requires the generation of a synthetic population. By drawing samples and artificially introducing non-response, the performance of various imputation methods can be rigorously benchmarked against the known "ground truth" of the synthetic data (see Spanish use case).

Another key methodological distinction lies in the preservation of data distributions. In early imputation, there is no strict requirement to maintain distributional consistency before and after imputation, as respondents and non-respondents often belong to inherently different distributions. In such scenarios, the objective is to maximize predictive accuracy for non-respondents by leveraging longitudinal data and external administrative sources from all units, and cross-sectional information from respondents.

Conversely, in post-collection imputation, a more conservative strategy is often preferred: preserving the distributional characteristics of the respondents. While certain traditional imputation methods are specifically designed to meet this requirement, standard Machine Learning (ML) models do not inherently guarantee distributional consistency unless they are intentionally overfitted, a practice that is generally undesirable. Consequently, some projects within this deliverable propose an additional post-processing step to ensure that the mean and statistical properties of the imputed values remain aligned with the original observations (see the Luxembourg use case).

Methodological Approach

The project implemented a standardized ML pipeline across all use cases, structured into five main phases:

1. **Data Preprocessing and Synthetic population generation:** Integration of survey microdata with administrative registries. In cases where the ground truth was unavailable, high-quality synthetic populations were generated to allow for design-based error evaluation.

2. **Feature Engineering:** Extraction of temporal dynamics (lags, rolling averages), spatial aggregations, and categorical encodings to maximize the predictive power of the models.

3. **Model Selection and Tuning:** Comparative analysis of state-of-the-art algorithms, including **Random Forest, XGBoost, LightGBM, and k-Nearest Neighbors (kNN)**, using Bayesian optimization for hyperparameter tuning.

4. **Predictions post-processing:** In specific applications, raw model outputs must undergo a post-processing stage. This is necessary either to reverse mathematical transformations applied to the target variable during preprocessing (e.g., back-transformation to the original scale) or to ensure that the imputed values satisfy logical and domain-specific constraints inherent to the statistical problem.

5. **Dual-Level Evaluation:** Models were assessed not only for micro-level predictive accuracy (RMSE, MAE) but also for their ability to preserve macro-level statistical properties and population aggregates.

1.3. Scope and structure

The primary objective of Deliverable 9.2 is to provide a comprehensive methodological assessment of ML techniques applied to post-collection imputation across six distinct use cases:

- **Tourism Statistics (Poland):** Imputing foreign tourist flows in accommodation establishments.

- **Price Statistics (Luxembourg):** Enhancing hedonic models for consumer electronics.

- **Labour Force Survey (Spain):** Predicting employment status using administrative linkages and synthetic populations.
- **Construction Statistics (Portugal):** Treating annual survey data with administrative census information.
- **Earnings Data (Slovenia):** Evaluating ML imputation in monthly employment earnings.
- **International Trade (Austria):** Correcting erroneous and missing values in trade micro-data.

2

Projects Description

In this chapter the six projects related to post-collection imputation are described. The notation could change among projects. First a brief summary of the projects:

Poland (PL-Accommodation): This project focuses on improving the estimation of missing values for foreign tourists in accommodation establishments, with a particular emphasis on country-of-origin information. The objective is to use ML to capture complex temporal and regional patterns that traditional methods may overlook. The target variables are the number of tourists and overnight stays, which are **numerical** values. The data is collected on a **monthly** basis and is characterized by strong seasonality.

Luxembourg (LU-Price): The goal of this use case is to enhance price index construction by imputing missing prices for products that are not available in all comparison periods, particularly in markets with high product churn like consumer electronics. The project explores ML as an alternative to semi-logarithmic hedonic regressions to estimate **numerical** item prices. The data, sourced from web-scraping and scanner data, follows a **monthly** periodicity.

Spain (ES-LFS): This project aims to predict the missing employment status of individuals in the Spanish Labour Force Survey (LFS) by linking survey microdata with Social Security administrative records. The study evaluates whether ML-based imputation provides more accurate population aggregates compared to traditional reweighting techniques. The target variable is employment status (employed, unemployed, or inactive), which is **categorical**. The survey is **quarterly**, while administrative sources are continuous and the analysis utilizes a synthetic population for design-based error evaluation.

Portugal (PT-Construction): This study investigates how ML models can assist in estimating missing stratification variables in the Annual Survey on Construction Enterprises. By leveraging the "Simplified Business Information" administrative census, the project seeks to impute the revenue on buildings by type of construction, a **numerical** target variable. The underlying data follows an **annual** periodicity and relies heavily on historical enterprise behavior and NACE classifications.

Slovenia (SI-Earnings): The primary objective of this project is to evaluate ML-based imputation methods using monthly employment earnings data supplemented with data from administrative and statistical registers. The evaluation focuses on predictive performance and operational efficiency for potential use in a production environment, with regular earnings per hour as the **numerical** target variable. The employment earnings data are collected and processed on a **monthly** basis.

Austria (AT-ITS): This project aims to enhance the quality of International Trade Statistics by imputing values identified as erroneous or as "placeholders" in import records. The models focus on two variables: weight, which is **numerical**, and the Combined Nomenclature (CN) code, which is a **categorical** target. The dataset consists of millions of micro-level observations collected on a **monthly** basis, requiring strategies to handle highly imbalanced classes in product codes.

2.1. Imputation of foreign tourist in accommodation establishments survey (PL) [PL-Accommodation]

2.1.1. Description

Accurate information on tourist flows is essential for monitoring tourism activity, supporting evidence-based policymaking, and producing reliable official statistics. Accommodation establishment surveys constitute one of the main sources of tourism data and provide information on the number of tourists, overnight stays, and visitor characteristics.

However, a recurring challenge in tourism statistics is the occurrence of missing or incomplete data, particularly regarding foreign tourists and their country of origin. While accommodation establishments generally report aggregate information, detailed reporting for foreign visitors is often incomplete or unavailable. Missing observations may result from non-response, reporting burden, operational constraints, or inconsistencies in data collection processes.

During the period 2018–2020, missing information on foreign tourists was addressed through statistical imputation procedures, with imputed observations representing approximately 7–11% of reported foreign tourist data. Although such methods enable the production of complete statistical outputs, traditional imputation approaches may not fully capture the strong seasonality, regional diversity, and complex temporal patterns that characterize tourism demand.

This project addresses the challenge of improving the estimation of missing values for foreign tourists in accommodation establishments, with particular attention to country-of-origin information. The objective is to investigate whether Machine Learning techniques can provide more accurate and robust imputations than conventional methods.

To evaluate their effectiveness, an experimental framework will be implemented in which artificial missing values will be introduced into complete datasets at levels of 10%, 20%, and 30%. Different imputation methods will then be applied and their outputs compared with the original observed values. The results will support the identification of the most suitable approach for handling missing tourism data and contribute to improving the quality and reliability of official tourism statistics.

2.1.2. Use of Machine Learning

Machine Learning methods will be applied to model and reconstruct missing observations of foreign tourist counts in accommodation establishment survey data. The analysis will focus on the ten countries generating the largest number of foreign tourists in Poland, allowing the models to learn country-specific tourism patterns and differences in travel behaviour.

The ML approach will exploit relationships across multiple dimensions of the data, including time, seasonality, regional characteristics, and historical tourism dynamics. By incorporating these dependencies, the models are expected to capture patterns that may not be fully represented by conventional imputation techniques.

Several ML algorithms will be implemented and evaluated to determine their ability to recover missing values under different levels of data incompleteness. Performance assessment will be based on comparisons between imputed and observed values, with particular attention to preserving temporal trends, seasonal variation, and regional consistency.

The resulting framework will provide evidence on the suitability of ML methods for supporting statistical production processes in tourism surveys and improving the quality of derived indicators for international tourism analysis.

2.1.3. Exploratory Data Analysis

Description of the input data:

The input data used in this project originate from the *KT-1: Report on the occupancy of tourist accommodation establishments*, which is an official monthly statistical survey collecting information on tourism activity in accommodation establishments in Poland.

The survey covers tourist accommodation establishments regardless of ownership structure, location, or type of activity, as well as establishments primarily intended for non-tourism purposes that are temporarily used by tourists (e.g., student dormitories or sports and recreation centres). Accommodation establishments are classified according to the NACE Rev. 2 classification and include the following activity groups:

- 55.1 – Hotels and similar accommodation,
- 55.2 – Holiday and other short-stay accommodation,
- 55.3 – Camping grounds, recreational vehicle parks and trailer parks.

Overall, the survey covers 20 types of tourist accommodation establishments.

The KT-1 survey is conducted monthly and follows a census approach. The survey covers all accommodation establishments in Poland that provide at least 10 bed places, regardless of ownership structure, and each eligible establishment is legally required to submit the report every month. In total, the survey includes approximately 14,859 establishments. Although the official reporting deadline is the 10th day following the reference month, reports are typically submitted throughout the entire month. The average unit response rate is approximately 80%, with seasonal fluctuations affecting reporting completeness.

The analysis focuses on item non-response, which occurs predominantly for variables related to foreign tourists. In particular, missing observations concern the number of foreign tourists and overnight stays classified by country of residence.

The dataset contains both categorical and count variables and includes information collected at the accommodation establishment level, such as:

- establishment characteristics (name, type, address, and category),
- months of operation,
- number of rooms and bed places,
- monthly occupancy indicators (number of tourists and overnight stays),
- occupancy indicators for foreign tourists by country of residence.

For the purpose of this project, the analysis is limited to foreign tourists originating from the ten countries with the highest number of tourist arrivals recorded in Poland. Restricting the scope to the most represented countries allows the study to focus on the largest and most stable tourism segments while preserving meaningful seasonal and regional patterns relevant for evaluating Machine Learning imputation methods.

Data Analysis:

An exploratory analysis of foreign tourism in accommodation establishments was conducted for the years 2018–2020. The analysis focused on temporal trends, regional differences, country-of-origin structure, and occupancy indicators in order to identify patterns relevant for missing data imputation.

Temporal patterns and seasonality

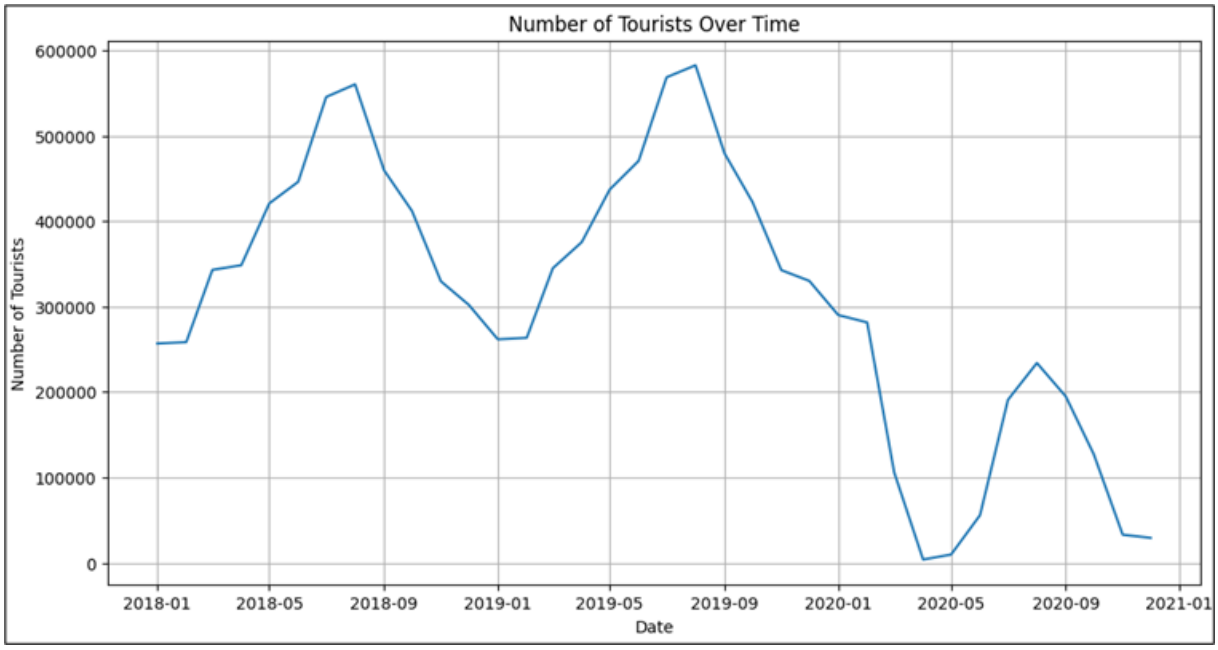


Figure 2.1 Number of tourists over time and regions

A strong seasonal pattern was observed in the total number of foreign tourists. In both 2018 and 2019, tourist arrivals increased gradually during spring and reached their maximum levels during the summer months, particularly in July and August. Similar behaviour was observed for the number of overnight stays, indicating stable seasonal tourism demand.

A major structural break occurred in March 2020 following the outbreak of the COVID-19 pandemic. International travel restrictions and lockdown measures resulted in a sharp decline in foreign tourist arrivals and overnight stays. Although tourism activity partially recovered during summer 2020, tourist volumes reached only approximately one quarter of the levels observed in the corresponding period of 2019. Additional declines were visible toward the end of 2020 due to renewed restrictions and temporary closure of accommodation establishments.

The average number of foreign tourists per accommodation establishment confirms these observations. During summer periods in 2018 and 2019, establishments accommodated approximately 20 foreign tourists on average, whereas in 2020 this value decreased by nearly half. This suggests not only lower tourism demand but also reduced operational activity of accommodation facilities.

Country-of-origin structure

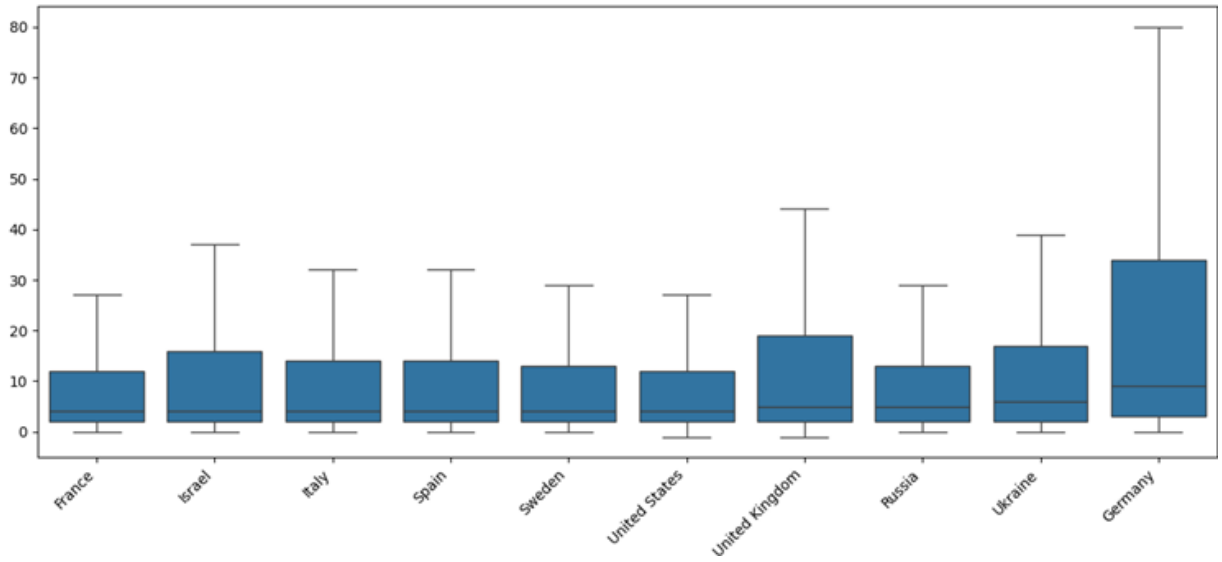


Figure 2.2 Distribution of number of foreign tourists by country in 2018

The boxplot above presents the distribution of the number of tourists for the 10 countries with the highest tourist arrivals in accommodation facilities. The highest values are observed for tourists from Germany and the United Kingdom. It should be noted that these figures represent the number of tourists per single accommodation establishment. The average number of tourists for other nationalities is relatively similar, with no major differences observed between most of the remaining countries.

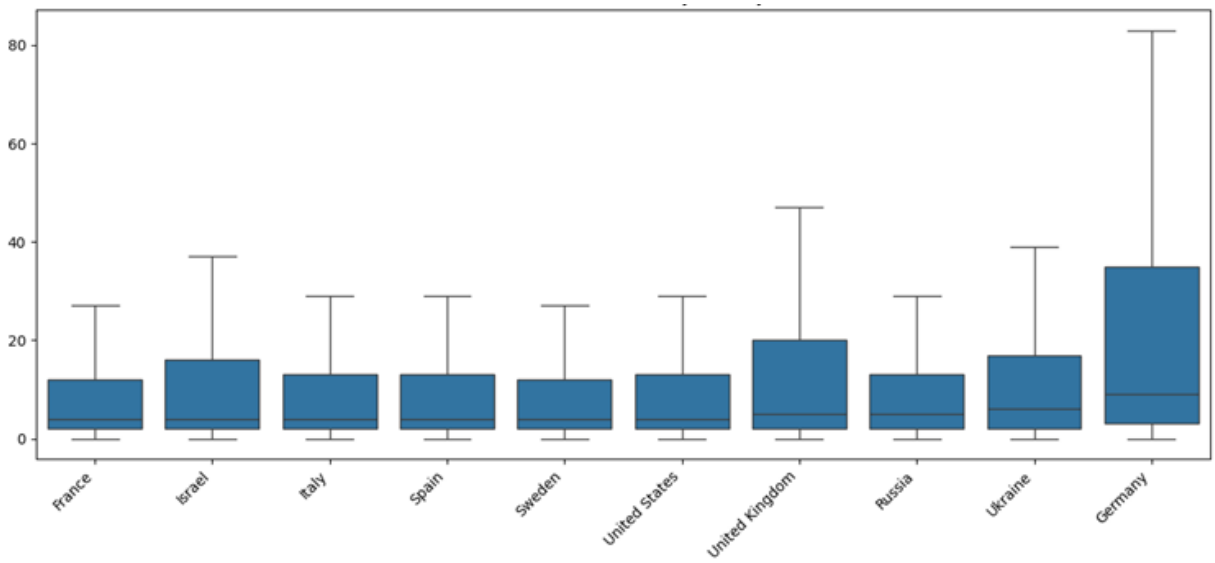


Figure 2.3 Distribution of number of foreign tourists by country in 2019

In 2019, the number of tourists from Germany also tends to reach the highest observed values. The United Kingdom is another country with similarly high levels of tourist arrivals. The average number of tourists from the remaining countries is at a comparable level. Only the mean number of tourists from Germany is slightly higher than that of other nations.

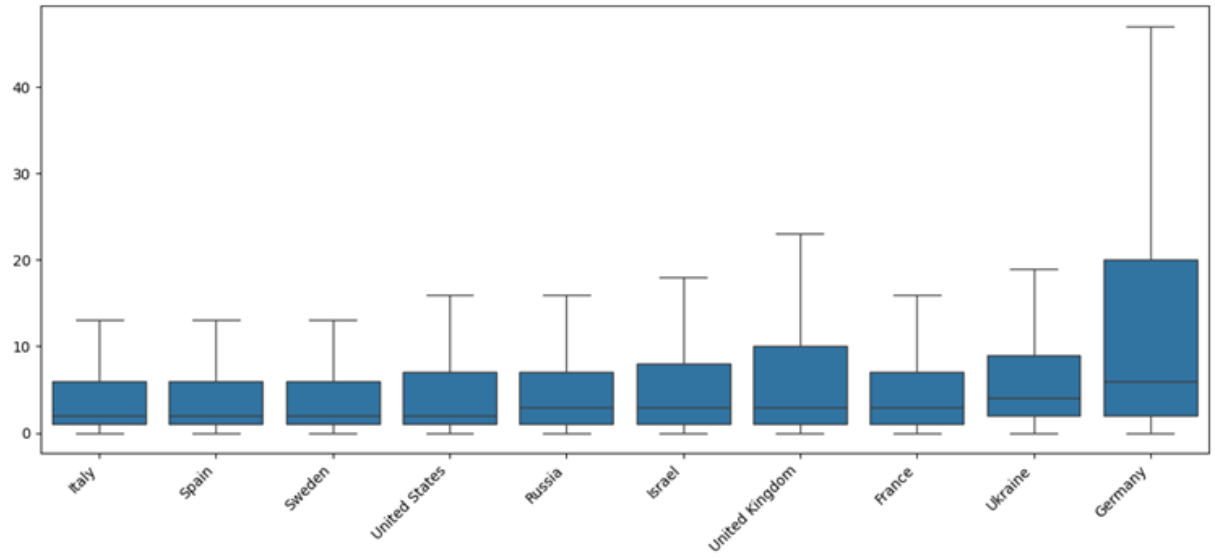


Figure 2.4 Distribution of number of foreign tourists by country in 2020

The highest number of tourists is also observed among visitors from Germany; however, it should be noted that the values are significantly lower than in the previous year. It is also visible on the scale that the values for the remaining countries are much lower compared to earlier years, which reflects the reduced tourist movement due to restrictions.

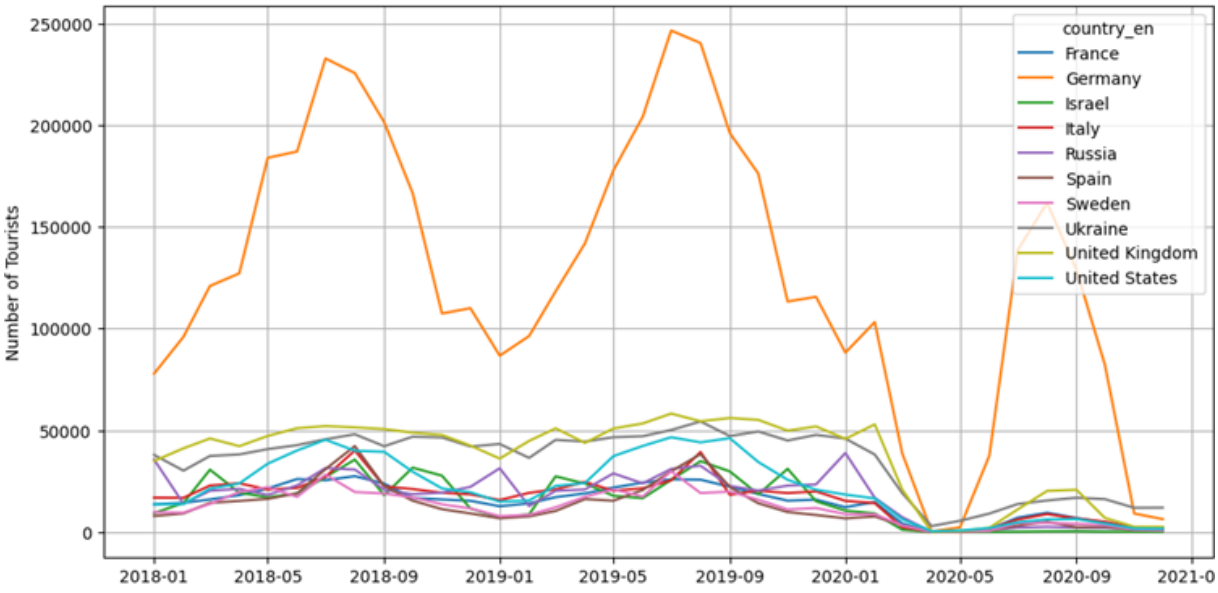


Figure 2.5 Number of Foreign Tourists by Country of Origin

When analyzing the values over the three-year period, it is clearly visible that tourists from Germany constitute by far the largest group visiting Poland. A distinct seasonality can also be observed for this group, as the number of German tourists reaches its peak during the summer months. In the final year of the dataset, these values decline significantly and even drop to zero in April 2020, due to the closure of accommodation facilities for visitors as a result of pandemic-related restrictions. The number of tourists from other countries is considerably lower. Although in most cases the highest values are also recorded during the summer months, the seasonal pattern is not as clearly pronounced as in the case of German tourists. A noticeable seasonality is observed only for tourists from Italy.

Table 2.1 Tourist Statistics by Country of Origin (2018–2020)

Country	Year	Count	Sum	Mean	Median	Std	Min	Max
France	2018	17302	233778	14	4	31	0	789
Germany	2018	37237	1836092	49	9	174	0	8401
Israel	2018	8923	260634	29	4	104	0	2152
Italy	2018	17929	271096	15	4	35	0	920
Russia	2018	18719	274663	15	5	43	0	2150
Spain	2018	12090	212901	18	4	51	0	1100
Sweden	2018	12316	200548	16	4	41	0	828
Ukraine	2018	28587	497764	17	6	46	0	1388
United Kingdom	2018	19108	555098	29	5	79	0	1639
United States	2018	14117	341006	24	4	99	0	3099
France	2019	17374	232346	13	4	31	0	1024
Germany	2019	38745	1912378	49	9	176	0	7846
Israel	2019	8936	255699	29	4	103	0	1899
Italy	2019	17985	264279	15	4	35	0	1011
Russia	2019	19490	290265	15	5	44	0	2126
Spain	2019	11959	198989	17	4	49	0	1360
Sweden	2019	12574	191215	15	4	38	0	823
Ukraine	2019	30832	556382	18	6	49	0	1382
United Kingdom	2019	20198	604950	30	5	79	0	1656
United States	2019	14902	373871	25	4	104	0	3101
France	2020	8692	64699	7	3	20	0	762
Germany	2020	24886	796547	32	6	146	0	8646
Israel	2020	1761	21920	12	3	42	0	854
Italy	2020	8923	64023	7	2	17	0	466
Russia	2020	7224	75204	10	3	43	0	2673
Spain	2020	5071	33347	7	2	13	0	261
Sweden	2020	5189	39335	8	2	20	0	346
Ukraine	2020	19386	206258	11	4	32	0	1277
United Kingdom	2020	10955	185700	17	3	52	0	2011
United States	2020	5752	68629	12	2	43	0	1028

The summary of descriptive statistics for the number of tourists by country of origin in the years 2018–2020 indicates clear differences both in the scale of tourist flows and their variability. In 2018–2019, the highest total values and averages were recorded for Germany (total approx. 1.8–1.9 million, mean 49), making it by far the most important source of inbound tourists. At the same time, this country exhibits very high variability (standard deviation approx. 170), indicating substantial fluctuations in arrival numbers. Relatively high average values are also observed for the United Kingdom and Israel (approx. 29–30), while countries such as France, Italy, and Spain show lower averages (approx. 13–18). Across all analyzed countries, medians are significantly lower than means (typically around 4–6), which indicates a strong right-skewed distribution and the presence of extreme values. This is further confirmed by very high maximum values (e.g., over 8,000 for Germany) alongside minimum values equal to zero. This suggests that most observations are concentrated at low levels, while the mean is inflated by occasional very high values. In 2020, a clear collapse in tourist traffic is observed across all countries. Total values and means decrease often by several tens of percent (e.g., for Germany to approx. 796 thousand and a mean of 32), while medians fall to the range of 2–4. This indicates a general reduction in the intensity of arrivals and a lower frequency of high-value observations. These changes are consistent with the global decline in mobility during the pandemic period.

Regional distribution

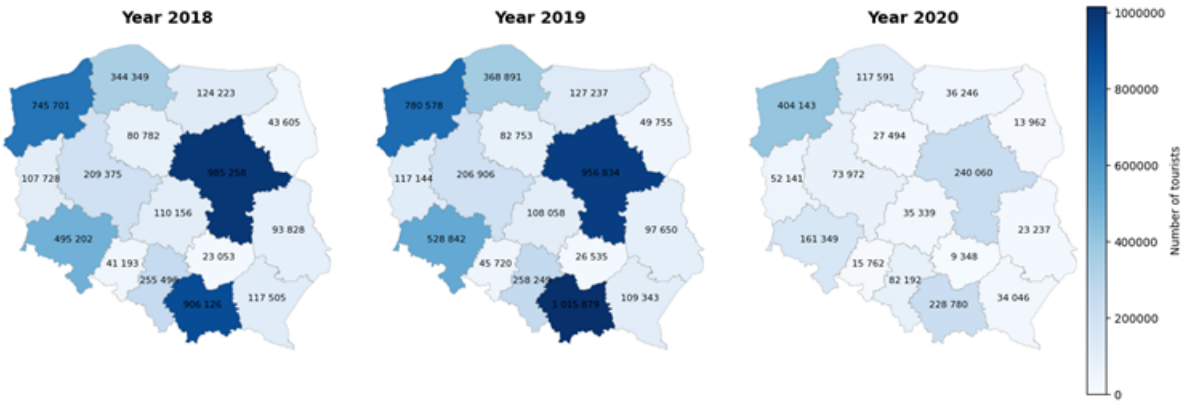


Figure 2.6 Number of Tourists Over Time

Foreign tourism activity showed substantial regional concentration. The highest numbers of foreign tourists were recorded in the Mazowieckie and Małopolskie voivodeships, mainly due to the attractiveness of Warsaw and Kraków as major urban and tourism centres. Together, these regions attracted more than 3 million foreign tourists annually in 2018 and 2019.

In 2020, regional tourism patterns changed considerably. Coastal regions and Lower Silesia gained relatively greater importance, likely reflecting changing travel preferences and the advantage of geographical proximity to neighbouring countries.

Implications for imputation modelling

The exploratory analysis demonstrates several characteristics that are directly relevant for the missing data imputation task. First, tourism data exhibit strong seasonality and substantial regional variation. Second, tourist counts differ considerably across countries of origin, resulting in heterogeneous distributions. Third, the COVID-19 pandemic introduced abrupt structural changes that may influence model performance.

These findings indicate that effective imputation models should incorporate temporal information, regional context, and country-specific behaviour in order to accurately reconstruct missing observations. The identified patterns support the use of Machine Learning methods, which are capable of modelling complex and non-linear relationships within tourism data.

2.2. Imputation of Price Statistics (LU) [LU-Price]

2.2.1. Description

Price indices are traditionally constructed by comparing the prices of identical products across two time periods. However, in markets with dynamic product assortments, this matched-model approach — limited to products available in both periods — can be insufficient. To address this, prices can be imputed for products that appear in only one of the two comparison periods.

Imputed prices are typically estimated using models that relate observed prices to a set of price-determining characteristics. A common method involves semi-logarithmic multiple regression. In this project, we explore the application of machine learning (ML) techniques to enhance the imputation of such imputed prices.

ML methods can be seen as an alternative to a traditional regression in order to improve accuracy. Both approaches eventually allow to estimate missing prices. The main objective of ML methods is to find the best possible prediction for a price, which can be learned from the available data. One could argue that a more accurate price imputation will lead to a more accurate price index.

Our use case involves data collected from the web or from scanner or transaction data. The dataset forms an imbalanced panel, where prices are missing in periods when a product is not sold. Alongside price data, a set of product features is available to describe each item. This real-world example helps us to validate the ML-based imputation approach and assess its effectiveness in supporting quality adjustment procedures within price statistics.

There have been various proposals for relying on ML approaches in connection with price indices [see e.g. Bajari et al., 2025, Calainho et al., 2024, Cafarella et al., 2023, Zeng and Rao, 2025]. This work follows-up on earlier work presented in May et al. [2024]. These papers typically use ML for prediction, but rarely tie it fully back to the price index theory.

2.2.2. Use of Machine Learning

In the field of Machine Learning (ML), practitioners typically adopt a data-driven perspective: models are designed to discover predictive relationships directly from the data with minimal reliance on prior theoretical constraints. This flexibility makes ML a promising candidate for improving price imputations, especially in product categories with short life cycles and rich item characteristics—such as consumer electronics.

However, the data-first nature of ML contrasts with the theory-driven paradigm that underpins official price statistics, where constant-quality measurement and imputation procedures are anchored in a well-established conceptual framework. The opposition between theory-driven and data-driven approaches in ML is also discussed in Daas and Puts [2023].

This raises an important question: does it make sense to use ML-based imputations within a framework that is fundamentally theory-based? One of the objectives of our study is precisely to investigate this issue. We revisit the theoretical foundations of price measurement to examine under what conditions ML models can produce imputations that are both empirically accurate

and conceptually aligned with constant-quality price index methodology.

We will thus evaluate the added value of ML according to two dimensions:

- **Conceptual assessment** We first built an analytical link to CPI theory or show how the imputations from an ML model replicate an exact hedonic index under standard index number theory conditions. The key conditions that we identified is that the ML model should have no bias: on average, the imputed prices from the ML model in each period should be equal to the observed prices. This condition should hold in each period.
- **Empirical assessment** We will evaluate the different imputation methods (ML models) at the individual price observation level, using standard ML metrics, and at the aggregated price index level.

2.2.3. Exploratory Data Analysis

Description of the input data:

STATEC is using alternative data sources for compiling the Consumer Price Index (CPI). This includes scanner data or data collected from the web. In this project, we are using data collected from a website of a major retailer selling consumer electronics. The data are collected since 2021 through an API. Prices and product characteristics are obtained in JSON files before being stored in local csv files. Data for around 10 000 products are collected each week. With the help of the categories used on the website, it is possible to map the individual products to product categories used in the CPI.

In this project, we are going to work with a dataset covering TVs. There are in total 342 TV models, each identifiable by a unique European Article Number (EAN code), over 13 months (June 2021 to June 2022).

Each TV model can be described with the help of 44 technical characteristics. In addition there is a monthly price.

The data used for this project is not the raw data obtained from the website. Some data cleaning, normalisation and imputations have been applied to the raw data in order to obtain the data used in this analysis.

Data Analysis:

This dataset exhibits a very high degree of churn. Every month new products appear, while others are not on sale any more.

In figure 2.7, each row in that table corresponds to an individual TV model, indicating its availability and tracking how its price evolves over time. The dataset covers a 13-month period, although not all TV models are available throughout the entire observation window. A blue color indicates that the model is on offer, whereas a white color indicates that the model is not available anymore on the website. The shade of the blue color illustrates the level of the price in the month compared to the average price over the entire life-cycle of the product. In general, there are high entry prices and low exit prices. This is illustrated by the fact that the blue shades are generally lighter at the beginning of each TV model's lifespan and darker toward the end.

We use in total 13 periods. Table 2.2 shows the number of items per period.

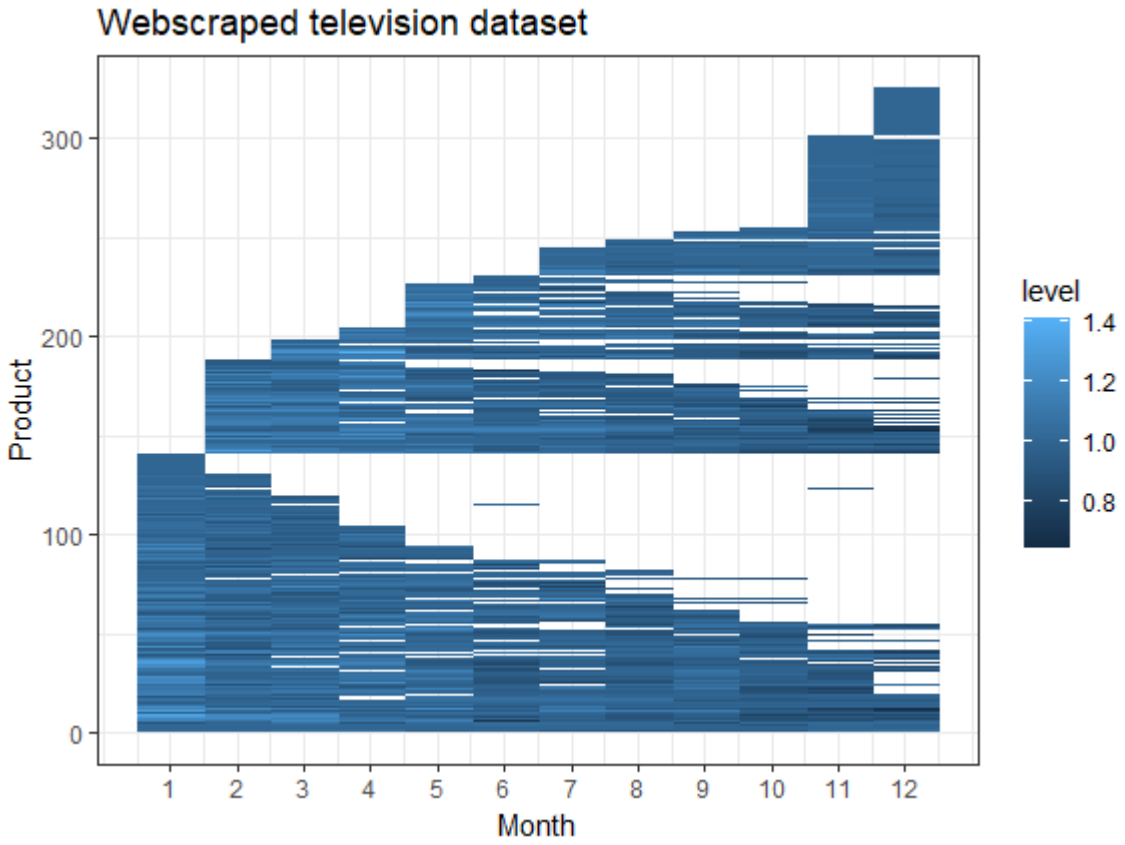


Figure 2.7 Availability and price levels of TV Models over the observation period

Period	Number of items
1	141
2	176
3	173
4	154
5	163
6	154
7	158
8	162
9	148
10	137
11	162
12	164
13	155
Average	157

Table 2.2 Number of TV models available in each period

The key assumption in price indices is to compare prices of the same *basket* of items over time. This is a challenge with high product churn, as seen in figure 2.7. We aim to solve the lack of matching through imputations. When comparing the price change between two periods (let us say $t1$ and $t2$), we impute a price in $t2$ for all the items available in period $t1$, and we impute a price in $t1$ for all the items available in period $t2$. This idea is illustrated in table 2.3. Note that imputations are applied to all items in the comparison period, including those for which an observed price is already available.

This allows us to calculate two constant basket aggregate price changes: an aggregated price change for the basket N_1, and an aggregated price change for the basket N_2.

$$I^1_{t1,t2} = \frac{\left(\Pi_{i \in N_1} \hat{p}_i^{t2}\right)^{\frac{1}{n_1}}}{\left(\Pi_{i \in N_1} p_i^{t1}\right)^{\frac{1}{n_1}}}$$

$$I^2_{t1,t2} = \frac{\left(\Pi_{i \in N_2} p_i^{t2}\right)^{\frac{1}{n_2}}}{\left(\Pi_{i \in N_2} \hat{p}_i^{t1}\right)^{\frac{1}{n_2}}}$$

The final price index is obtained as the average of these two price changes. The theoretic foundations of this price index are further discussed in section 5.2.

$$I_{t1,t2} = \sqrt{I^1_{t1,t2} * I^2_{t1,t2}}$$

Basket	Observed prices	Imputed prices
N_1 (set of items available in $t1$)	p_i^{t1} $(\forall i \in N_1)$	$\hat{p}_i^{t2}$ $(\forall i \in N_1)$
N_2 (set of items available in $t2$)	p_i^{t2} $(\forall i \in N_2)$	$\hat{p}_i^{t1}$ $(\forall i \in N_2)$

Table 2.3 Imputations of baskets

2.3. Imputation for non-response in Labour Force Survey using synthetic data (ES) [ES-LFS]

2.3.1. Description

The Spanish Labour Force Survey (EPA) is a continuous household survey produced by the National Statistics Institute (INE) in compliance with Regulation (EU) No. 2019/1700 (the Integrated European Social Statistics framework) and Commission Implementing Regulation 2019/2240. The EPA aims to measure the evolution of labour market activity among the population residing in Spain through their relationship with employment and unemployment. Thus, the core target variable is *employment status*, i.e., "the classification of each individual of working age according to their labour situation in the reference week, distinguishing between employed (those who have worked for at least one hour in exchange for pay, profit, or family gain, or who are temporarily absent from their job), unemployed (those who are without work, actively seeking employment, and available to start work), and inactive (those who are not in the labour force, including students, retirees, homemakers, or persons unable to work)." The survey covers a wide range of additional variables, including occupation, economic activity sector, hours worked, type of contract, duration of job search, and previous employment history.

The aim of this project is not only to predict the missing employment status of individual respondents, but ultimately to improve the estimation of labour market aggregates such as the number and proportion of employed, unemployed and inactive individuals. Two alternative estimation strategies are considered.

The first strategy corresponds to the current methodology used in official statistics. Aggregate quantities are estimated using only responding units and non-response adjusted calibration weights:

$$\hat{\theta}_I = \sum_{i \in S'} f_i' y_i,$$

where S' denotes the set of respondents and f_i' are the calibrated weights corrected for non-response.

The second strategy uses imputed values for the missing target variable and estimates aggregates from the complete sample:

$$\hat{\theta}_{II} = \sum_{i \in S} f_i y_i^*,$$

where S denotes the original sample, f_i are the standard calibration weights and y_i^* represents either the observed or the imputed employment status.

The key question addressed by this project is therefore whether $\hat{\theta}_I$ or $\hat{\theta}_{II}$ provides more accurate estimates of population aggregates.

In official survey methodology, estimator quality is commonly assessed through the design-based mean squared error

$$N\text{MSE}(\hat{\theta}) = \sum_{s \in \mathcal{S}} p(s) \left(\hat{\theta}(s) - \theta \right)^2,$$

where $\mathcal{S}$ denotes the set of all possible samples under the sampling design, $p(s)$ is the probability of selecting sample s, and θ is the true population parameter.

However, after data collection only one realized sample is available, making it impossible to directly evaluate the design-based error of competing estimation procedures. To overcome this limitation, the project relies on synthetic populations that mimic the main characteristics of the real Labour Force Survey population. These synthetic populations allow repeated sampling

experiments and Monte Carlo simulations to be performed, providing a controlled framework in which the design-based performance of both estimation strategies can be compared.

The workflow of the project is therefore as follows: first, a synthetic population is generated by combining Labour Force Survey microdata with auxiliary administrative information. Second, repeated samples are drawn from this synthetic population and different non-response mechanisms (MCAR, MAR and MNAR) are introduced. Third, both estimation strategies are applied: the current reweighting and calibration approach, and the imputation-based approach. Finally, the resulting estimates are compared against the known population values, allowing the design-based performance of each methodology to be assessed through Monte Carlo simulation. Consequently, the development and validation of a high-quality synthetic population constitutes a fundamental component of the project methodology.

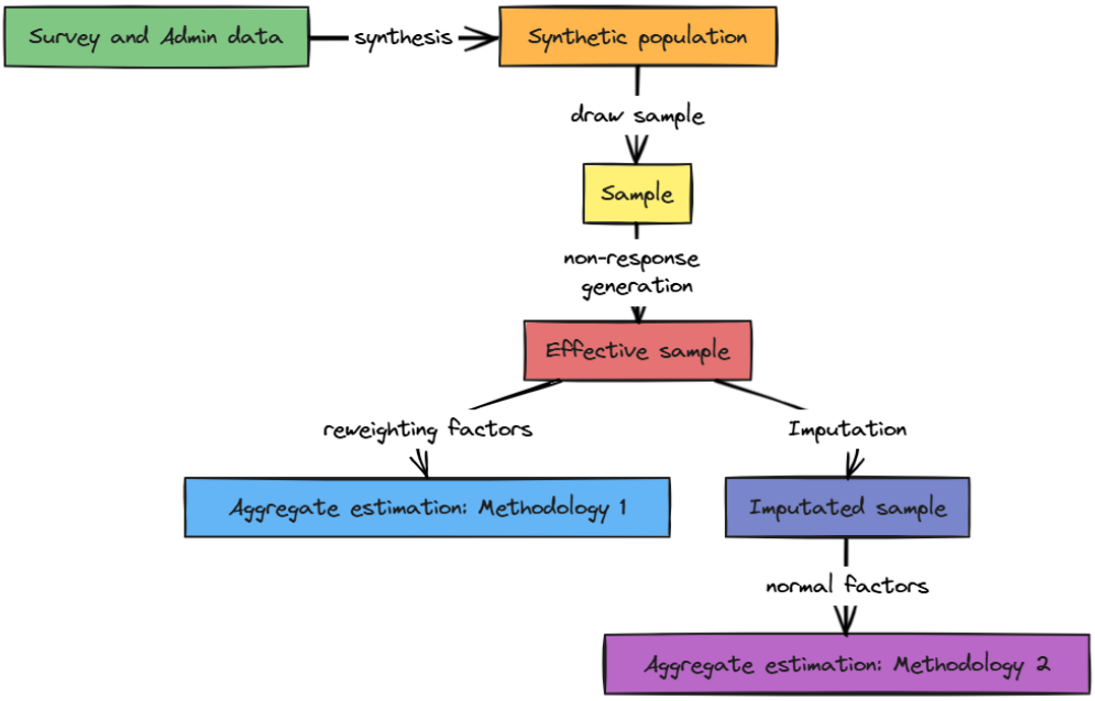


Figure 2.8 Overall workflow of the project.

2.3.2. Use of Machine Learning

Machine learning plays two complementary roles in this project.

First, it is used in the construction of the synthetic population. Since the design-based performance of alternative estimation procedures cannot be evaluated from a single observed sample, a realistic synthetic population is required to perform repeated sampling experiments. Several variables in the synthetic population are generated using machine learning models, allowing complex relationships present in the original data to be reproduced. The resulting synthetic population provides a controlled environment in which design-based quantities such as bias, variance and mean squared error can be approximated through Monte Carlo simulation.

Second, machine learning is used for the imputation of missing unit-level values of the target variable, namely employment status. Traditional imputation procedures commonly used in survey statistics are often based on relatively simple donor-selection rules or regression models.

In contrast, machine learning algorithms can exploit complex relationships between the target variable and a large set of auxiliary variables, including non-linear effects and interactions that may be difficult to specify explicitly.

In this project, the imputation models are trained using both survey variables and administrative registry information. The availability of high-quality auxiliary data from Social Security records is expected to improve the prediction of employment status, particularly in situations where non-response is related to labour market characteristics. The ultimate objective is not only to improve the accuracy of individual imputations, but also to assess whether these improvements translate into more accurate estimates of population aggregates.

2.3.3. Exploratory Data Analysis

Description of the input data:
The input data consists of a linkage between publicly available Labour Force Survey microdata and internally available Social Security administrative records. The survey data provide demographic and socioeconomic information, while the administrative sources contain detailed information on labour market participation, benefit receipt, and Social Security affiliation. The combination of both sources creates a rich set of predictors that can potentially explain employment status.

Data Analysis:
An exploratory analysis was conducted to evaluate the relationship between the available predictors and the target variable. As expected, several administrative variables showed a strong association with employment status. In particular, Social Security affiliation, employment solicitor status, retirement pension receipt, disability pension receipt and other state benefits were found to be highly informative predictors. This result suggests that model-based imputation methods may be able to recover missing employment status values with a high degree of accuracy.

By contrast, most survey variables exhibited only moderate or weak marginal associations with employment status. Nevertheless, variables such as age, sex, nationality, region and household composition may still contribute useful predictive information when combined with administrative records. For this reason, both survey and administrative variables were considered during the feature selection process.

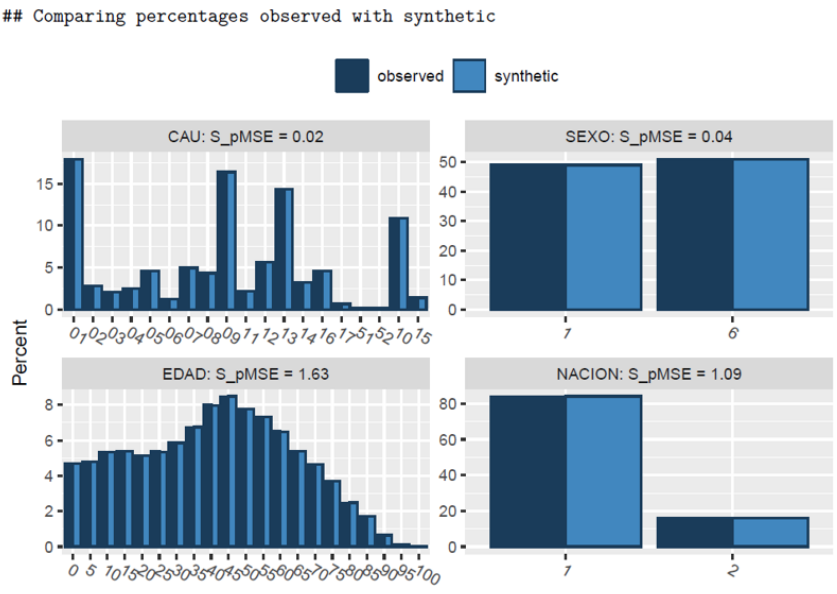
The dataset used in the project consists of a merge of publicly available Labour Force Survey microdata with auxiliary Social Security registry data. This dataset has already been cleaned and edited by the Labour Market department.

A synthetic population that replicates fundamental demographic distributions was then generated using the "simPop" R package. Extreme Gradient Boosting was used to synthesize most of the variables in the dataset. Synthesis utility was measured through graphical analysis and the pMSE metric, and was found to be satisfactory. The pMSE (Propensity Score Mean Squared Error) is a metric used to evaluate the quality of synthetic populations by measuring how distinguishable synthetic data are from real data. The procedure combines the real and synthetic datasets into one, creates a binary indicator (0 for real, 1 for synthetic), and trains a classifier—typically a logistic regression or CART model—to predict whether each record is real or synthetic based on all other variables. The pMSE is then computed as the average squared difference between the predicted probabilities of being synthetic and the expected baseline probability (usually 0.5 when the datasets are equally sized). Lower values indicate

better synthetic quality, as the classifier cannot easily distinguish synthetic records from real ones.

The univariate evaluation (see figure 2.9) demonstrates an exceptional alignment between the observed and synthesized distributions. As illustrated in the bar charts and histograms for core demographic variables, including region (CAU), sex (SEXO), age (EDAD), and nationality (NACION). The observed empirical distributions (dark blue) and their synthetic counterparts (light blue) show near-perfect visual concordance. This fidelity is confirmed by the standardized utility table, where the standardized propensity score mean squared error (S_pMSE) remains remarkably close to zero across all seventeen attributes. The values range from negligible discrepancies in geographic and administrative indicators (e.g., $S_pMSE = 0.016$ for CAU and $S_pMSE = 0.007$ for DEMANSEPE) to minor deviations in continuous age groups ($S_pMSE = 1.63$) and educational activities ($S_pMSE = 1.38$). These consistently low values indicate that the marginal distributions were replicated with negligible bias, ensuring that the synthetic population retains the fundamental demographic and administrative profile of the original survey.

The multivariate relationship assessment (see figure 2.10) evaluates the preservation of pairwise associations through the two-way S_pMSE correlation matrix. The vast majority of pairwise cells display very pale tones ($S_pMSE < 1.0$), demonstrating that the XGBoost-based synthesis successfully captured the complex joint dependencies and cross-sectional correlations without distorting the underlying structure. The few localized darker cells (reaching values above 7.5) correspond exclusively to pairs of variables governed by quasi-deterministic or structural definitions, such as the relationship between current student status (16.ESTUDIA) and inactive youth status (17.NINI), or specific administrative unemployment registrations (06.DEMANSEPE and 12.PAROSEPE).




```
##
## Selected utility measures:
##           pMSE   S_pMSE df
## CAU          0 0.016265 18
## SEXO         0 0.037734  1
## EDAD         0 1.633612  4
## NACION       0 1.087128  1
## AFISS        0 0.942319  1
## DEMANSEPE    0 0.007025  1
## OFEMSEPE     0 0.241068  1
## PJUBILA      0 0.083036  1
## PINCAPAC     0 0.072302  1
## OTRASP       0 0.025630  1
## OFEMP        0 1.217727  5
## PAROSEPE     0 0.446618  1
## OPI          0 0.424654  3
## SITUP        0 0.508831  7
## ESTUDIOS     0 0.167600  9
## ESTUDIA      0 1.376364  2
## NINI         0 0.455273  2
```

Figure 2.9 Univariate utility evaluation: comparison of observed and synthetic marginal distributions for selected demographic variables (CAU, SEXO, EDAD, NACION) alongside standardized S_pMSE utility measures across all variables

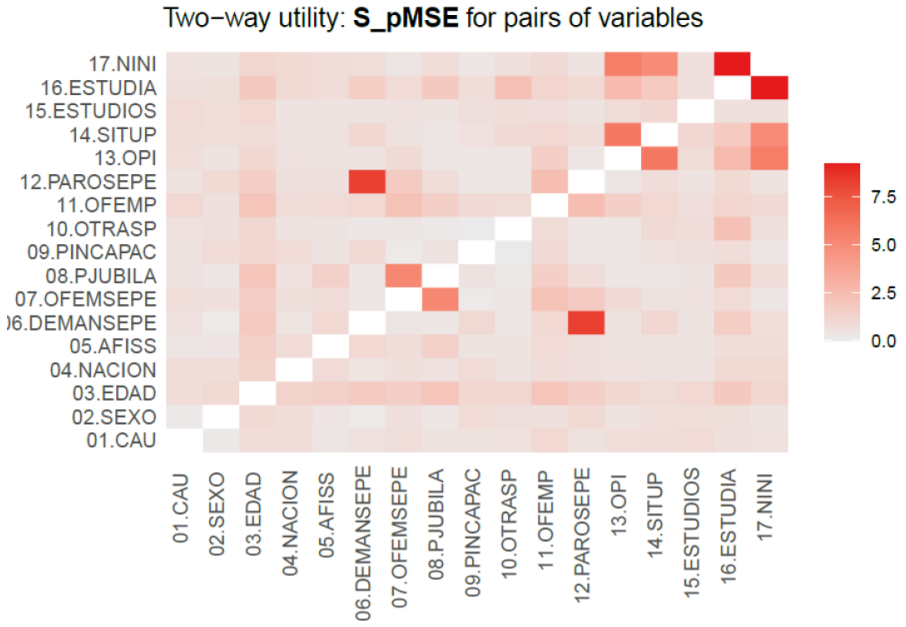


Figure 2.10 Two-way utility heatmap: pairwise standardized propensity score mean squared error (S_pMSE) assessing the preservation of bivariate associations across all synthesized survey and administrative variables

2.4. ML Treatment of the Annual Survey on Construction Enterprises Using Admin Data (PT) [PT-Construction]

2.4.1. Description

Each year, a Survey on Construction Enterprises is conducted to gather critical industry insights. The responses collected are used to build a stratification model, enabling estimations beyond the sampled businesses. However, for this methodology to be effective, all variables used in the stratum definition must be complete, yet some remain missing. This introduces challenges related to non-response and partial non-response, which must be properly addressed to ensure data quality and reliable analysis. The current version of the sample is not used for building the strata. Of particular interest is the Revenue on Buildings by Type of Construction (QF185001), our target for the current prediction exercise. The reference period is 2023. All data for training, testing and other forms of validation always refers to 2023.

To mitigate these gaps, a valuable administrative data source can be leveraged: the Simplified Business Information (IES), collected by the Tax Administration. Unlike survey-based data collection, IES is a census-based system that undergoes rigorous validation through the NSI's Integrated Business Account System (SCIE). SCIE consolidates enterprise-level data, contributing to a broader understanding of sectoral growth and economic trends. When data from this source, from previous years is included in the dataset, it never references the year directly but instead the relative year. For example, for 2022, $n-1$ is used, because 2023 is our reference year n.

This administrative dataset provides essential business characteristics, including geographical classification based on the Nomenclature of Territorial Units for Statistics (NUTS), economic activity sector categorized by the Statistical Classification of Economic Activities in the European Community (NACE), as well as financial indicators such as turnover and employment figures. Beyond general business attributes, IES also includes construction-specific financial data, such as expenditures on building materials, costs associated with building construction, and investments in infrastructure-related projects.

The current value editing is proposed by rules and consists of retrieving last years answer. This is similar to a memory exercise. The hypothesis is that learning from past behaviour could be better than just memorizing the past. The administrative source being used the Simplified Business Information (IES), collected by the Tax Administration is a census-based system that undergoes rigorous validation through the National Statistical Institute's (NSI) Integrated Business Account System (SCIE) and that provides ultimately very similar variables to which a high correlation is expected.

Given the economic structure of the data, a strong relationship is anticipated between the administrative variables and the target, Revenue on Buildings by Type of Construction. Preliminary descriptive analyses have confirmed this correlation, reinforcing the suitability of these variables for predictive modelling.

This is always at enterprise level. The current method is based on using the administrative variable for the current year (I187701 – administrative data proxy for construction value) and deriving a growth factor by comparing it with the previous year (n-1), or, when unavailable, the year before (n-2). This growth factor is then applied to the target variable from the most recent available year, within predefined bounds, to obtain the edited value for the current year (Revenue on Buildings by Type of Construction - QF185001).

A key limitation of this approach is its reliance on historical values of the target variable ($n-1$ or $n-2$). When such values are missing, the method cannot be directly applied for that enterprise. Given the simplicity and structure of this process, it is reasonable to expect that a machine learning approach could learn this relationship and extend it to cases where historical target values of the enterprise are not available.

2.4.2. Use of Machine Learning

The project target is to evaluate how machine learning (ML) models can assist in estimating missing stratification variables, improving data completeness and reliability. Through extensive data exploration, valuable insights are gained in profiling construction enterprises, enabling a deeper understanding of industry dynamics. Additionally derived variables are available that capture interactions between NACE classifications (ECAE) and turnover, as well as key distinctions between the two types of existing questionnaires. These questionnaires are specifically designed for larger enterprises (Type A) and smaller enterprises (Type B), based on their number of employees.

2.4.3. Exploratory Data Analysis

We are lacking in the administrative source information about some enterprises. This should not occur IES should be census like but some 0s are showing up. Some contacts with the data provider were scheduled but in some cases is not possible to get the data.

The variable ESTRATO (derived variable described below) had groups with very few observations thus after some trial it was decided to use ECAE (NACE) as explained in point 1.

It was observed that when focusing solely on the totals from QF185 and QF186 (survey questions), enterprises typically responded to one or the other, suggesting that different types of enterprises may exhibit distinct behaviours. To capture this, two classes — Class X and Class Y — were created to inform the model. However, the relationship between these classes and the reported revenue was not clearly established. As an alternative to categorical classification, a second hypothesis could be considered: calculating a proportion ranging from 0 to 1 to reflect the extent to which each enterprise is more focused on the building sector (QF185) versus infrastructure and other construction works (QF186 & QF187).

Description of the input data:

Surveys and other data sources:

- Annual Survey on Constructions Enterprises (IAEC);
- Administrative data from Simplified Business Information (IES);
- Additional geographical/economical data from enterprise register;
- Transaction data (e-invoices aggregated monthly by supplier and buyer including their respective NACE)

Data Analysis:

The main source for the ML algorithm is the Annual Survey on Constructions Enterprises (IAEC). The reference year is 2023. On that year, 5076 data points should have been collected but 78 were missing. Despite this small percentage (1.5%) it is important to have the complete data due to the stratum approach that will follow (described above in section 2.4.1).

The number of data points will be approximately 5000 every year. The proportion that requires imputation is stable.

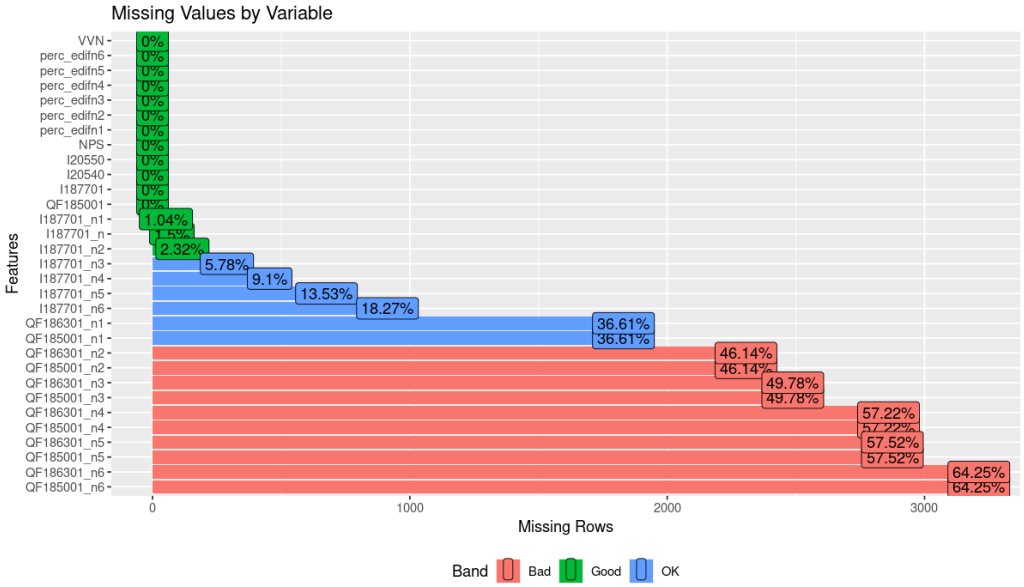


Figure 2.11 Missing Values by Variable

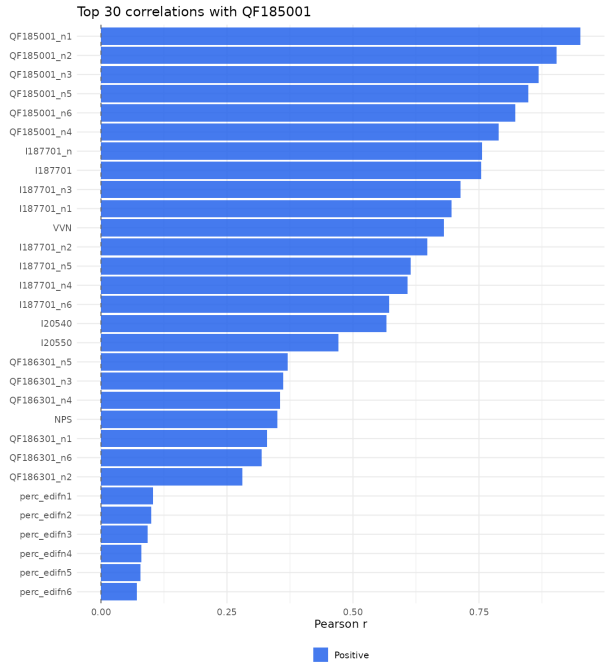


Figure 2.12 Top 30 correlations with Target variable

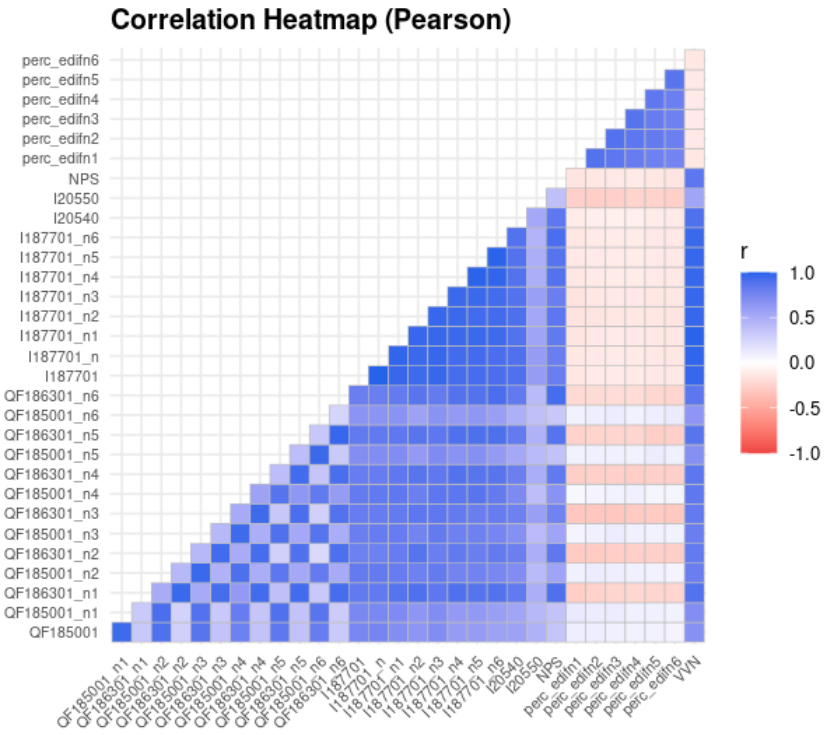


Figure 2.13 Correlation heatmap

QF186301 block (num_QF186301 + _n1 to _n6): near-perfect internal correlation maybe redundant, keep only 1 representative perhaps during testing (depending on the model chosen).

I187701_n block (_n, _n2, _n4, _n6): also highly intercorrelated could be reduced to 1 representative on testing.

perc_edifn1 to edifn6: moderate mutual correlation (lighter orange), less extreme than the numeric blocks, it is worth exploring 2–3 of these for non-linear patterns.

Although the Pearson correlation coefficient measures linear correlation and may not be suitable for all types of variables, all variables in the dataset were retained for ease of application.

2.5. Evaluating Machine Learning Imputation Methods Using Employment Earnings Data (SI) [SI-Earnings]

2.5.1. Description

The project carried out by the Statistical Office of the Republic of Slovenia (SURS) aims to evaluate machine learning (ML) imputation methods using monthly employment earnings data. The project focuses in particular on assessing ML-based approaches with respect to predictive performance, robustness, transparency, and interpretability.

The monthly employment earnings data are currently processed within the SURS in-house generic system for data editing and imputation. While the generic system code itself is open source, its current implementation relies on a non-open-source programming language. The production process incorporates a comprehensive set of edit rules for identifying and correcting inconsistencies, errors, and implausible values, together with established imputation procedures. In this context, the project also explores the use of open-source technologies for ML-based imputation and relevant operational aspects, including computational efficiency, code parametrisation, required manual intervention, traceability, reusability, and reproducibility.

While monthly earnings are the main measure published in monthly earnings statistics, their level depends directly on the amount of working time and is also affected by absences. Regular earnings per hour was therefore used as the target variable for the ML-based imputation experiment to distinguish variation in the level of earnings from variation due to differences in working time. For the evaluation, a simulation-based approach was applied in which missing values were artificially introduced into regular earnings per hour under different missingness mechanisms and at varying missingness rates. This setup enables the performance of the imputation methods to be assessed, as the original values remain available for direct comparison.

2.5.2. Use of Machine Learning

ML is used in the project to investigate an alternative approach to the imputation of missing values in employment earnings data. Regular earnings per hour is used as the numerical target variable, while a set of variables describing employees, their employment, and employers is used as input to the models. Variables that are direct components of the target variable are excluded from the predictors to avoid information leakage.

Several ML algorithms were initially evaluated for imputation, including tree-based ensemble methods and neural network-based approaches. Based on predictive performance and operational considerations, LightGBM was selected for further experimentation. The models are trained on validated observations from the reference-month dataset, with a separate model fitted for each combination of missingness mechanism and missingness rate.

The ML-based imputation approach is evaluated using artificially introduced missing values under different missingness mechanisms and missingness rates. Since the original values are retained for evaluation, the imputed values can be compared directly with the true values. Predictive performance is assessed using several measures covering prediction error, bias, explained variance, and the preservation of variability and distributional characteristics. The results are also benchmarked against traditional imputation methods.

The use of ML is considered not only from the perspective of predictive performance but also with regard to its potential application in statistical production. The project therefore examines operational efficiency. The ML-based approach is envisaged as a potential complement to existing imputation procedures, in this case particularly for newly missing values in the current reference month for which historical information is not yet available. It also has the potential to reduce manual intervention and reliance on manually defined rules, as ML models can learn relationships between variables directly from the data.

2.5.3. Exploratory Data Analysis

Description of the input data:
The dataset consists of monthly earnings, their components, and other relevant variables. Earnings data are published and also serve as input for other surveys. Statistical survey *Earnings* consists of three main data releases. *Earnings of Persons in Paid Employment by Legal Persons* is published on a monthly basis and it represents official monthly data on earnings in Slovenia. *Earnings of Persons in Paid Employment by Registered Natural Persons* is also published monthly. *Structure of Earnings Statistics* is published annually. It presents data on the amount of average monthly earnings of persons in paid employment by selected geographic and socio-demographic characteristics of persons in paid employment (sex, age, education, occupation, territorial unit) and characteristics of employers (activity, sector, territory). The users of the data include policy-makers, researchers, businesses and the general public.

Observation units are individuals who, in the observed month, received earnings and/or non-refundable wage compensation paid by the employer. The dataset provides full population coverage, encompassing approximately 815,000 persons. The ML algorithms were tested using data for a single reference month, namely October 2024. In addition to earnings data, the dataset includes contextual variables related to legal entities and individuals, which are relevant for both data editing and imputation.

The two primary data sources are:

- Administrative data from forms used to calculate withholding tax returns on employees' income from employment, submitted by employers. SURS obtains these data monthly from the Financial Administration of the Republic of Slovenia and subsequently processes them for statistical purposes.
- Data on persons in employment, extracted monthly from the Statistical Register of Employment maintained by SURS. The main source for the register is administrative data on registrations, changes, and deregistrations related to compulsory insurance, collected by the Health Insurance Institute of Slovenia and obtained by SURS.

An additional statistical source is:

- The Statistical Business Register of Slovenia (SPRS), maintained by SURS and updated monthly using data from the Business Register of Slovenia, provided by the Agency of the Republic of Slovenia for Public Legal Records and Related Services.

Other administrative sources include:

- Data from the administrative data collection entitled Record of Temporary or Permanent Absence from Work Due to Illness, Injury, Care, Escort and Other Reasons provided monthly by the National Institute of Public Health.

- Data on recipients of parental allowance rights, provided monthly by the Ministry of Labour, Family, Social Affairs and Equal Opportunities.

- Data from the Information System for the Transmission and Analysis of Data on Earnings, Other Payments, and the Number of Employees in the Public Sector, provided monthly by the Agency of the Republic of Slovenia for Public Legal Records and Related Services.

- Data from the Central Population Register, provided monthly by the Ministry of the Interior.

Data Analysis:
Earnings data are published with a time lag of two months $(t-2)$. The target population of employees is derived from the Statistical Register of Employment, which also provides contextual variables related to individuals that have already undergone prior editing and validation. Unit non-response for earnings data is mainly due to the fact that employers are legally allowed to submit or revise their reports after the reference month, i.e. after the data have already been obtained and processed by SURS. In addition, employers may subsequently correct previously submitted erroneous data.

The earnings data are statistically processed on a monthly basis using the SURS in-house generic system for data editing and imputation. Erroneous data and unit non-response are treated through a combination of systematic corrections and imputation methods. Systematic corrections are based on a comprehensive set of more than 1,000 edit rules designed to identify and correct inconsistencies and errors in the reported data. Examples of such errors and inconsistencies include incorrect reporting by salary components, incorrect reporting of working hours, delayed reporting of extraordinary payments, reporting under the wrong local kind-of-activity unit, inconsistent data across different earnings components, and similar issues.

The currently applied imputation methods include logical imputation, historical imputation, and nearest-neighbour hot-deck imputation. These methods are used to address erroneous values and unit non-response. To ensure the suitability of the donor population and exclude unsuitable donors in nearest-neighbour hot-deck imputation, generalized linear models (GLMs) with logarithmic transformations of selected variables are applied. Unit non-response varies slightly across months but generally remains relatively low. For October 2024, the reference month used for testing the ML methods, it amounted to 1.4%. Although monthly earnings are the primary measure of interest in the monthly earnings statistics, their level depends directly on the amount of working time and is also affected by absences. Regular earnings per hour was therefore used as the target variable for the ML-based imputation experiment to distinguish variation in the level of earnings from variation related to working time. Information on working time and absences was nevertheless retained among the predictor variables, as it may provide relevant information for predicting regular earnings per hour. Regular earnings per hour is a derived variable calculated as the sum of base salary, performance pay, supplements (e.g., seniority bonuses), business performance bonuses, and salary compensation paid by the employer, expressed per hour worked.

The distribution of regular earnings per hour is right-skewed, with a pronounced concentration of observations around the level corresponding to the statutory minimum wage. This concentration reflects the fact that a substantial number of employees receive earnings at or close to the minimum wage level. Above this level, the density gradually decreases as regular earnings per hour increase, resulting in a long right tail towards higher earnings. Overall, the

target variable exhibits a distinctly asymmetric distribution.

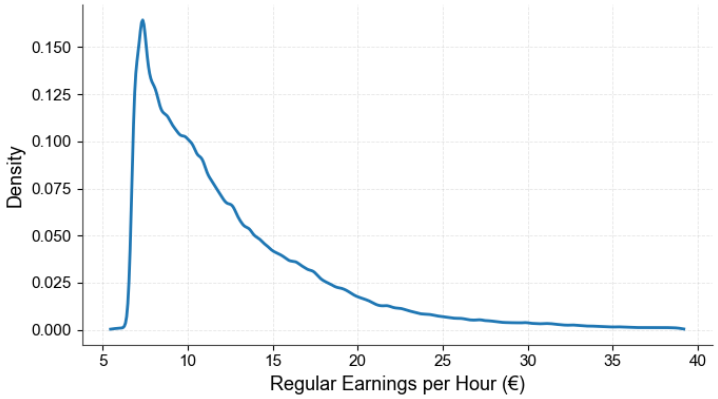


Figure 2.14 Distribution of Regular Earnings per Hour in the Dataset Used for the ML Experiment

Note: Values above the 99th percentile are excluded from the visualization for clarity.

2.6. Imputation of International Trade Statistics (AT) [AT-ITS]

2.6.1. Description

In this project, we aim to impute values in international trade data. The objective is to enhance the quality of trade statistics by imputing reliable values. Specifically, we focus on two variables: *weight* a numerical value, and *CN* (Combined Nomenclature), a categorical variable representing a standardized coding system used in the European Union for classifying goods in customs and international trade. The international trade dataset comprises all imports from the European Union to Austria, including information on product codes, country of origin, reporting country, weight, and quantity. Each month, approximately two million micro-level observations are recorded. Each unit, i.e. import into Austria, is recorded and reported to Statistics Austria. This data is used to produce multiple statistical products, therefore ensuring high data quality is essential. In this project, we aim to improve the reliability and overall quality of international trade statistics by imputing records or values that have been identified as likely erroneous and imputing them, rather than imputing all missing values.

2.6.2. Use of Machine Learning

We aim to impute values for the two variables *weight* and *CN* using machine learning (ML) models. The imputed values should be as accurate, while avoiding the introduction of additional bias into the data. Within this project we compare and evaluate several ML approaches, including k-nearest neighbors (kNN), Random Forest [Breiman, 2001], XGBoost [Chen and Guestrin, 2016], and a transformer-based language model [Vaswani et al., 2017]. The most suitable model is identified not only based on standard evaluation metrics, such as accuracy and error measures, but also with regard to computational efficiency (e.g., runtime) and its ability to preserve the underlying data distribution. Computations are run using the R Programming language [R Core Team, 2026], the different imputation methods are applied using the R packages VIM [Kowarik and Templ, 2016], keras [Allaire and Chollet, 2026] and tensorflow [Allaire and Tang, 2025].

2.6.3. Exploratory Data Analysis

Description of the input data:

Our dataset comprises all imports from the European Union to Austria for the year 2024, including information on product codes, country of origin, reporting country, weight, and quantity. Each month contains approximately two million observations, providing a large volume of data suitable for training machine learning models. For the purpose of our experiments, we restrict the analysis to a subset of the data, specifically the months of January and February.

Though the dataset does not contain missing values per se, a substantial number of entries were identified as placeholders and are considered incorrect. For instance, in many records, the reported quantity of units does not correspond to the recorded weight. For example, a record showing 1,000 units imported with a total weight of 1 kg likely contains an erroneous weight value, which must be corrected through imputation.

The data contains the following variables:

- Weight of the import used directly and grouped into classes
- CN of the import with different degrees of granularity (2 digit and 4 digit code)
- Country of origin and reporting country as ISO-3 Code or aggregated geographically

- Value (in €) of the imported goods.

The variables are described in more detail in the Appendix.
Data Analysis:

Concerning the variable CN, the categorical variable for classifying goods in international trade, there are over 1333 four-digit codes, and 98 two-digit codes in total. The classes are highly imbalanced, which means that some CN codes appear much more frequently than others, and a naive model could be biased toward predicting the most common classes. To illustrate this point the data used for this experiment, namely the January and February data, contains only 1024 four digit codes and 97 two-digit codes. Furthermore more than 50% of the data records correspond to over 60 four-digit codes, e.g. 5% of available four-digit codes. Similarly the 10 most common two-digit codes make up more than 50% of the data.

For variable weight the distribution is also highly skewed with a vast majority of imported products weighing only very little and a few products with very extreme weight. In the data used for the experiment almost 50% of the records had a weight up to 2.5 kg and only 5% of the data showed a weight larger than 1000kg.

This imbalance can make it harder for the model to correctly predict the less frequent CN codes, potentially reducing the overall predictive accuracy, especially for the rare categories. This can be taken into account during the model selection process, by comparing evaluation metrics as well as the target and imputed distribution. To address this, common strategies include:

- Resampling techniques: oversample rare classes or undersample dominant ones.
- Class weighting: assign higher loss weights to rare classes during training.
- Evaluation metrics: use metrics that account for imbalance (e.g., macro F1-score) instead of overall accuracy.

In this experiment we relied on using macro F1-scores. Although treating rare categories with care can be important even more care needs to be taken when imputing majority classes, as their impact on final statistics is likely much higher.

3

Data preprocessing

3.1. Data preparation and editing

Data preparation constitutes a fundamental phase of the Machine Learning (ML) pipeline, ensuring that the input datasets are consistent, clean, and structurally sound for algorithmic processing. In the context of official statistics, this involves transforming raw administrative records and survey microdata into a standardized format.

Key procedures include the identification and treatment of outliers, which are often detected using the Interquartile Range (IQR) method or Z-score thresholds. For a variable x, an observation is typically flagged if: $x_i < Q_1 - 1.5 \cdot IQR \quad$ or $\quad x_i > Q_3 + 1.5 \cdot IQR$.

Furthermore, data preparation involves handling data type conversions (e.g., string to datetime for time-series analysis) and the synchronization of unique identifiers (IDs) to enable successful linkage between survey and administrative sources.

The principle of "Garbage In, Garbage Out" (GIGO) serves as a fundamental axiom in machine learning, asserting that the predictive integrity and generalization capabilities of a model are strictly bounded by the quality of the training data. Despite the development of increasingly sophisticated algorithmic architectures, the learning process remains susceptible to noise, bias, and incompleteness inherent in the input datasets. Research indicates that neglecting data quality leads to "data cascades"—compounding downstream issues that result from technical debt in data pipelines, which can ultimately compromise the reliability and safety of the system [Sambasivan et al., 2021]. Consequently, rigorous data curation and preprocessing are not merely auxiliary tasks but are essential determinants of the scientific validity of any machine learning framework.

Data editing, which encompasses the systematic detection and correction of errors, inconsistencies, and outliers, constitutes a critical phase in the machine learning pipeline to ensure the integrity of input datasets. The process of data cleaning directly mitigates the risks associated with noisy or incomplete data, which can otherwise lead to biased parameter estimation and suboptimal model convergence. Systematic data editing transforms raw, heterogeneous data into a structured format that preserves the underlying statistical signal while filtering out stochastic noise. As emphasized by Chai et al. [2023], the efficacy of learning algorithms is fundamentally constrained by the quality of the data; therefore, robust cleaning frameworks are essential to resolve data quality issues—such as missing values and integrity constraint violations—before any analytical modeling occurs.

In official statistics, this step of the process is essential and there is deep research and implementations to be used [Pannekoek et al., 2013]. As argued by Van der Loo and De Jonge [2018], data cleaning should be treated as a formal statistical discipline rather than an ad-hoc preprocessing step, as the validity of any subsequent model inference is fundamentally dependent on the integrity of the input data. By implementing rigorous validation rules and correction methods, researchers can mitigate the impact of measurement errors and noise, thereby ensuring that the learning algorithm captures the true underlying signal instead of artifacts of poor data quality [see e.g. van der Loo and de Jonge, 2016, Van der Loo and De Jonge, 2018].

3.2. Generation of synthetic data

The generation of synthetic data is primarily motivated by the need to provide researchers and statisticians with high-quality microdata while strictly adhering to data protection regulations, such as the General Data Protection Regulation (GDPR) [European Union, 2016]. Ensuring the absolute confidentiality of synthetic data remains a significant challenge, as there is an inherent trade-off between the statistical utility of the dataset and the risk of information disclosure [Raab et al., 2016]. High-fidelity synthetic microdata may inadvertently leak sensitive attributes or be vulnerable to membership inference attacks, particularly when the generative models overfit the original sensitive records [Drechsler, 2011].

In the context of official statistics, synthetic data serves as a crucial tool for methodological development, allowing for the testing of complex algorithms, such as post-collection imputation, without exposing sensitive information [Reiter, 2003]. Furthermore, as demonstrated in the Spanish Labour Force Survey use case within this project, synthetic populations provide a known "ground truth." This allows for a rigorous evaluation of imputation performance by comparing imputed values against the original values in the synthetic population, a process that is impossible with real missing data where the true values are unknown [Drechsler, 2011].

The process of generating synthetic data typically involves modeling the joint distribution of the original dataset and drawing new observations from the estimated model. A widely recognized framework for this task is the **simPop** package in R, which is specifically designed for the simulation of complex synthetic populations based on sample data. The `simPop` methodology follows a modular, sequential approach that allows to keep hierarchical structures and dependencies present in the real population. For example: it first replicates the household structure and then simulates individual-level categorical and continuous variables using stochastic models, such as multinomial logistic regression or decision trees [Templ et al., 2017].

To ensure that synthetic data is fit for purpose, its quality must be rigorously evaluated through utility and disclosure risk metrics. The goal is to achieve high "statistical representativeness," meaning that the synthetic data should yield results similar to the real data for the same statistical analyses. A standard metric for measuring this quality is the **Propensity Score Mean Squared Error (pMSE)**. This measure evaluates the indistinguishability of the synthetic data by training a classifier to distinguish between real and synthetic records; a low pMSE indicates that the synthetic data successfully mimics the multivariate distributions of the original dataset [Snoke et al., 2018]. Additionally, quality is assessed by comparing marginal distributions and correlation matrices between the real and synthetic datasets to ensure that the underlying statistical relationships are preserved.

If we combine the real and synthetic datasets and train a classifier to predict whether a record is synthetic ($\hat{p}_i$), the pMSE is defined as:

$$pMSE = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - c)^2 \tag{3.1}$$

where c is the expected probability (usually 0.5 for equal-sized datasets). Lower values of pMSE indicate higher synthetic quality, meaning the ML model cannot distinguish between real and artificial records.

Furthermore, the methodological exploration of synthetic data within the AIML4OS project is further complemented by the work conducted in Work Package 13. Specifically, the forthcoming deliverable **D13.1** provides a Proof of Concept (PoC) for generating synthetic datasets in official statistics using advanced AI/ML techniques [AIML4OS Project, 2026b]. This deliverable includes functional Python and R code designed for real-world use cases, moving beyond theoretical simulation to practical implementation. A core focus of this work is the dual-objective quality assessment, which rigorously evaluates both the *privacy preservation* of the generated data and its *statistical utility*. By providing a standardized framework for these assessments, D13.1 ensures that synthetic datasets are not only safe for public dissemination but also remain analytically representative of the original sensitive microdata.

3.3. Non-response Mechanism Analysis

Understanding the nature of missing data is crucial for selecting an appropriate imputation strategy. We analyze the response indicator R, where $R_i = 1$ if the unit i responds and $R_i = 0$ otherwise. The literature defines three primary mechanisms [see e.g. Little, 1988, Haziza, 2009]:

1. **Missing Completely at Random (MCAR):** The probability of missingness is independent of both observed data (X) and unobserved data (Y).

$$P(R|X,Y) = P(R) \tag{3.2}$$

2. **Missing at Random (MAR):** The missingness depends only on the observed auxiliary variables.

$$P(R|X,Y) = P(R|X) \tag{3.3}$$

3. **Missing Not at Random (MNAR):** The probability of missingness depends on the value of the missing variable itself.

$$P(R|X,Y) \neq P(R|X) \tag{3.4}$$

3.3.1. Validation Requirements and Evaluation Criteria

Regardless of the assumed mechanism, the imputation procedure must be rigorously validated to ensure that it does not compromise the statistical integrity of the microdataset. Specifically, the evaluation framework should ensure that the procedure does not introduce:

i) **Bias in Parameter Estimation:** The procedure must preserve the accuracy of the characteristics of interest, such as population totals, means, and ratios.

ii) **Bias in Marginal Distributions:** The imputed values should replicate the distributional shape and variability of the target variables, avoiding the "variance compression" often associated with deterministic ML point estimates.

iii) **Bias in Joint Distributions:** The procedure must preserve the associations and relationships between the imputed target variable and the predictor variables used in the model.

3.3.2. Methodological Assessment in Practice

The use cases presented in this report focus primarily on the first two criteria (i and ii). While the preservation of relationships (iii) has not been explicitly assessed in all projects, it represents a critical area for further investigation. Concerns have been raised in some ML-based scenarios regarding the potential distortion of the joint distribution between the imputed variable and its predictors.

Evaluating whether an imputation procedure preserves these associations is relatively straightforward and can be incorporated into the evaluation framework with limited additional effort. Crucially, under the **MCAR assumption**, such an assessment can be performed directly on the available data without the need for complex simulation studies.

In contrast, while the same evaluation criteria (i, ii, and iii) remain relevant under **MAR and MNAR** assumptions, the validity and performance of the ML imputation cannot be assessed using the observed data alone. In these more complex settings, the evaluation must rely on **simulation studies** (such as the synthetic population approach used in the Spanish use case) or other methods based on known reference values to ensure that the model correctly accounts for the underlying missingness mechanism [see e.g. Haziza, 2009, Snoke et al., 2018].

Summary of Non-response Scenarios by project

To evaluate the robustness of Machine Learning models, the projects within this deliverable simulated various missingness scenarios, ranging from random data gaps to systematic non-response patterns. The following summary details the specific mechanisms addressed in each use case:

- **Poland (PL-Accommodation):** The focus is on the impact of **missingness rates** rather than specific stochastic mechanisms. Gaps were introduced at **10%, 20%, and 30%** levels. This allowed for a stress-test of the stability of kNN and ML models across different levels of data incompleteness, particularly during the structural break caused by the COVID-19 pandemic.
- **Luxembourg (LU-Price):** Missingness in this use case is characterized as **structural MAR** driven by "product churn." Prices are missing in periods when a product is not on sale. The analysis focused on the ability of the model to handle an imbalanced panel where the availability of an item is correlated with its life cycle (entry and exit prices) [May and Lamboray, 2026].
- **Spain (ES-LFS):** This project conducts the most extensive simulation of mechanisms using a synthetic population. It evaluated **MCAR, MAR, and MNAR** scenarios. Under the MAR assumption, missingness was modeled as a function of observed auxiliary variables (e.g., nationality and region). In the MNAR scenario, the probability of non-response was directly linked to the target variable (employment status), specifically targeting the "unemployed" category to test the ability of the model to recover difficult-to-predict minority classes [Haziza, 2009].
- **Portugal (PT-Construction):** The project addresses **item non-response** in administrative data and the specific challenge of **missing historical values**. The mechanism is treated as MAR, where the missingness of the current year's revenue is often tied to the absence of

historical survey responses ($n-1$ or $n-2$), needing an ML approach that can generalize beyond simple growth-factor rules.

- **Slovenia (SI-Earnings):** The Slovenian project implements a targeted simulation of **MCAR, MAR and MNAR** mechanisms at varying rates (1%, 5%, 10%, and 20%).
  - *MAR:* Missingness was assumed to be related to the employer, where smaller enterprises had a higher probability of unit non-response.
  - *MNAR:* The probability of missingness depended directly on the level of earnings, with higher-income individuals being less likely to report, thereby testing the performance of the model on the distribution's tail.
- **Austria (AT-ITS):** This project simulates all three mechanisms to address both numerical (weight) and categorical (CN code) variables:
  - *MCAR:* A purely random masking of values.
  - *MAR:* Missingness probability was increased for specific geographical country groups (e.g., "Eastern Europe" and "Southern Europe").
  - *MNAR:* For the numerical variable, missingness increased with the weight of the import; for the categorical variable, it was linked to specific high-frequency CN codes.

To evaluate the missing data mechanism in a dataset, it is essential to distinguish between Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR), as the validity of the subsequent imputation depends on these assumptions. According to the methodological framework of official statistics, MCAR and MAR are considered "ignorable" mechanisms because the likelihood of the data can be modeled using observed information, whereas MNAR is "non-ignorable" and can lead to systematic bias if not explicitly addressed [de Waal et al., 2011].

In practice, the evaluation begins by testing the MCAR assumption through a comparison of the distributions of auxiliary variables between respondents and non-respondents. If significant differences are found—for instance, if the response rate varies across different age groups or regions—the MCAR hypothesis is rejected in favor of MAR, assuming that the available auxiliary data can explain the missingness pattern [Little, 1988].

While MAR can be justified by the survey design and the inclusion of relevant predictors, the MNAR mechanism is inherently non-identifiable from the observed data alone. Therefore, its detection relies on "red flags" and substantive knowledge of the variables. As highlighted in the provided technical documentation, indicators of MNAR include high non-response rates in sensitive variables (e.g., income), systematic patterns in extreme values, or cases where results are highly sensitive to the choice of imputation model. To manage the risks associated with MNAR, National Statistical Institutes (NSIs) often employ sensitivity analyses. This involves generating multiple imputation scenarios—such as shifting imputed values upward or downward by a fixed percentage—to evaluate the robustness of the final estimates against departures from the MAR assumption [Haziza, 2009].

In this work, logistic regression and propensity score models are used to assess which mechanism best describes the dataset, as model-based imputation is most effective under MAR conditions.

3.4. Feature engineering

Feature engineering aims to extract maximum predictive power from the available variables. In the projects described hereafter, three main techniques are applied:

- **Temporal Features:** For time-series data (e.g., tourism or prices), lag variables are created: $L^k y_t = y_{t-k}$ and rolling statistics (moving averages) over windows of size w:

$$MA_{t,w} = \frac{1}{w} \sum_{j=0}^{w-1} y_{t-j} \tag{3.5}$$

- **Spatial and Hierarchical Aggregations:** Inclusion of group-level means and standard deviations (e.g., average income by municipality) to provide the model with contextual information.
- **Categorical Encoding:** Transformation of qualitative variables. For high-cardinality features, *Impact Encoding* is used, replacing a category with the mean of the target variable for that group.

Feature engineering in the context of post-collection imputation for official statistics aims to transform raw survey and administrative data into a high-dimensional feature space that captures complex dependencies. Across the various projects detailed in this deliverable, four main strategies emerge: the modeling of temporal dynamics, spatial and hierarchical aggregations, domain-specific economic ratios, and advanced categorical encoding.

3.4.1. Temporal and Autoregressive Features

For datasets with a time-series component, such as tourism occupancy and price statistics, capturing historical persistence is critical. As seen in the Polish and Portuguese use cases, models incorporate autoregressive structures by generating lag variables ($L^k y_t = y_{t-k}$) and rolling statistics. Specifically, moving averages over windows of 3, 6, 9, and 12 periods are used to capture both short-term fluctuations and long-term seasonal cycles [Beck et al., 2021]. This approach allows the algorithms to account for structural breaks and trend shifts, such as those observed during the COVID-19 pandemic.

While the inclusion of lags and rolling statistics provides significant predictive power, modern official statistics production benefits from feature engineering approaches that are more closely aligned with the formal statistical properties of time series. Moving beyond a "black-box" approach toward a more interpretable and theoretically grounded framework ensures that ML models respect the underlying stochastic processes of the data [Hyndman and Athanasopoulos, 2018].

Time-Series Decomposition and Component Extraction

A key strategy for enhancing model interpretability is the explicit decomposition of the target variable into its constituent components: trend, seasonality, and residuals. Rather than allowing the ML algorithm to implicitly learn these patterns, features can be engineered using methods such as **STL (Seasonal-Trend decomposition using Loess)** or classical moving average decomposition [Cleveland et al., 1990]. By providing the model with explicit seasonal indices or detrended values, the modeling framework becomes more transparent, allowing statisticians to verify if a prediction is driven by a long-term economic trend or a recurring seasonal peak. This approach is particularly relevant for the tourism and price use cases, where seasonal fluctuations are a dominant characteristic of the data.

Stationarity and Structural Breaks

To maintain predictive performance during periods of high volatility, feature engineering must also account for non-stationarity and structural breaks. As observed in the Polish and United Kingdom use cases, the COVID-19 pandemic introduced abrupt shifts that traditional lag structures struggled to capture. A theoretically grounded approach involves the use of **intervention variables** (binary indicators for specific events) and **differencing transformations** ($\Delta y_t = y_t - y_{t-1}$) to achieve stationarity [Hyndman and Athanasopoulos, 2018]. By incorporating these statistically grounded features, the model can distinguish between regular noise and fundamental changes in the data-generating process, leading to a more robust imputation framework that remains reliable even under anomalous economic conditions.

3.4.2. Spatial and Hierarchical Aggregations

To address spatial heterogeneity, several projects implement vertical and horizontal aggregations. In the Austrian international trade and Polish tourism use cases, raw variables are aggregated into higher-level categories—such as grouping 27 countries into 6 geographical regions or calculating municipality-level standard deviations. These features provide the model with contextual information regarding local tourism intensity or regional trade patterns, effectively reducing the noise inherent in micro-level observations [Templ et al., 2017].

3.4.3. Domain-Specific Ratios and Growth Proxies

A sophisticated approach identified in the Portuguese construction survey involves the creation of interpretable economic ratios. By constructing features such as the ratio between target variables and enterprise turnover, or growth proxies defined as the ratio between consecutive historical values, the models can compare units of vastly different economic scales on a common basis. This ensures that the ML models do not merely "memorize" past values but learn the underlying growth dynamics of the sector.

3.4.4. Categorical Encoding and Transformation

Given that official statistics rely heavily on qualitative data (NACE codes, occupation types, etc.), the projects employ diverse encoding techniques. While low-cardinality variables are typically handled via One-Hot Encoding, high-cardinality features—such as specific product identifiers in the Luxembourg price use case—utilize *Impact Encoding* (Target Encoding). This technique replaces a category with the mean target value for that group, significantly reducing dimensionality while preserving predictive power. Furthermore, log transformations are systematically applied to skewed numerical targets, such as income or prices, to stabilize variance and satisfy the orthogonality properties required for statistical validity [May and Lamboray, 2026].

In the following section, a brief description of the data preprocessing applied in each project is provided below. In summary, regarding feature engineering:

PL-Accomodation : Lags (1-12), Moving Averages (3-12), Regional/Category aggregations (mean, SD, CV).

LU-Price : Log transformation, Impact Encoding (high cardinality), One-Hot Encoding (low cardinality), Time Dummies.

ES-LFS : Linkage of demographic survey data with administrative registry data (Social Security, Pensions).

PT-Construction : Economic ratios (Target/Turnover), Growth proxies (Current/Previous year), Interaction terms.

SI-Earnings : Derived variables from horizontal/vertical rules, scaling for Neural Networks.

AT-ITS : Weight per unit calculation, CN code granularity reduction (4-digit to 2-digit), Geographical country grouping.

3.5. Target Variable Transformations and Methodological Implications

While a frequently cited advantage of Machine Learning (ML) methods is their ability to capture complex, non-linear relationships without requiring explicit variable transformations, practical applications in official statistics often necessitate their use. Factors such as extreme right-skewness or heteroscedasticity—common in economic indicators like prices or earnings—may still justify the application of a **logarithmic transformation** to stabilize variance and improve model convergence.

However, the use of log-transformations introduces a significant methodological challenge during the inference phase. It is important to stress that if a model is trained on $\ln(Y)$ and the resulting predictions are naively back-transformed using the exponential function ($\exp(\widehat{\ln Y})$), the resulting values generally estimate the **conditional median** rather than the **conditional mean** of Y [Miller, 1984]. In the context of official statistics, where population totals and means are the primary outputs, this "back-transformation bias" can lead to a systematic underestimation of the target variable. This issue is well-documented in classical regression theory but remains equally applicable to non-parametric and ML-based methods whenever a non-linear transformation is applied to the target variable [Miller, 1984].

3.6. Practical information from the projects

All the regressors used in each project with a proper explanation is detailed in the tables of the appendix A. Some explanations about the use of these techniques are explained in the following.

3.6.1. Imputation of foreign tourist in accommodation establishments survey (PL)

Data preparation

No extensive data preprocessing was performed prior to model development. The dataset originates from an official statistical survey and is already provided in a structured and standardised format, which reduces the need for additional transformations.

Only basic preprocessing steps were applied, including verification of data types and conversion of variables representing dates into a proper datetime format. This step was necessary to enable the creation of additional time-dependent variables.

No standardisation or scaling was applied during initial data preparation; however, explanatory variables were standardized later within the modeling pipeline for regularized regression models.

More advanced preprocessing techniques, such as outlier removal, smoothing, or data transformation, were intentionally avoided in order to preserve the original characteristics of the dataset, including seasonality, regional variation, and effects of external events.

Since the objective of the project is to evaluate imputation methods under realistic conditions, maintaining the original data structure was considered more appropriate than applying extensive preprocessing.

Feature engineering

Tourism data exhibit strong seasonality and regional heterogeneity, therefore the original dataset was extended with additional explanatory variables designed to capture these dependencies.

The reporting period variable was first transformed into a date representation and observations were ordered chronologically within each spatial unit. This enabled the construction of time-dependent variables.

To capture temporal dynamics and seasonal behaviour, rolling statistics (moving averages) were generated using windows of 3, 6, 9 and 12 periods. These variables represent both short-term changes and long-term seasonal patterns. Shorter windows reflect recent fluctuations, whereas longer windows capture annual cycles.

Autoregressive features were also created by generating lag variables from 1 to 12 periods. These variables describe historical dependencies and provide the model with information about previous tourist activity. Since tourism demand is highly dependent on historical observations, lag features were expected to provide substantial predictive value.

To account for spatial heterogeneity, additional aggregated variables were created for multiple territorial levels and accommodation categories. These variables include regional averages, standard deviations and coefficients of variation and allow the models to capture local tourism intensity and variability.

Checking correlation

For the selected 5 countries, correlations were computed between the dependent variable (number of tourists) and variables created through feature engineering

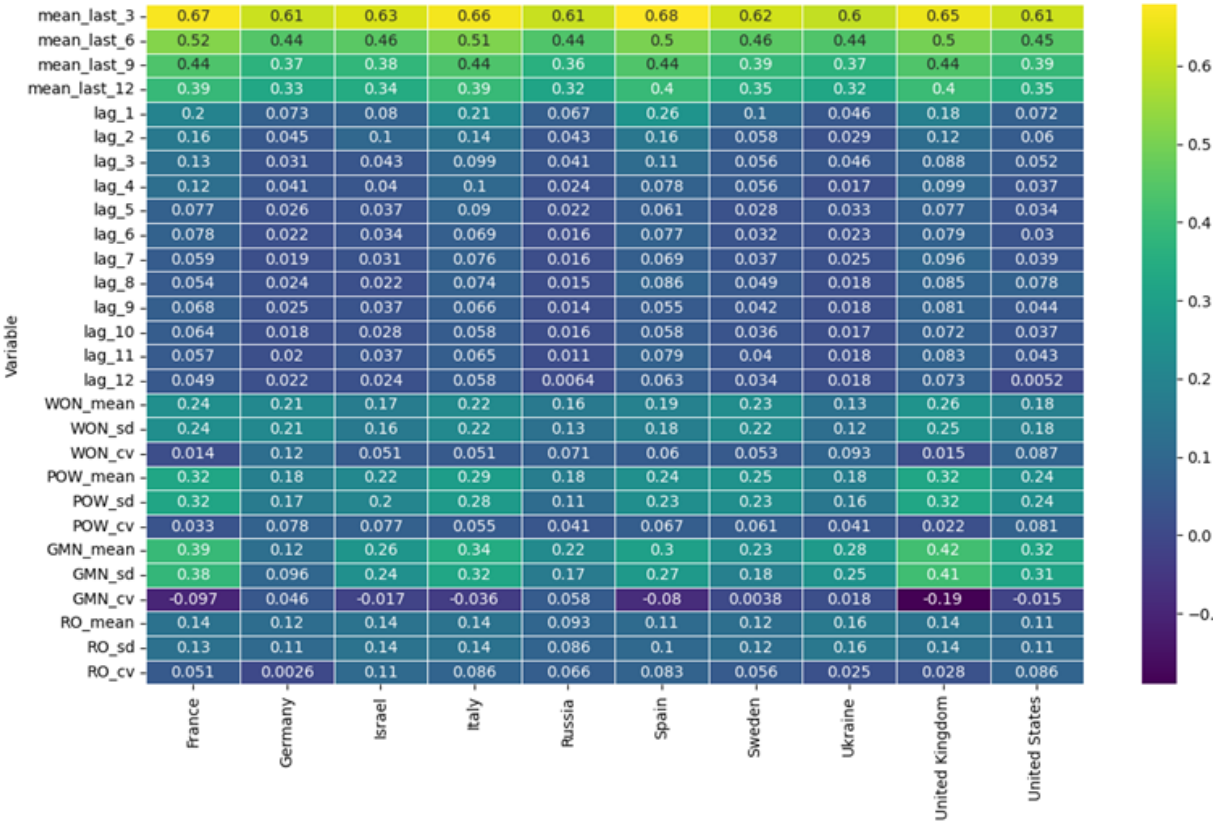


Figure 3.1 Correlation between number of tourists variables created through feature engineering

The variables most strongly correlated with the dependent variable are the 2-, 6-, 9-, and 12-period moving averages. For some countries, a correlation is also observed between the average values at the municipality level and the corresponding standard deviation within the same municipality

Feature selection and model interpretation

Feature selection was performed using regularized linear regression models to identify the most relevant predictors of tourist numbers while reducing multicollinearity among explanatory variables.

Due to strong correlations between temporal variables (particularly lagged values and moving averages), regularization techniques were applied. Ridge regression was used for coefficient shrinkage and model stabilization, without performing feature selection. The regularization strength was selected using cross-validation.

To identify the most important predictors, Elastic Net regularization was additionally applied. By combining L1 and L2 penalties, this approach enables both coefficient shrinkage and automatic variable selection by shrinking less relevant coefficients to zero.

The analysis was conducted separately for each country using chronologically ordered training and testing datasets. Model performance was evaluated using Root Mean Squared Error (RMSE).

Final variable selection was based on non-zero coefficient values, which identified the most informative predictors of tourist flows. Across all countries, autoregressive variables (lags) and moving averages were consistently the most important predictors, indicating strong temporal dependence in tourist arrivals.

To improve interpretability, results are summarized at the level of feature groups rather than individual variables.

Table 3.1 Summary of selected feature groups by country (Elastic Net / Ridge results)

Country	Features	Selected feature groups
Germany	22	Moving averages, extended lag structure (1–11), multiple spatial indicators (WON, POW, GMN, RO)
Russia	19	Moving averages, lag structure, spatial variability indicators
Ukraine	18	Multi-scale moving averages (3–12), lag structure (1–8), spatial indicators
United Kingdom	20	Moving averages, lag structure (1–11), spatial variability measures
United States	20	Moving averages, long lag structure (1–12), spatial aggregates

Overall, the results show that temporal dynamics are the dominant drivers of tourist flows across all countries. In particular, lagged values and moving averages consistently emerge as the most important predictors, confirming strong persistence in tourism demand over time.

In addition, spatial variables capturing the level and variability of tourist flows in other countries frequently contribute to model performance, indicating the presence of international interdependencies and spillover effects between tourism markets.

3.6.2. Imputation of Price Statistics (LU)

Data preparation

The raw web-scraped dataset was preprocessed to improve data quality. Products without associated technical characteristics were removed, as they do not provide informative predictors for price estimation. Features with no variability were discarded, and highly correlated or structurally related variables (e.g. alternative measures of product dimensions) were eliminated to reduce multicollinearity. In addition, redundant features encoding similar information were removed to limit dimensionality.

Missing values in product characteristics were limited and were manually complemented using external sources. Overall, these steps resulted in a consistent dataset. The final set of features used in this project can be found in the Annex A.2

The target variable of the ML model is the price of an item i in a given period t. For each price observation $p_{i,t}$, we know the corresponding time period and a set of price-determining product characteristics. We assume that the product characteristics are constant over time.

The list of features used in this project can be found in the Annex A.2.

The input data is transformed in order to obtain the final features to be used in the ML models. These pre-processing steps are included in a pipeline. The pipeline applies the specified pre-processing steps on a given train set, avoiding any data leakage from the test set into these transformations. After pre-processing, the ML model will be applied. Finally, in a post-processing step, the resulting predictions from the ML model will be transformed back into the original scale. Thereby, we make sure that the performance measures to evaluate the ML model will be calculated using the initial price scale, and not a log-transformed scale. At the same time, in the training step, the ML model will be estimated using log-transformed prices.

In particular, the pipeline is composed of the steps outlined below. A graphical representation of this pipeline can be found in figure 3.2.

Pre-processing

- Log transformation of the price (the target variable).
- Impact encoding computes the difference between the average target value within a specific time period and the overall average target value across the dataset. We use impact encoding for categorical variables with a high number of modalities.
- One hot encoding transforms categorical variable into dummy indicators. We use one-hot encoding for categorical variables with a small number of modalities.
- We remove the variable that identifies the item, such as the GTIN or any shop internal code

ML Model

- Estimation of the ML model (the base learner).
- Combining the predictions of the ML model of the test set with the features.

- Keep only the predictions of the ML model and the one-hot encoding of the time variables (i.e. the time dummies).
- Apply a regression to correct the predictions of the ML model in order to satisfy the orthogonality property. We explain in section 5.2 why this step is necessary.

Post-processing

- Back transformation (exponential) of the values obtained by the ML model.

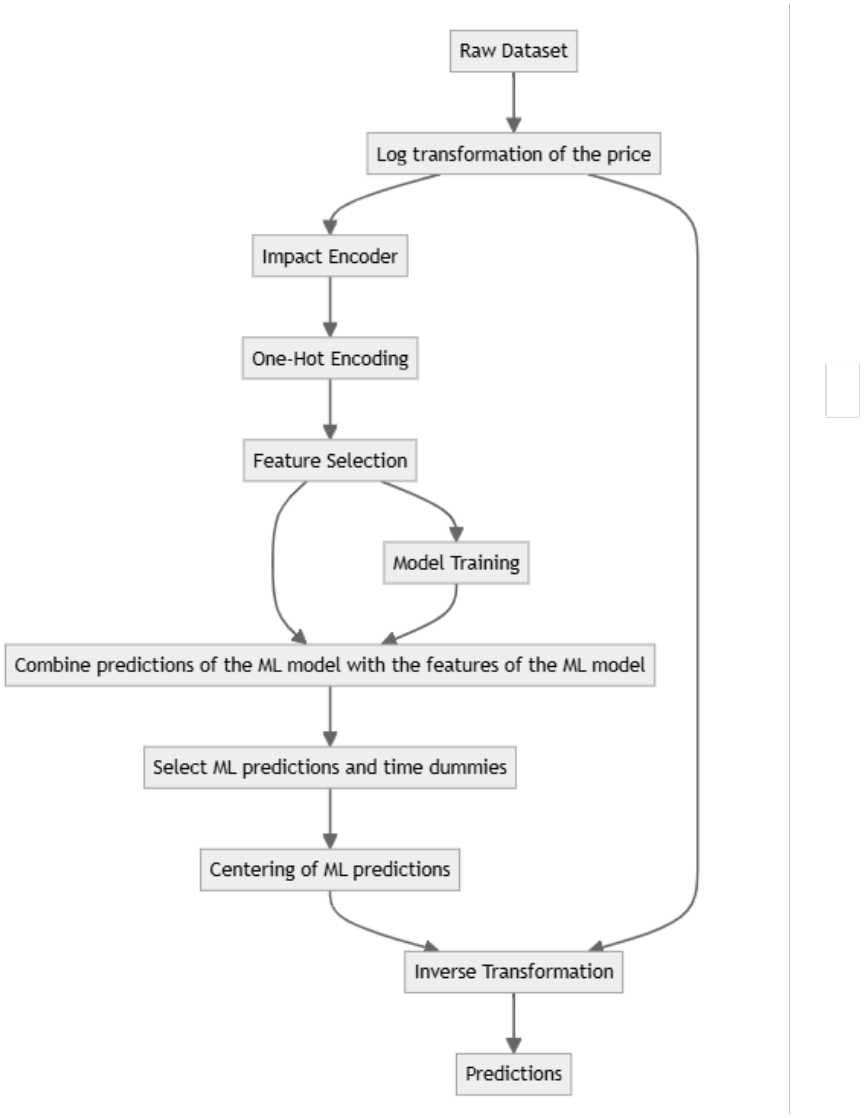


Figure 3.2 The pipeline

A well-known issue when working with log-transformed targets is retransformation bias: since the model is trained to predict $\mathbb{E}[\ln(p_{i,t})|z_i]$, a naive exponentiation of this prediction does not recover the conditional mean of $p_{i,t}$. Because $\exp(\cdot)$ is a strictly convex function, Jensen's inequality implies $\mathbb{E}[p_{i,t}|z_i] \geq \exp\left(\mathbb{E}[\ln(p_{i,t})|z_i]\right)$, so the naive back-transformed prediction systematically understates the conditional mean price whenever residuals have positive variance; instead, one recovers the conditional median of $p_{i,t}$ whenever the residuals $\varepsilon_{i,t}$ are symmetric in log-space. This issue is well known in the hedonic regression literature, where the choice

between a linear and a log-linear specification has long been recognized to carry implications for the bias of the resulting price estimates [Diewert, 2003], and for which corrections have been proposed both for the general regression prediction case [Duan, 1983] and specifically for dummy-variable coefficients in semi-log models [see e.g. Kennedy, 1981, van Garderen and Shah, 2002]. In practice, the hedonic price index literature has generally found this bias to be small in magnitude and has often refrained from applying an explicit correction [Hill, 2013]. Following this common practice, we do not apply a smearing correction [Duan, 1983] in this work, though we flag it as a natural refinement for future work.

An important element of the pipeline is the machine learning model itself. We first train a standard ML model as a base learner, and then apply a learned correction using a regression model. This last step is necessary as we explain in section 5.2 that the ML model should have no bias: on average, the imputed prices from the ML model in each period should be equal to the observed prices. This setup can be viewed as a form of stacked learning, where two models are composed in sequence: an arbitrary ML model followed by a linear correction model that adjusts the initial predictions.

The correction is based on a regression in which the observed log prices serve as the dependent variable, while the explanatory variables consist of time dummies and the ML model's predicted log prices (the base learner) on the training set.

Note that the model used to make these predictions have been obtained from the same train set. In general, this is not a good practice to predict the price for the same units that have also been used to train the model. However, in this specific case, this approach ensures that the final predictions satisfy the orthogonality property (restricted to the train set). If $\hat{p}_i^t$ denotes the final predictions of item i in period t, the pipeline ensures that:

$$\sum_{i \in N_t \cap TRAIN} \left(\ln(\hat{p}_i^t) - \ln(p_i^t) \right) = 0 \qquad \forall t \tag{3.6}$$

3.6.3. Imputation for non-response in Labour Force Survey using synthetic data (ES)

Data preparation

This project focuses on using external registry information along with available demographic survey data to predict the occupational status of individuals. Thus, in our model, we only use unit-level variables selected by evaluating model performance and analyzing correlations, as shown in the figure 3.3 below.

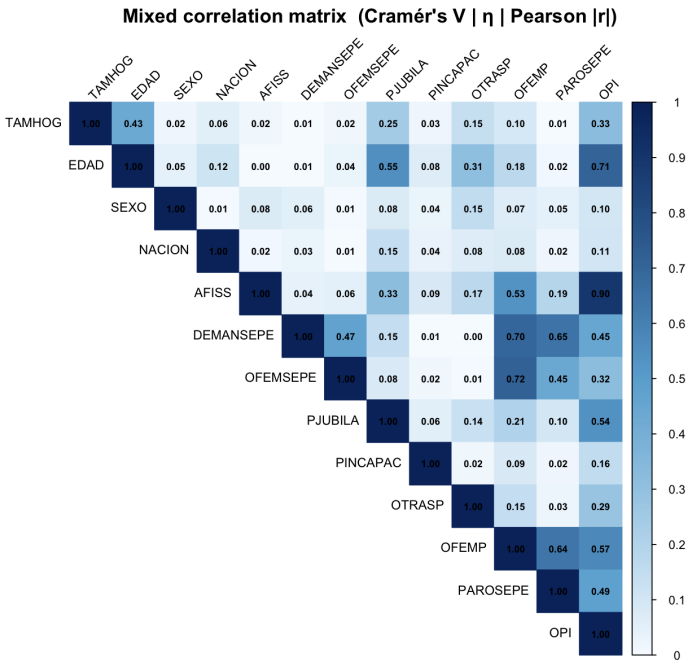


Figure 3.3 Mixed correlation matrix among the set of variables (survey predictors, administrative variables, and the target occupational status (`OPI`)), computed using Cramér's V (categorical–categorical), correlation ratio η (categorical–numerical), and absolute Pearson's $|r|$ (numerical–numerical)

Feature engineering

The final selected features are:

- Demographic characteristics: age, sex, nationality, region, household size.
- Registry data: Social Security affiliation, employment solicitor status, retirement pension receipt, disability pension receipt, and other state subsidy receipt.

Some variable transformations were tested, but none proved useful for the model.

Except for age, all featured predictors are categorical. The R packages used to evaluate different imputation models can handle these categorical predictors directly, so no numerical encoding is required.

The final set of predictors used for the imputation of employment status combines demographic variables and administrative registry information. Specifically, the model includes age, sex, nationality, region, and household size, as well as variables derived from Social Security administrative records, such as affiliation status, employment solicitor registration, receipt of retirement or disability pensions, and other state subsidy indicators. With the exception of age, all predictors are categorical and are directly handled by the machine learning implementations used in this study without the need for additional numerical encoding or feature transformation.

Non-response Mechanism Analysis

An additional analysis was conducted to understand the structure of non-response in the Labour Force Survey and its implications for the choice of estimation strategy. In particular, a binary response indicator was constructed, taking value 1 if the unit responds and 0 otherwise. This indicator was then modelled as a function of the available survey and administrative variables in order to assess whether non-response can be explained by observed characteristics.

The results indicate that non-response is, to a very large extent, close to independent of the observed covariates. In other words, most of the available variables show very limited explanatory power for response propensity. This suggests that the non-response mechanism is close to MCAR, with only a very weak departure towards MAR.

A slight but consistent relationship is observed with nationality and autonomous community, which emerge as the most relevant predictors of response behaviour. However, even in these cases, the magnitude of the effects is small, indicating that non-response is only weakly structured by observable characteristics.

Taken together, the analyses suggest that the non-response mechanism in the Labour Force Survey lies between MCAR and weak MAR, rather than being strongly ignorable or heavily dependent on the outcome variable. According to evidence from the simulation study and existing methodological literature, in this type of setting model-based imputation does not typically provide systematic gains over calibration-based reweighting for the estimation of population aggregates.

3.6.4. ML Treatment of the Annual Survey on Construction Enterprises Using Admin Data (PT)

Data preparation

- Standardize and validate unique identifiers (e.g. business IDs) to ensure successful merges. Remove or flag duplicate IDs.

- Handle mismatches due to typos, inconsistent formatting, or outdated records. Standardize variable names and formats (e.g., values reported in different units).

- Identify where data is missing in one or both sources (cases with zeros were detected).

- Ensure numeric fields are properly typed (e.g., revenue as float, date as datetime).

- Strip extra characters or spaces from string variables.

- Create harmonized or derived variables (e.g., ESTRATO = CAE + TQ (size indicator) + EVVN).

- Insert data from previous years to enable the algorithm to emulate memory mechanisms.

Several machine learning techniques are being applied ranging from simple regressions to regularized regressions, and the more promising being Random Forest and XGBoosts. The value needed is predicted.

The same period, annually, will be adopted in both cases.

Feature engineering

No explicit feature selection step was performed, as regularization techniques — particularly Lasso — were expected to handle this implicitly by shrinking or eliminating less relevant predictors. This approach is well-suited to high-dimensional settings and avoids the need for manual variable selection.

However, in practice, the performance of regularized linear models was inferior to that of more flexible methods such as Random Forest and XGBoost. This suggests that the underlying relationships between predictors and the target variable are not purely linear or additive, and may involve interactions or non-linear effects that are better captured by tree-based models.

As a result, feature selection was effectively addressed through model choice rather than as a separate preprocessing step, with tree-based methods inherently performing variable selection during training.

Feature engineering focused on deriving interpretable ratios and growth indicators aligned with the existing methodology. In particular, the variable perc_hab_n1 was constructed as the ratio between the target variable in the previous year (QF185001_n1) and the enterprise turnover (VVN), acting as a normalized measure of housing-related activity relative to the economic scale of the enterprise. This allows the model to compare units of very different sizes on a common basis.

Additionally, a growth proxy (crescimento) was defined as the ratio between consecutive historical values (QF185001_n1 / QF185001_n2), capturing temporal dynamics similar to the trend used in the current rule-based method.

These derived features were complemented with standard preprocessing steps, including median imputation for missing values, treatment of invalid or extreme values, and log-transformations to reduce skewness. Categorical variables were encoded using one-hot encoding, and selected interaction terms between key predictors were introduced to capture non-linear relationships. Finally, predictors with little or no variability were removed to improve model stability and robustness.

3.6.5. Evaluating Machine Learning Imputation Methods Using Employment Earnings Data (SI)

Data Preparation

The monthly data preparation process, which is already well established, includes data integration, identification and recalculation of individuals' main employment, and derivation of new variables. Prior to model training, the data were subjected to the standard validation and editing procedures implemented in the production system. These procedures comprise a comprehensive set of validation rules designed to identify and correct inconsistencies, errors, and implausible values. Only records and variable values that successfully passed the validation checks and were not modified during the data editing process, representing approximately 86% of all records, were used for training the ML models. This ensured that the models were trained exclusively on high-quality and reliable data.

Missing values were artificially introduced for the target variable, regular earnings per hour, under different missingness mechanisms (MCAR, MAR, and MNAR) and rates (1%, 5%, 10%, and 20%). Under the MAR mechanism, the probability of missingness was assumed to be related to employers, reflecting employer-level unit non-response, with smaller employers having a higher probability of non-response. Under the MNAR mechanism, the probability of missingness was assumed to depend directly on the value of the target variable itself, with higher earnings associated with a greater likelihood of missingness.

Data preprocessing steps were applied depending on the requirements of the specific algorithm. In particular, neural network-based models required scaling of numerical variables and encoding of categorical variables, while XGBoost and Random Forest models required only encoding. In addition, LightGBM models with a logarithmic transformation of the target variable were also evaluated. For these models, retransformation bias was addressed by applying Duan's smearing estimator during back-transformation to the original scale. The final model selected for the analysis, however, was fitted directly to the untransformed target variable.

Feature engineering

For the prediction of regular earnings per hour, only variables that are not direct components of earnings were considered as predictors. This ensures that the model does not rely on variables that are mechanically related to the target variable, thereby avoiding circularity or information leakage in the imputation process. For the construction of derived (horizontal rule) and aggregated (vertical rule) variables, a reusable component was developed.

3.6.6. Imputation of International Trade Statistics (AT)

Data Preparation

For our experiments, we use trade data recorded over a two-month period, resulting in approximately 4.1 million observations. In production, likely erroneous records are identified and imputed. The validity of outliers must also be checked and imputed if they are determined to be erroneous or implausible. Outliers can arise from data entry mistakes, measurement errors, or unusual but valid events. To maintain data quality and model performance, such values should be carefully assessed: valid extreme values can be retained, while invalid ones should be corrected by imputation.

During this project, we artificially introduce missing values according to the mechanisms missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). For MCAR we take a random sample from all data to imitate missingness. Missing values following the MAR setting are injected by increasing the probability for missingness for imports from country groups "Eastern Europe", "Southern Europe" and "Others". For artificially generating values that are MNAR, we increase the probability of missingness with increasing weight of an import, and two-digit CN codes in $\{39, 40, 61, 62\}$. This allows us to create multiple experimental dataset, varying both the type of missingness and the proportion of missing values, which ranges from 1% to 20%.

Additionally, we draw samples of different sizes from the full dataset to evaluate the effects of sample size on our analyses. This is done for both the categorical and the numerical variable separately. In total, each combination of type of missingness, rate of missingness, dataset size, and missing variable is repeated 50 times. For each setting, we compute the evaluation metrics for each run and report their mean or median.

Feature engineering

The data variables are either categorical or numeric. For a subset of variables we need to do the following transformations:

- *weight*: The weight variable to be imputed is the weight per unit, which we calculate before imputation.
- *CN2*: The 2-digit code is derived from the 4-digit CN code.
- *country*: We group 27 different countries to 6 country groups. These are "Western Europe", "Eastern Europe", "Northern Europe", "Southern Europe", "Western Asia" and "Other".

4

Model selection process

The model selection process in post-collection imputation is designed to identify the optimal algorithmic configuration that balances predictive accuracy with statistical robustness. Across the projects, this is achieved through rigorous data splitting, systematic hyperparameter optimization, and multi-level quality assessments.

4.1. Algorithmic Taxonomy and Candidate Models

The model selection process follows a benchmarking logic where Machine Learning (ML) algorithms are compared against traditional statistical baselines. The candidate models evaluated across the use cases can be categorized into three families:

- **Baseline and Linear Models:** Including *Closest Price* (naive imputation), *Linear Regression*, and *Generalized Linear Models (GLMs)*. These serve as the benchmark for stability and interpretability [Eurostat, 2022].

- **Tree-Based Ensemble Methods:** This is the core focus of the deliverable, including *Random Forest*, *XGBoost*, and *LightGBM*. These models are selected for their ability to capture non-linear interactions and handle high-cardinality categorical data (e.g., NACE or product codes) without manual feature specification [Zeng and Rao, 2025].

- **Experimental and Proximity Models:** Including *k-Nearest Neighbors (kNN)* and *Deep Learning* architectures like *TabNet* or *Transformers*. While kNN is used for its localized estimation, Transformers are evaluated for their potential in high-dimensional trade data, despite their higher computational cost [Bischl et al., 2024].

Special attention should be given to Transformer architectures due to their significant potential and strong performance depending on the specific use case. The Transformer is a neural network architecture that fundamentally transformed modern Artificial Intelligence, first introduced by Vaswani et al. [2017]. While early implementations were computationally intensive and time-consuming, recent advancements have enabled the development of foundational models tailored for tabular data. Although these models have not been evaluated within the current scope of this project, their promising results make them well worth highlighting. A prominent example is TabPFN Hollmann et al. [2023], a pre-trained Transformer designed for rapid supervised classification on tabular data. This model performs optimally on small (approximately 2,000 instances) to medium-sized datasets (around 10,000 instances), where inference remains computationally reasonable despite substantial memory requirements. Nonetheless, scaling to large tabular datasets is not yet feasible.

4.2. Data splitting workflow

To ensure the generalizability of the imputation models and prevent overfitting, various cross-validation (CV) strategies are employed depending on the data structure.

- **K-Fold Cross-Validation:** Standard 5-fold or 10-fold CV is used in the Slovenian and Austrian use cases to provide robust estimates of predictive performance. In the Portuguese case, this is extended to *repeated 10-fold cross-validation* (3 repeats) with stratification on the target variable to ensure a balanced distribution across folds.
- **Time-Series and Item-Level Splitting:** For longitudinal data, such as in the Polish and Luxembourgish projects, data is split chronologically. Luxembourg implements an innovative item-level CV where the n last or n first observations of a product's life cycle are reserved for testing, respecting the temporal dependencies and within-product correlations inherent in price statistics [Bischl et al., 2024].
- **Monte Carlo Simulation:** The Spanish use case utilizes a Monte Carlo approach, testing imputation models repeatedly across multiple samples drawn from a synthetic population to approximate design-based quantities like bias and variance.

4.2.1. Cross-validation

We utilize K-fold cross-validation. The dataset $\mathcal{D}$ is partitioned into K subsets. The model is trained on $K-1$ folds and tested on the remaining fold. For longitudinal data, a **Time-Series Split** is preferred, where the training set only includes observations prior to the test set to respect chronological order.

The selection of an appropriate cross-validation framework is a critical decision that must be strictly aligned with the underlying structural characteristics of the dataset. As detailed in the methodological guidelines developed for the AIML4OS project, National Statistical Institutes must distinguish between independent cross-sectional data and dependent structures, such as time series or hierarchical groupings [AIML4OS Project, 2026a]. For instance, while standard K-fold cross-validation is suitable for independent units, longitudinal datasets require temporal splitting techniques—such as expanding or rolling windows—to prevent the model from inadvertently utilizing future information to predict past values. By matching the validation strategy to the nature of the data (e.g., using stratified splits for imbalanced classes or group-based splits for household data), the framework ensures that the performance estimates are unbiased and representative of the actual production environment [AIML4OS Project, 2026a].

4.2.2. Hyperparameter tuning

Hyperparameter tuning is a critical stage in the model selection process, as the performance of Machine Learning algorithms—such as XGBoost, Random Forest, and LightGBM—is highly sensitive to the configuration of their internal parameters (e.g., learning rate η, maximum tree depth, or the number of estimators). Unlike model weights, these hyperparameters are not learned directly from the training data and must be optimized externally to achieve the best balance between predictive accuracy and model generalization.

The objective of hyperparameter tuning is to find the optimal configuration θ^* within a predefined parameter space Θ that minimizes a specific loss function (such as RMSE or MAE) evaluated through cross-validation:

$$\theta^* = \arg\min_{\theta \in \Theta} \text{Loss}(\text{Model}(\theta)) \tag{4.1}$$

Across the use cases described in this deliverable, the approach to hyperparameter tuning was tailored to the specific nature of the data and the complexity of the algorithms:

- **Systematic and Space-Filling Grids:** In projects such as the Portuguese construction survey, tuning was performed using *space-filling grids*. This approach allows for a structured exploration of the hyperparameter space, ensuring that the combinations of parameters are distributed efficiently to capture the performance of the model landscape without the exhaustive computational cost of a full grid search. That project searches the best parameters for Random Forest (e.g., *mtry*, *min_n*) and XGBoost (e.g., *learning rate*, *tree depth*).
- **Regularization Path Optimization:** For linear models utilizing regularization (e.g., Ridge, Lasso, or Elastic Net in the Polish project), the optimal regularization strength (λ) and the L1/L2 mixing parameters were selected by evaluating a wide range of penalty values directly within the cross-validation pipeline.
- **Bayesian Optimization:** For use cases involving more complex search spaces or requiring high computational efficiency, such as the Luxembourgish price statistics and the Slovenian earnings data, *Bayesian Optimization* was employed. This strategy uses a probabilistic surrogate model to guide the search, focusing on the most promising regions of the parameter space. Specifically, the **Optuna** library was utilized in these projects to automate the optimization process and identify the best-performing configurations for tree-based ensemble models.

Additionally, *early stopping* is applied in boosting models (LightGBM, XGBoost) to halt training when validation error ceases to improve, further mitigating the risk of learning noise.

By employing these diverse strategies, each project ensured that the selected models were not only optimized for their specific target variables but also remained robust against overfitting in a production environment.

4.3. Post-processing and Constraint Satisfaction

In the production of official statistics, Machine Learning models must satisfy specific constraints to ensure that imputed microdata are logically consistent, statistically unbiased, and theoretically sound. While standard ML algorithms prioritize minimizing a global loss function (e.g., MSE), they do not inherently respect domain-specific requirements, such as non-negativity or the preservation of statistical moments. Consequently, a post-processing stage is essential to align raw model outputs with the operational standards of National Statistical Institutes (NSIs).

4.3.1. Numerical Variables: Bias Correction and Logical Bounds

When dealing with numerical targets such as prices, earnings, or tourist counts, post-processing addresses two primary challenges: enforcing physical boundaries and mitigating mathematical biases introduced by data transformations.

Enforcing Domain and Logical Constraints

Most economic indicators are strictly non-negative ($y_i > 0$). To enforce this, pipelines often employ target transformations during preprocessing (e.g., log-transformation). Post-processing then involves the corresponding inverse transformation (e.g., exponential) to ensure that the final imputed values remain within the valid economic domain of the variable.

Mitigating Back-Transformation Bias

A significant methodological challenge arises when a model is trained on $\ln(Y)$. Naively back-transforming predictions using the exponential function generally estimates the *conditional median* rather than the *conditional mean* of Y, leading to a systematic underestimation of population totals [Miller, 1984]. To ensure that imputed values remain valid for aggregate estimation, two primary remedies are considered in this deliverable:

- **Duan Smearing Estimator:** A non-parametric approach that adjusts back-transformed values by the average of the exponentials of the training residuals, avoiding specific distributional assumptions [Duan, 1983].
- **Empirical Calibration:** This involves fitting a secondary model of the observed values on the back-transformed predictions to realign the scale of the imputed data with the original observations.

The Luxembourg Case: Integrating Theory and Calibration

The Luxembourgish price statistics use case (LU-Price) provides a sophisticated example of simultaneous constraint satisfaction. To remain consistent with Consumer Price Index (CPI) theory, the model must satisfy the **orthogonality property**, ensuring it remains unbiased within each time period t:

$$\sum_{i \in N_t \cap \text{TRAIN}} (\ln(\hat{p}_i^t) - \ln(p_i^t)) = 0, \quad \forall t \tag{4.2}$$

By implementing a *stacked learning* approach, the raw ML predictions are fed into a secondary linear correction model using time-dummies. This "re-centering" step serves a dual purpose: it satisfies the theoretical requirements of index number construction and effectively performs the empirical calibration necessary to mitigate back-transformation bias [see e.g. May and Lamboray, 2026, International Monetary Fund et al., 2025].

4.3.2. Categorical Variables: Preserving Distributional Diversity

For categorical variables, post-processing focuses on avoiding "mode-reversion"—the tendency of ML models to over-predict majority classes at the expense of population diversity.

Stochastic Imputation vs. Deterministic Classification

Standard ML classifiers typically rely on the **most-voted-class criterion**. While this maximizes micro-level accuracy, it often distorts marginal distributions by suppressing minority classes. To preserve the distributional integrity of the microdataset, it is recommended to take into account as possibility the procedure to **draw values randomly from the estimated conditional probability distribution** provided by the model.

This stochastic approach follows the rationale of traditional **random hot-deck imputation**. As demonstrated in the Austrian international trade use case (AT-ITS), sampling from predicted probabilities is essential for maintaining the representativeness of high-cardinality variables like product codes, where capturing rare categories is critical for macro-level validity [Haziza, 2009].

Handling Non-Probabilistic Outputs via PPS

A further challenge occurs with algorithms that produce ordinal rankings or raw scores rather than calibrated probabilities. In such cases, these outputs can be interpreted as **size measures** within a **Probability-Proportional-to-Size (PPS)** random selection framework. By treating the model's score as a surrogate for the relative likelihood of a category, NSIs can implement

a selection mechanism where the probability of imputing a specific class is proportional to its score. This allows for the integration of diverse ML architectures while ensuring that the final microdataset remains probabilistic and representative of the inherent variability in the population.

4.4. Quality evaluation of the model

Model performance is assessed at both the unit level (micro) and the aggregate level (macro), ensuring that the results are fit for the production of Official Statistics. In addition to the metrics shown in [AIML4OS WP9. European Commission, 2026] for both numerical and categorical variables at micro and macro level, we show here some specific metrics used in the projects of this document.

4.4.1. Micro-level evaluation

For continuous variables (prices, earnings, tourist counts), the following metrics are used: RMSE, MAE and LRMSE (Used in price statistics).

- **Log Root Mean Squared Error (LRMSE):**

$$LRMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\ln\frac{y_i}{\hat{y}_i}\right)^2}, \tag{4.3}$$

  where y_i is the true value and $\hat{y}_i$ is the imputed value

In multi-class classification problems with highly imbalanced classes, such as the imputation of CN codes in the Austrian trade use case, relying solely on accuracy can be misleading. Therefore, the F1-score—the harmonic mean of precision and recall—is calculated using two different averaging strategies to capture different aspects of model performance (micro and macro level).

The micro-average aggregates the contributions of all classes to compute the average metric. It calculates the total number of True Positives (TP), False Positives (FP), and False Negatives (FN) across all categories:

- **Micro-averaged F1-score ($F1_{micro}$):**

$$Precision_{micro} = \frac{\sum TP_i}{\sum TP_i + \sum FP_i}, \quad Recall_{micro} = \frac{\sum TP_i}{\sum TP_i + \sum FN_i} \tag{4.4}$$

$$F1_{micro} = 2 \cdot \frac{Precision_{micro} \cdot Recall_{micro}}{Precision_{micro} + Recall_{micro}} \tag{4.5}$$

In the AT-ITS project, $F1_{micro}$ reflects the **overall predictive accuracy** of the model. Since it gives equal weight to every individual observation, it is dominated by the performance on the most frequent CN codes (the majority classes).

4.4.2. Macro-level evaluation

The macro-level evaluation ensures that the imputation process preserves the statistical properties of the population and produces reliable aggregates. This is critical for National Statistical Institutes (NSIs), as the ultimate goal of post-collection imputation is to minimize the impact of missing data on published indicators.

Distributional Consistency

To verify that the imputation does not distort the underlying data structure, the distribution of imputed values is compared against the observed distribution. While the **Kolmogorov-Smirnov (K-S) test** is a widely recognized non-parametric tool for this purpose, its standard application assumes independent and identically distributed (i.i.d.) data from simple random samples.

In the context of complex survey designs or administrative registries where i.i.d. assumptions may not hold, the projects within this deliverable utilize complementary approaches:

- **Visual Inspection:** Overlaid density plots and histograms are used to detect "variance compression" or artificial spikes in the distribution (as seen in the Slovenian earnings use case).
- **Propensity Score Metrics:** As implemented in the Spanish use case, the **pMSE** (Propensity Score Mean Squared Error, Equation 3.1) is used to evaluate the indistinguishability of the distributions. This metric is particularly robust for high-dimensional data as it does not rely on the same parametric constraints as the K-S test [Snoke et al., 2018].

Aggregate Stability and Error Frameworks

The methodology for evaluating the impact on totals and means varies according to the nature of the data source:

- **Sample Surveys:** For projects based on probabilistic sampling (e.g., the Spanish Labour Force Survey), the **Design-based Mean Squared Error (MSE)** is the primary metric. It accounts for the randomness introduced by both the sampling design and the imputation model:
$$MSE(\hat{\theta}) = \sum_{s \in S} p(s)(\hat{\theta}(s) - \theta)^2 \tag{4.6}$$
where θ is the true population parameter and $\hat{\theta}$ is the estimate derived from the imputed sample [Haziza, 2009].
- **Administrative Data and Censuses:** For use cases involving administrative records or census-like data (e.g., the Portuguese construction survey or Austrian trade statistics), the sampling fraction is often near unity, and the "sampling error" is negligible. In these scenarios, the evaluation shifts toward a **model-based perspective**, focusing on the **imputation bias** and the preservation of sub-population structures. The stability is assessed by comparing the final population totals against benchmarks or historical trends to ensure that the ML model has not introduced systematic shifts in the registry [Beck et al., 2021].

Categorical Integrity and Rare Class Performance

For categorical target variables, such as the product codes in the Austrian trade use case, macro-level evaluation focuses on the model's ability to preserve the diversity of classes. In this context, the **Macro-averaged F1-score ($F1_{macro}$)** is employed. This metric computes the F1-score independently for each class i and then takes the arithmetic mean across all C classes:

$$F1_{macro} = \frac{1}{C} \sum_{i=1}^{C} F1_i \tag{4.7}$$

Unlike the micro-average (which is dominated by the most frequent categories), $F1_{macro}$ treats all classes equally. For official statistics, this metric is critical as it evaluates the model's ability to correctly impute **rare categories** (minority classes). As noted in the results for the

Austrian project, a large discrepancy between micro and macro F1-scores indicates that the model performs well on common trade goods but struggles to accurately identify less frequent products, which could lead to an over-simplification of the published categorical aggregates [Bischl et al., 2024].

Theoretical Constraints

Finally, in specific domains such as price statistics, the macro-level evaluation includes verifying the **orthogonality property** to ensure that ML-based imputations remain unbiased within each period and conceptually aligned with index number theory [May and Lamboray, 2026].

4.4.3. Robustness Analysis: Missingness Levels and Mechanisms

A critical component of the model selection process is the evaluation of model robustness under stress-test conditions. Instead of evaluating models on a single static dataset, the methodology introduces artificial gaps to simulate real-world non-response scenarios:

- **Missingness Rates:** Models are systematically tested under varying levels of data incompleteness, typically at **10%, 20%, and 30%**. This allows for the assessment of the "break-point" where an algorithm's predictive integrity begins to deteriorate significantly.
- **Missingness Mechanisms:** Following the theoretical framework of Rubin, models are evaluated under **MCAR, MAR, and MNAR** assumptions. This is particularly evident in the Spanish and Slovenian use cases, where the choice of the final model depends on its performance in MNAR scenarios, which are the most challenging for official statistics production [Little, 1988].

The final model is selected not only for having the lowest error at a 10% missing rate but for maintaining the most stable performance as the missingness increases.

4.5. Practical information from the projects

To provide a clear overview of the experimental setups across the different use cases, table 4.1 summarizes the structural characteristics of the datasets and the specific composition of the training sets used in each project. This comparison highlights the methodological diversity within the work package, particularly regarding how each National Statistical Institute (NSI) balances the use of current reference data against historical information to train their respective Machine Learning models.

The table reveals two distinct strategies concerning temporal depth in the feature engineering. Projects such as Poland (PL) and Portugal (PT) leverage multi-year datasets and explicitly incorporate historical dynamics into their feature space through the use of lags or growth factors in the regressors. In contrast, the Slovenian (SI) and Austrian (AT) use cases adopt a more localized approach, focusing on training sets restricted to the reference month or a very narrow window. This diversity ensures that the deliverable evaluates ML performance across a wide spectrum of data availability scenarios, from data-rich longitudinal panels to restricted cross-sectional snapshots.

Table 4.1 Summary of Data Structure and Training Set Composition by Project

Project	**Periodicity**	**Available Data**	**Training Set Units**	**Historical Regressors**
PL	Monthly	2018 – 2020	Rolling 12-month windows	Yes
LU	Monthly	June 2021 – June 2022	13 months	No
ES	Quarterly	1 Quarter (+ Synthetic Pop)	1 Quarter	No
PT	Yearly	3 years	1 year	Yes
SI	Monthly	October 2024	1 month	No
AT	Monthly	2 months	Subset of the 2 months	No

4.5.1. Imputation of foreign tourist in accommodation establishments survey (PL)

4.5.1.1. Performance Evaluation of Machine Learning Models

Machine learning models for forecasting international tourist arrivals

Accurate forecasting of international tourist arrivals is a challenging task due to the complex and nonlinear nature of tourism demand, which is influenced by multiple economic, social, political, and environmental factors. Traditional statistical approaches may have difficulties in capturing such complex dependencies; therefore, machine learning methods were applied to evaluate their ability to model patterns in monthly tourist arrival data and generate reliable forecasts.

The predictive performance of five machine learning algorithms, namely Random Forest (RF), Gradient Boosting Regression (GBR), Decision Tree (DT), K-Nearest Neighbors (kNN), and Extreme Gradient Boosting (XGB), was evaluated using monthly data describing the number of international tourists from the five largest source countries: Germany, Russia, Ukraine, the United Kingdom, and the United States. The selected algorithms represent different machine learning approaches, including tree-based models, ensemble learning methods, and distance-based techniques, enabling a comprehensive comparison of forecasting capabilities (see section 5.1 and appendix B).

K-Nearest Neighbors Regression (kNN) is a non-parametric algorithm that predicts future values based on the similarity between observations. The model identifies the closest historical observations in the feature space and estimates the target value using information from these neighboring cases. Due to its simplicity and flexibility, kNN can capture local patterns in tourism data without assuming a predefined functional relationship.

Decision Tree Regression (DT) is a supervised learning algorithm that divides data into smaller subsets based on decision rules derived from input variables. The resulting tree structure provides an interpretable representation of the relationship between predictors and tourist arrivals. Although individual decision trees may be sensitive to data variations, they provide an important benchmark for evaluating the performance of more advanced ensemble models.

Random Forest (RF) is an ensemble learning algorithm based on multiple decision trees. The model generates a set of independent trees using randomly selected subsets of observations and predictors, and the final prediction is obtained by averaging the results of individual trees. This structure improves model stability, reduces the risk of overfitting, and allows RF to effectively capture nonlinear relationships present in tourism demand data.

Gradient Boosting Regression (GBR) is an iterative ensemble method that combines multiple weak prediction models into a stronger forecasting model. Unlike Random Forest, where trees are constructed independently, GBR builds trees sequentially, with each new tree focusing on reducing the errors generated by previous models. This allows the algorithm to achieve high predictive accuracy and efficiently model complex relationships between variables.

Extreme Gradient Boosting (XGB) is an optimized gradient boosting algorithm designed to improve both prediction accuracy and computational efficiency. The model incorporates additional regularization techniques that help control model complexity and prevent overfitting. Due to its ability to handle nonlinear relationships and complex datasets, XGB is widely used in forecasting applications involving structured socioeconomic and time-series data.

To investigate the robustness of the models under incomplete data conditions, three levels of artificially introduced missing values (10%, 20%, and 30%) were considered. This approach enabled the evaluation of model reliability under different levels of information loss and simulated realistic situations where tourism statistics may contain missing observations.

The analysis was conducted separately for each country and missing-data scenario. This approach enabled a detailed comparison of model performance under different data conditions and facilitated the identification of the most robust forecasting algorithms for predicting international tourist arrivals.

Evaluation Metrics

Model performance was assessed using three complementary statistical measures: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Bias (%). Together, these metrics provide a comprehensive evaluation of predictive performance by measuring the magnitude of prediction errors, sensitivity to large deviations, and the presence of systematic overestimation or underestimation. These metrics quantify different aspects of prediction quality and together provide a comprehensive evaluation of model performance.

Root Mean Squared Error (RMSE) measures the average magnitude of prediction errors while assigning greater weight to larger deviations. Lower RMSE values indicate higher prediction accuracy, particularly when large prediction errors are undesirable. Mean Absolute Error (MAE) measures the average absolute difference between observed and predicted values. Unlike RMSE, MAE assigns equal importance to all prediction errors regardless of their magnitude. Consequently, lower MAE values indicate better predictive performance.

Bias evaluates the systematic tendency of a model to overestimate or underestimate the total number of international tourists. It compares the sum of predicted values with the sum of observed values and is calculated as

$$Bias(\%) = \frac{\sum_{i=1}^{n} \hat{y}_i - \sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} y_i} \times 100. \tag{4.8}$$

A Bias value close to zero indicates the absence of systematic prediction error, whereas positive and negative values indicate overestimation and underestimation, respectively.

The simultaneous use of RMSE, MAE, and Bias provides a balanced assessment of prediction quality. While RMSE is particularly sensitive to large prediction errors, MAE reflects the average magnitude of deviations, and Bias identifies potential systematic tendencies in the forecasts.

The analysis was conducted separately for each country and missing-data scenario. This approach enabled a detailed comparison of model performance under different data conditions and facilitated the identification of the most robust forecasting algorithms.

4.5.1.2. Hyperparameter optimization of the selected models

The comparative analysis of the machine learning models showed that predictive performance varied across the country-specific datasets. Based on the initial model comparison, Extreme Gradient Boosting (XGB) and Gradient Boosting Regressor (GBR) were selected for further hyperparameter optimization. Grid Search was performed separately for each country and for each of the two selected algorithms. The optimized models were subsequently evaluated on independent test sets.

The optimization process was performed using **5-fold cross-validation**. For each candidate model, the training dataset was divided into five subsets, allowing the model to be trained and validated five times using different validation folds. This approach provided an estimate of

model performance on unseen validation data and reduced the dependence of the evaluation on a single train-validation split.

The objective of the modelling was to predict the number of tourists visiting Poland according to their country of origin. The analysis was conducted separately for five countries: Germany, Russia, the United Kingdom, the United States, and Ukraine. For each country, regression models were developed using a set of explanatory variables selected individually for that country.

Two gradient boosting regression algorithms were employed: Extreme Gradient Boosting (XGBoost, XGBRegressor) and Gradient Boosting Regressor (GBR). These models were selected because of their ability to capture nonlinear relationships and model complex interactions between explanatory variables in regression problems.

Hyperparameter optimization was performed using Grid Search (GridSearchCV) with 5-fold cross-validation. The optimization criterion was the negative root mean squared error (`neg_root_mean_squared_error`), ensuring that the hyperparameter configuration yielding the lowest RMSE during cross-validation was selected.

For the XGBoost model, the following hyperparameter grid was explored:

Table 4.2 Hyperparameter grid used for XGBoost.

Hyperparameter	Candidate values
`n_estimators`	200, 400
`learning_rate`	0.05, 0.10
`max_depth`	4, 6
`min_child_weight`	1, 3
`subsample`	0.8, 1.0
`colsample_bytree`	0.8, 1.0
`gamma`	0, 0.2

In total, 128 hyperparameter combinations were evaluated for the XGBoost model.

For the Gradient Boosting Regressor, the following hyperparameter grid was considered:

Table 4.3 Hyperparameter grid used for Gradient Boosting Regressor.

Hyperparameter	Candidate values
`n_estimators`	100, 200
`learning_rate`	0.05, 0.10
`max_depth`	3, 5
`subsample`	0.8, 1.0
`min_samples_split`	2, 5

Overall, 32 hyperparameter combinations were evaluated for the Gradient Boosting Regressor.

Following hyperparameter optimization, the best-performing configuration for each country–model combination was retrained using the optimal hyperparameters and subsequently evaluated on an independent test set containing artificially generated missing observations. Model performance was assessed using three evaluation metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Bias.

The comparison of predictive performance across the five analyzed countries indicated that the Gradient Boosting Regressor (GBR) achieved lower RMSE and MAE values than XGBoost (XGB) on the independent test set for all five countries. The differences in Bias between the two models were relatively small for most datasets, although a larger negative Bias was observed for

the GBR model for the United Kingdom. Based on the test-set RMSE and MAE results, GBR was selected as the final model for forecasting the number of tourists by country of origin.

The complete results for each model, including the error metrics (RMSE, MAE, and Bias) and the corresponding prediction plots, are presented in the Appendix.

The following tables provide a detailed synthesis of the model optimization and evaluation results for the Polish tourism use case. Table 4.4 presents the optimal hyperparameter configurations identified through the Grid Search procedure for each country–model combination. Table 4.5 presents the corresponding cross-validation and independent test-set performance metrics.

Table 4.4 Optimal Hyperparameter Configurations by Country and Model

Country	Model	LR (η)	Depth	Esti	MinChildW	Subsamp	Colsamp	Gamma	MinSplit
Germany	XGB	0.10	4	400	1	0.8	0.8	0.0	–
Germany	GBR	0.10	5	200	–	0.8	–	–	2
Russia	XGB	0.10	4	400	3	1.0	1.0	0.0	–
Russia	GBR	0.10	5	200	–	1.0	–	–	5
Ukraine	XGB	0.05	6	400	3	1.0	0.8	0.2	–
Ukraine	GBR	0.10	5	200	–	1.0	–	–	2
United Kingdom	XGB	0.10	4	400	1	1.0	0.8	0.2	–
United Kingdom	GBR	0.10	5	200	–	0.8	–	–	5
United States	XGB	0.05	4	400	1	1.0	0.8	0.0	–
United States	GBR	0.10	5	200	–	0.8	–	–	2

Table 4.5 Summary of Model Performance Metrics

Country	**Model**	$RMSE_{CV}$	$RMSE_{test}$	MAE_{test}	$Bias_{test}$ **(%)**
Germany	XGB	35.99	36.59	6.48	-0.11
Germany	GBR	22.59	19.81	4.68	-0.05
Russia	XGB	10.74	14.64	1.59	-0.90
Russia	GBR	7.68	9.27	1.43	-0.83
United Kingdom	XGB	13.38	18.93	3.85	-0.69
United Kingdom	GBR	11.34	15.38	3.52	-2.59
United States	XGB	19.03	14.92	3.50	0.68
United States	GBR	18.59	11.53	2.74	-0.34
Ukraine	XGB	6.90	6.32	1.49	-0.03
Ukraine	GBR	6.25	5.85	1.42	0.12

To ensure the readability of the summary table, several abbreviations have been used for the hyperparameter headers. Specifically, **LR (η)** refers to the *learning_rate*, while **Depth** and **Esti** correspond to *max_depth* and the total number of trees (*n_estimators*), respectively. **MinChildW** represents *min_child_weight*, **Subsamp** denotes the *subsample* parameter, **Colsamp** denotes the fraction of features subsampled per tree (*colsample_bytree*), and **Gamma** corresponds to the *gamma* parameter. Finally, **MinSplit** refers to *min_samples_split*.

The optimization results indicate that GBR achieved lower test-set RMSE and MAE values than XGB for all five countries considered. The largest difference was observed for Germany, where the test RMSE was 19.81 for GBR compared with 36.59 for XGB. For Russia, the United Kingdom, the United States, and Ukraine, the differences between the two models were smaller but consistent in terms of both RMSE and MAE. Based on the test-set RMSE and MAE results, GBR was selected as the final model for forecasting the number of tourists by country of origin.

The absolute value of the test-set Bias was below 1% for both models in Germany, Russia, the United States, and Ukraine. The only exception was the GBR model for the United Kingdom, for which the Bias reached -2.59%. Thus, the test-set Bias was relatively close to zero for most country–model combinations, while the largest deviation was observed for the GBR model for the United Kingdom.

The comparison between cross-validation and test-set RMSE values shows that the differences between validation and test performance varied across countries and models. For GBR, the test RMSE was lower than the cross-validation RMSE for Germany and the United States, while it was higher for Russia, the United Kingdom, and Ukraine. For XGB, the test RMSE was higher than the cross-validation RMSE for Germany, Russia, and the United Kingdom, and lower for the United States and Ukraine. These differences indicate that model performance varied between the cross-validation and independent test sets, but the results do not suggest a consistent pattern of substantial deterioration in test-set performance across the models and countries.

It should be noted that direct comparison of RMSE values between countries is not appropriate because the scale of tourist arrivals differs substantially across countries. Consequently, model performance should be interpreted jointly using RMSE, MAE, and Bias, while taking into account the characteristics and magnitude of the target variable for each country. Within each country, however, the comparison between XGB and GBR provides a meaningful basis for model selection.

4.5.2. Imputation of Price Statistics (LU)

The approach can be applied to different base learners. We implemented the following learners:

- Linear regression
- Random Forest
- XGBoost

In addition, we developed a customized learner that we call "Closest price". This is a naive learner that uses as prediction the nearest available price of the same item. This model can serve as an additional benchmark or baseline model. It only relies on the time and item identifiers and does not require any additional features. This trainer is illustrated in figure 4.1. Suppose that an item is available in periods 3 to 5. The observed price in period 5 is used to impute the price in periods 6 and 7 (carry forward), whereas the price of observed price in period 3 is used to impute the prices in periods 1 and 2 (carry backward). Note that this learner is applied with respect to a given train set. If there is no price for an item in the train set (i.e. all price observation of that item are in the test set), the learner imputes for that item as a fallback solution the average price across all observations in the train set.

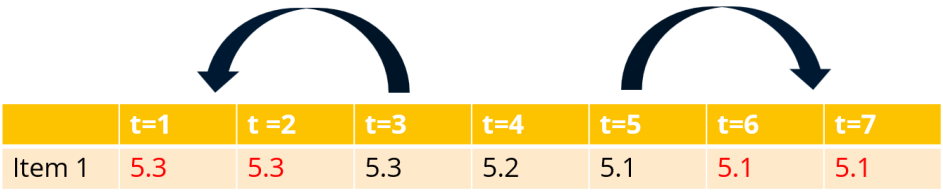


Figure 4.1 The closest price learner

ML models are evaluated using cross-validation techniques. In order to create train-test splits and obtain realistic model performance evaluations, we adapted the resampling approach to our use case. The prices data can be seen as a panel data: there are repeated measurements (price observations) across time. In addition, the panel is highly unbalanced. Because of product churn, prices are only available in some periods. We designed an innovative strategy that respects both the temporal structure of the data and the within-product correlations. It follows a two-step approach:

1. We apply K-fold cross validation at the item level.
2. For each fold and for each item in that fold, we select either the n last or the n first observations of an item to form the test set

The parameters of this method is the number of folds (k) and the number of price observations at the end or the beginning of an item's life-cycle to be considered (n). In the illustration in figure 4.2, we set k=3 and n=2. Although the resampling method is heuristic, it is consistent with cross-validation techniques commonly used in time series settings, where forecasts are typically made at the end of a time series. Because the periods during which items are available vary, we apply this approach separately for each item. After conducting an initial cross-validation step to select the item's price series, we then perform the item-specific resampling accordingly.

Many machine learning models depend on a set of hyperparameters that must be carefully selected. To address this, we perform Hyperparameter Optimization (HPO) where candidate

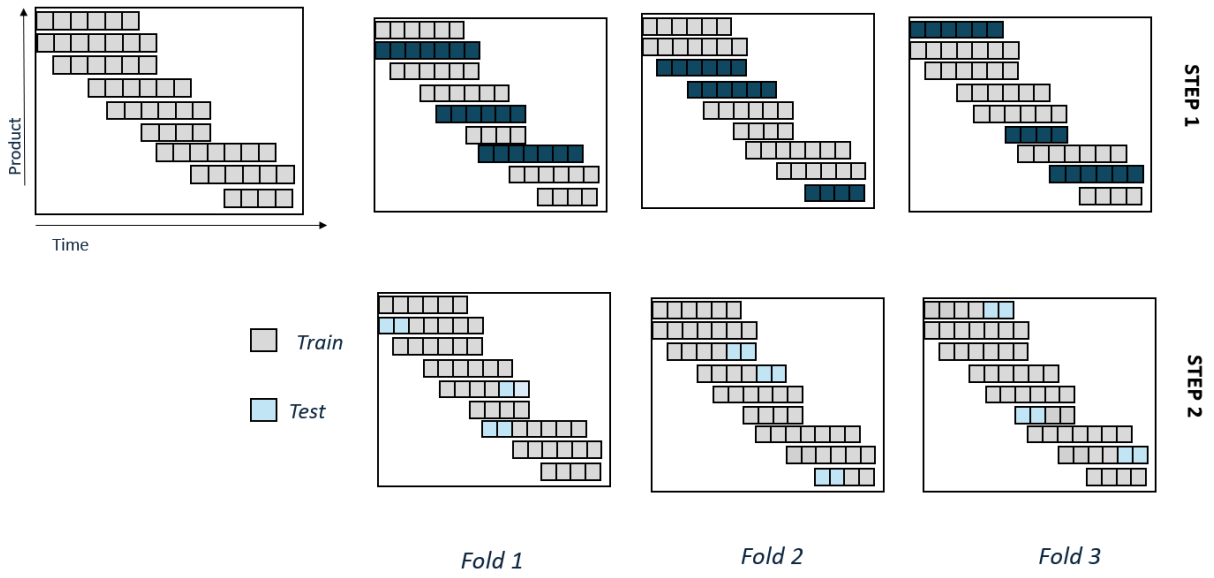


Figure 4.2 The resampling method

parameter sets are assessed based on their cross-validated performance. This step is conducted upfront, before model evaluation. Once optimal hyperparameters are determined, we train models on various train-test splits using the optimal hyperparameters in order to evaluate model performance.

This setup is a simplification from a pure nested re-sampling strategy [Bischl et al., 2024]. Nested re-sampling is a way to separate hyperparameter tuning from final model evaluation, so you get an unbiased estimate of performance. Nested re-sampling uses two layers of re-sampling loops. The outer loop is used for model evaluation (test set). Each fold in the outer loop is the further re-sampled, called the inner loop, in order to fine-tune hyperparameters (validation set). This idea is illustrated in figure 4.3. In theory we would recommend nested re-sampling, but for these initial experiments we applied a simplified workflow.

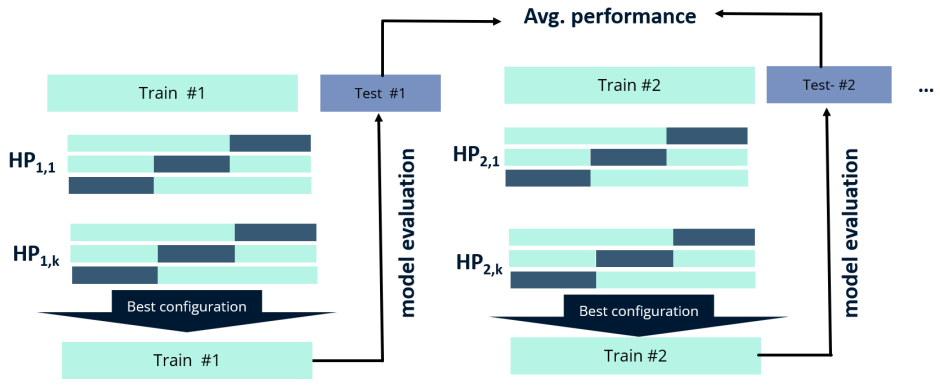


Figure 4.3 Nested re-sampling

We use a Bayesian optimization strategy to identify a set of hyperparameters that perform well for the data at hand, within a predefined search space. As a starting point, the Bayesian algorithm selects a set of hyperparameters that fall within the search space. The algorithm then

moves on to the next set of hyperparameters and evaluates them accordingly. The algorithm heuristically continues trying out potential combinations of hyperparameters that progressively improve the performance.

Let us suppose that all observations have been divided into a train s_{train} and a test s_{test} set. We define by $\hat{p}_{i,t}$ the prediction of the price for an item i in period t included in the test set s_{test}, with features z_i, using an ML model trained on the train set s_{train}.

The predicted price $\hat{p}_{i,t}$ can then be compared with the actually observed price $p_{i,t}$ in order to evaluate performance of the model. Performance of a model can be assessed using various aggregated measures. In line with the proposals made in Steurer et al. [2021], we propose to use the following metrics:

- Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{|s_{test}|} \sum_{(i,t) \in s_{test}} (p_{i,t} - \hat{p}_{i,t})^2}$$

- Mean Absolute Error (MAE)

$$MAE = \frac{1}{|s_{test}|} \sum_{(i,t) \in s_{test}} |p_{i,t} - \hat{p}_{i,t}|$$

- Mean Absolute Prediction Error (MAPE)

$$MAPE = \frac{1}{|s_{test}|} \sum_{(i,t) \in s_{test}} |\frac{p_{i,t}}{\hat{p}_{i,t}} - 1|$$

- Mean Prediction Error (MPE)

$$MPE = \frac{1}{|s_{test}|} \sum_{(i,t) \in s_{test}} \left(\frac{p_{i,t}}{\hat{p}_{i,t}} - 1\right)$$

- Coefficient of Dispersion (CoD)

$$CoD = \frac{1}{|s_{test}|} \sum_{(i,t) \in s_{test}} |\frac{p_{i,t}}{\hat{p}_{i,t}} / med(\frac{p}{\hat{p}}) - 1|$$

- Log Root Mean Squared Error (LRMSE)

$$LRMSE = \sqrt{\frac{1}{|s_{test}|} \sum_{(i,t) \in s_{test}} \left(\ln\left(\frac{p_{i,t}}{\hat{p}_{i,t}}\right)\right)^2}$$

- Max-Min Percentage Error Range (MMPER)

$$MMPER = \frac{1}{|s_{test}|} |\{(i,t) \in s_{test} : \frac{\max(p_{i,t}, \hat{p}_{i,t})}{\min(p_i^t, \hat{p}_{i,t})} > 1.1\}|$$

An additional perspective to assess the quality of imputation models is to analyse how their micro-level prediction errors propagate to the final price index. In particular, the bias–variance decomposition of prediction errors provides a useful conceptual framework: different ML models may exhibit different trade-offs between bias and variance at the level of individual price imputations, and these differences translate in a non-trivial way to the aggregated index.

However, translating this decomposition into practical tools for model selection and evaluation remains an open question. In particular, further research is needed both to develop feasible estimators of index-level bias and variance in realistic settings, and to better understand, from a theoretical perspective, how different modeling choices affect the bias–variance trade-off at the aggregate level.

4.5.3. Imputation for non-response in Labour Force Survey using synthetic data (ES)

Various imputation methods were compared using the "simputation" and "mice" packages. The seven tested methods were Random Forest, CART, Sequential Hot Deck, K-Nearest Neighbors (with K=5 and K=10), and MICE (with both Random Forest and CART). For the kNN classifier, the imputed category is determined by majority voting among the K nearest neighbours. Following preliminary experiments, we considered K=5 and K=10, as these values consistently provided the best predictive performance. Missing values were generated on repeated samples extracted from the synthetic population at various missing percentages and under different non-response assumptions (MCAR, MAR, and MNAR). The imputation models were then tested repeatedly across all these samples in a Monte Carlo process. Across most variations, Random Forest (single imputation) showed the best average performance.

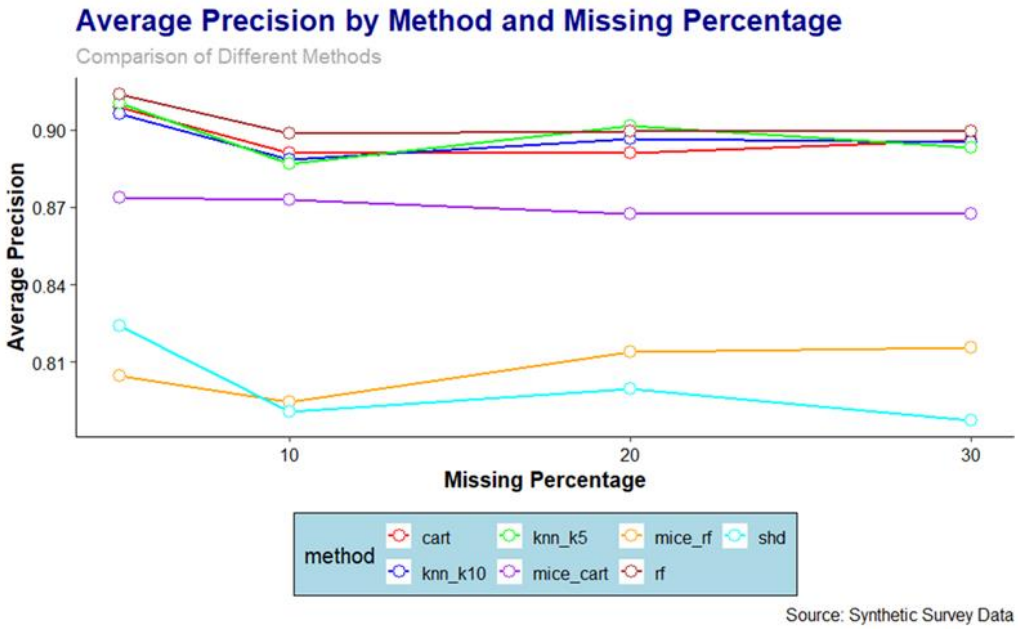


Figure 4.4 Evolution of average imputation precision across different missing data percentages by method

The table 4.6 reports the main classification performance metrics obtained from the Random Forest imputation model, which was ultimately selected as the preferred method based on its overall performance across all simulation scenarios.

In particular, the results are reported separately for each category of the employment status variable (inactive, employed, and unemployed), highlighting the differences in predictive performance across classes, especially for the minority class (unemployed), where lower sensitivity is observed. As expected, employment status is easier to predict for the employed category than for unemployment, reflecting the fact that unemployment is typically harder to explain using administrative data.

Table 4.6 Per-class performance of the final class-weighted Random Forest

Metric	INACTIVE	EMPLOYED	UNEMPLOYED
Sensitivity	0.80	0.91	0.88
Specificity	0.98	0.98	0.90
Pos Pred Value	0.95	0.97	0.39
Neg Pred Value	0.90	0.94	0.99
Precision	0.95	0.97	0.39
Recall	0.80	0.91	0.88
F1	0.87	0.94	0.54
Prevalence	0.35	0.43	0.07
Detection Rate	0.28	0.39	0.06
Detection Prevalence	0.30	0.41	0.15
Balanced Accuracy	0.89	0.95	0.89

After choosing Random Forest as our imputation method, a Random Forest classifier was trained and systematically tuned using the eleven predictors available in the sample: age (`EDAD`), sex (`SEXO`), nationality (`NACION`), Social Security affiliation (`AFISS`), and a set of benefit- and situation-related indicators (`PAROSEPE`, `DEMANSEPE`, `OFEMSEPE`, `OTRASP`, `PINCAPAC`, `PJUBILA`), along with an age–affiliation interaction term (`AFISS_EDAD`).

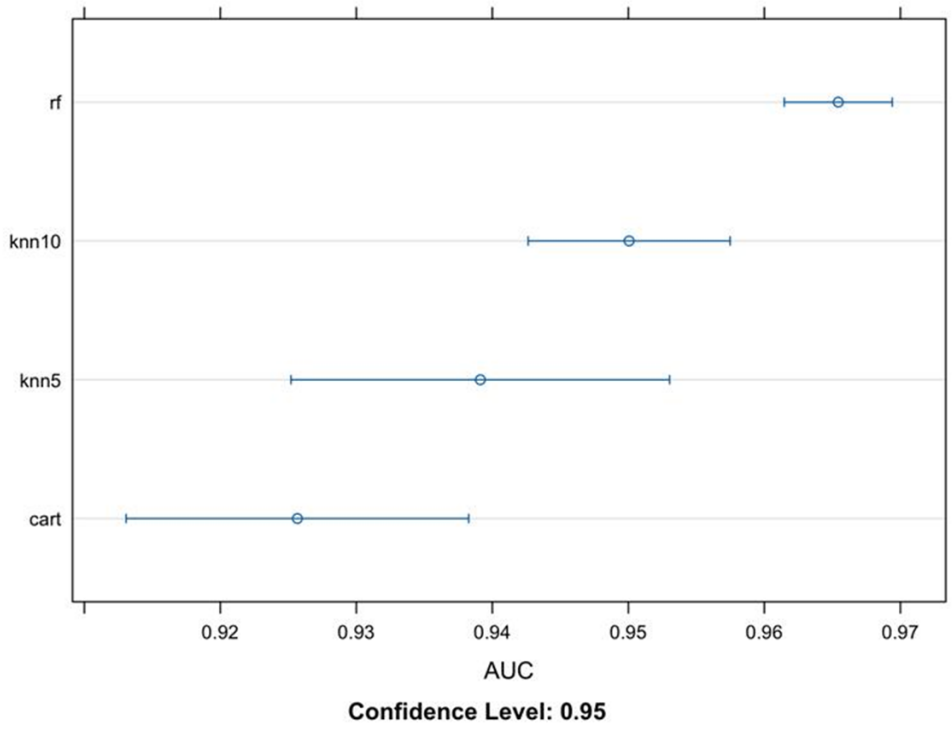


Figure 4.5 AUC point estimates and 95% confidence intervals by imputation method

The dataset was split into training (75%) and test (25%) sets using stratified sampling on the target variable, preserving the original class proportions in both partitions. A baseline model was first fit using standard Random Forest defaults (`ntree = 500`, `mtry` $= \sqrt{p}$) to serve as a reference point for evaluating the benefit of tuning.

Hyperparameter tuning was then performed via grid search with repeated cross-validation (5 folds, 3 repeats), exploring combinations of `mtry` (number of variables considered at each split), split rule (Gini impurity vs. extremely randomized trees), and minimum node size. Given the class imbalance in OPI—the *PARADO* (unemployed) category represents only around 10% of the sample—overall accuracy was deliberately avoided as the tuning objective, since a model that largely ignores the minority class can still achieve high accuracy by defaulting to the majority categories. Instead, the grid search optimized **Mean F1**, which weights each class's classification performance equally regardless of its size, ensuring the tuning process did not implicitly sacrifice PARADO detection in favor of the majority classes. The number of trees was tuned as a separate step, selecting the value that maximized PARADO's F1-score specifically, since this metric balances precision and recall for the class of greatest interest.

Beyond hyperparameter tuning, two additional strategies were evaluated to address the class imbalance and assess model parsimony:

1. **Feature selection**: a reduced model using only the six most important predictors (ranked by permutation importance) was fit and compared against the full-variable model, to check whether a simpler, more interpretable model could match performance.

2. **Class weighting**: a weighted variant was fit using inverse-frequency class weights, which penalizes misclassification of the minority class (PARADO) more heavily during training itself, rather than relying solely on post-hoc evaluation to catch imbalance-driven bias.

Model comparison across all variants used a battery of metrics beyond raw accuracy: balanced accuracy, macro-averaged F1-score, macro-averaged multiclass AUC (Hand and Till generalization), and PARADO-specific sensitivity and F1, so that minority-class performance could be assessed directly rather than being obscured by aggregate figures.

The reduced six-variable model underperformed relative to the full-variable model across the imbalance-sensitive metrics, indicating that the variables excluded from the top-six ranking—despite lower individual importance scores—still contributed meaningfully to distinguishing between OPI categories, particularly for the minority class. This model was therefore discarded in favor of retaining the full predictor set.

The class-weighted model produced the clearest improvement in minority-class detection. By penalizing PARADO misclassification during training, sensitivity for this category improved substantially relative to both the standard and unweighted tuned models, without materially degrading performance on the majority classes. Given that correctly identifying the unemployed population is of primary substantive interest, this improvement in balanced detection was prioritized over marginal gains in overall accuracy, and the class-weighted model was selected as the final specification.

The final model configuration is summarized in Table 4.7.

Table 4.7 Final Random Forest hyperparameters

Parameter	**Value**
`mtry`	4
`min.node.size`	1
`splitrule`	Gini impurity
`num.trees`	500
Class weighting	Inverse class frequency

4.5.4. ML Treatment of the Annual Survey on Construction Enterprises Using Admin Data (PT)

GLMNET family tuning was performed with grid on penalty and mixture.

Hyperparameter tuning was performed for both Random Forest and XGBoost models using space-filling grids. For the Random Forest model, the number of trees (500 – 3000), the number of predictors sampled at each split (mtry), and the minimum number of observations per node (min_n) were tuned.

For the XGBoost model, a broader set of hyperparameters was optimized, including the number of trees (500 – 2000), learning rate (0.001 – 0.01), tree depth (3 – 5), minimum node size (10 – 50), loss reduction (gamma), subsampling rate (0.6 – 0.9), and mtry. These ranges were chosen to favor more regularized and conservative models.

Model performance was evaluated using repeated 10-fold cross-validation (3 repeats), with stratification on the target variable (Revenue on Buildings by Type of Construction - QF185001) to ensure balanced distribution across folds, using the 4000 data points.

All the longitudinal data is included in the regressors so each unit is completely described in only one observation.

Model selection metric/criteria were MSE, R2 and MAE.

Two best-performing models (Random Forest and XGBoost) were selected based on cross-validated performance and used as reference models for cases where no historical observations were available. For these cases, predictions from both models were compared to ensure robustness.

The best XGBoost configuration used 653 trees, a maximum depth of 5, learning rate of 0.0054, and subsampling of 0.778, with an optimal mtry of 11 and minimum node size of 18. The best Random Forest model used 1706 trees, mtry of 11, and minimum node size of 25.

Given the apparent simplicity of the current method without ML, it was expected that linear models (heavily regularized) would provide competitive results. To enhance predictive power, additional features from multiple administrative sources were incorporated into the dataset. Regularization techniques were applied to allow less informative predictors to be shrunk or excluded from the model. However, the regularized linear models (glmnet family) underperformed when compared to more flexible approaches such as XGBoost.

There are still no results on the drift.

The current exercise aims to replace a more traditional method currently in use. In broad terms it consists on using data from previous years to impute the missing data. This can only be used when data from previous years exists and it is not used directly but affected by a growth factor calculated using administrative data. This process was explained before in section 2.4.1.

The traditional method without ML pipeline took around two days so our target was two hours and that is achievable in the current server. Not computational reduction but operational improvement due to the pipeline.

Table 4.8 Model specifications and hyperparameter tuning grids.

Model	Engine	Hyperparameters tuned	Grid size
Random Forest	ranger	trees: 500–3000 mtry: data-dependent min_n: 5–40 importance: impurity	30
XGBoost	xgboost	trees: 500–2000 learn_rate: 0.001–0.01 tree_depth: 3–5 min_n: 10–50 loss_reduction: 0–2 sample_size: 0.6–0.9 mtry: data-dependent nthread: 1	50
Elastic Net	glmnet	penalty: 10^{1}–10^{20} mixture: 0.2, 0.4, 0.8	150

4.5.5. Evaluating Machine Learning Imputation Methods Using Employment Earnings Data (SI)

Initially, several algorithms were tested. Among ensemble methods based on decision trees, XGBoost, LightGBM, CatBoost, and Random Forest were evaluated, all using default hyperparameters and yielding similar error results. In addition, neural network-based approaches were also tested, including classical feedforward neural networks and TabNet, which is specifically designed for tabular data. LightGBM was selected for further testing over XGBoost, CatBoost, Random Forest, NN and TabNet because it achieved comparable predictive performance while offering reasonably fast processing, simple implementation, and native handling of categorical features without the need for manual encoding.

The input features consist of 19 variables, primarily categorical, with some ordinal and continuous variables. Feature importance derived from the LightGBM model (gain-based importance) shows a strong concentration of predictive power in a relatively small number of variables. The most influential predictors are variables on occupation (ISCO-08), educational attainment, employer, economic activity (NACE Rev. 2), work experience, size of employer (in terms of the number of employees) and age of employee. Variables with moderate importance include institutional sector, field of education, municipality of residence, sex and municipality of workplace. Other variables (absence from work, type of employment, hours worked, citizenship, shift work, first employment, type of employer) show relatively low importance, suggesting that they contribute little additional predictive value beyond the core variables.

A separate model is trained for each combination of missingness rate and mechanism. In practice, a separate model would also be trained for each reference-month dataset. In the present study, only a single reference-month dataset (October 2024) is used for model development, hyperparameter optimisation and final model training. No historical reference months are included in the training process, and therefore no longitudinal validation is performed. This design reflects the intended operational use of the method, where machine learning-based imputation is envisaged as a complement to the existing historical trend-based imputation procedures. The model is intended to impute only newly missing values in the current reference month, for which historical information is not yet available. A modular experimental framework was designed with the intention of enabling LightGBM-based imputation across different datasets using reusable components.

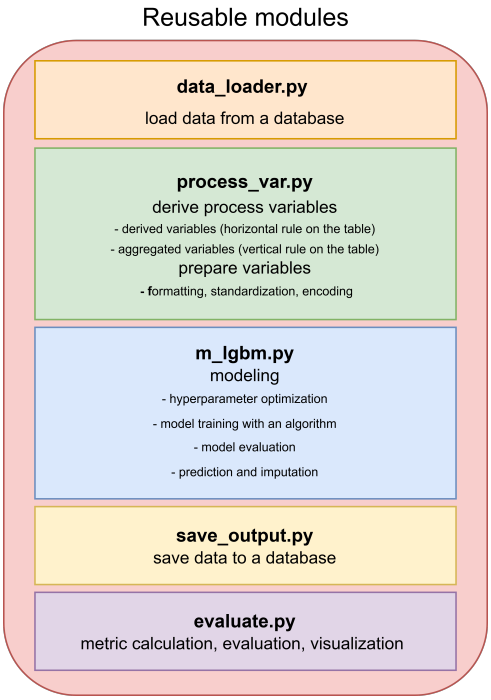


Figure 4.6 Experimental Framework for ML-Based Imputation

The Optuna library was used for hyperparameter tuning and optimisation with a custom-built search space. To reduce computational requirements, hyperparameter optimisation was performed on a random sample of 250,000 observations from the reference-month dataset. The selection of hyperparameters is adjusted to balance predictive performance and processing time, representing a trade-off between predictive performance and computational speed.

Hyperparameters and algorithm

- Target variable: regular earnings per hour.
- Features: 19 variables, mostly categorical, with some ordinal and continuous variables.
- Objective: regression.

Hyperparameters explored with Optuna (`n_trials = 100`, `num_threads = -1`) over a predefined search space:

- `num_leaves` $\in \{63, 127, 255\}$
- `max_depth` $\in \{6, 8, 10\}$
- `min_data_in_leaf` $\in \{100, 200, 400, 800, 1200, 2000\}$; candidates depend also on sample size.
- `learning_rate` $\in \{0.05, 0.07, 0.10\}$
- `min_gain_to_split` $\in \{0.0, 0.05, 0.10\}$
- `min_sum_hessian_in_leaf` $\in \{1e{-}3, 1e{-}2\}$
- Regularisation: `lambda_l1`, `lambda_l2` $\in [0.0, 5.0]$ with step 0.5.
- Subsampling: `feature_fraction` $\in \{0.6, 0.7, 0.8, 0.9\}$, `bagging_fraction` $\in \{0.6, 0.7, 0.8, 0.9\}$, and `bagging_freq` $\in \{1, 3\}$.

Post hoc sweep after Optuna for `min_data_in_bin` $\in \{10, 20, 50, 100\}$.
Other hyperparameters applied to Optuna search and model training:

- `boosting = gbdt`
- `early_stopping_rounds = 100`
- `boost_from_average = True`
- `max_boost_round = 10000`
- `cv_nfold = 5`
- `force_col_wise = True`
- `first_metric_only = True`
- `rare_fraction = 0.00003`; for each fold, a rare-category mapping was built using a relative frequency threshold computed from the training split only and subsequently applied to both the training and validation data.
- `verbose = 100`

Hyperparameters applied to final training for reproducibility:

- `deterministic = True`; `seed`, `feature_fraction_seed`, `bagging_seed`, `data_random_seed`, `drop_seed`, and `num_threads = 1`.

Cross-validation and early stopping are used as the primary mechanisms to ensure model stability and to limit the impact of noise in the data. The five-fold cross-validation is performed exclusively within the October 2024 reference-month dataset. During each cross-validation run, the available observations are randomly shuffled and partitioned into five approximately equal folds using a fixed random seed to ensure reproducibility. In each iteration, four folds are used for model training, while the remaining fold serves as the validation set. This process is repeated five times so that each fold is used once for validation and four times for training. The average validation RMSE across the five folds provides a robust estimate of predictive performance and reduces the influence of a particular train-validation split. During hyperparameter optimisation, the five-fold cross-validation procedure is used to evaluate candidate hyperparameter configurations. After the optimal hyperparameters have been selected, the same cross-validation procedure is applied during the final model training using the selected hyperparameters together with the remaining fixed hyperparameters. Deterministic training settings are additionally enabled at this stage to ensure full reproducibility of the final model.

Early stopping is applied within each cross-validation fold by monitoring the validation error and terminating the boosting process when no improvement is observed for 100 consecutive iterations. This prevents unnecessary boosting iterations, reduces the risk of overfitting, and limits the impact of noise in the data. Since cross-validation evaluates model performance across multiple data splits, it further ensures that the model captures stable patterns rather than patterns specific to a particular subset of the data. During model training, particular attention is paid to the comparison between training and validation performance. The difference between training RMSE and validation RMSE is monitored to detect potential overfitting. The absence of large discrepancies between these metrics indicates that the model generalises well to unseen observations and does not excessively learn noise from the training data.

Special attention is given to the handling of categorical variables, where rare categories below a predefined relative frequency threshold (`0.00003`) are grouped. For each cross-validation iteration, the rare-category mapping is constructed using only the training subset and is subsequently applied to both the training and validation data, thereby preventing information leakage. In addition, a post hoc adjustment of the `min_data_in_bin` parameter is performed after the hyperparameter optimisation. This contributes to more stable splits and reduces sensitivity to noise in the data. Residual analysis can be performed to evaluate the remaining unexplained variability, with the variance and distribution of residuals providing an indication of noise and model fit.

Following the final model training, predictive performance is evaluated on datasets with artificially introduced missing values. Since the original values are known before masking, the imputed values can be directly compared with the true values using the selected evaluation metrics. These datasets are reserved exclusively for the final evaluation of imputation accuracy and are not used during hyperparameter optimisation or model training.

4.5.6. Imputation of International Trade Statistics (AT)

As stated in subsection 3.6.6 we drew multiple samples from the original data and generated artificial missing values which were created using MCAR, MAR and MNAR mechanisms. For each sample the imputation models were trained on the subset that did not contain these artifical missing values subsequently used to predict them. Because the missing values were artificially generated, we can compare each imputed value with its true value. This allows us to compute evaluation metrics such as accuracy, F1-score (macro and micro), precision, and recall for categorical variables, and mean absolute deviation (MAD), mean squared error (MSE), and root mean squared error (RMSE) for numeric variables.

In addition, we consider computational performance (e.g., run-time) as well as the agreement between the distributions of the target and imputed values for our model selection process. The evaluation of distributions is done by plotting and comparing true and imputed distributions. This is an important part of model evaluation, as we not only want the results to be accurate on a micro-level, but also representative on the macro-level.

During our model selection process, we consider multiple models and various hyperparameter settings. We primarily use the R package VIM [Kowarik and Templ, 2016] for imputation. We compare the performance of more traditional regression and donor based models (kNN, hotdeck, generalized linear models, robust (generalized) linear models) to ML/AI models (random forest, xgboost, and transformer). Using the VIM function `vimpute()`, hyperparameters are tuned before training the final model on the data. Depending on the model and dataset size, between 10 and 20 predefined hyperparameter settings are considered. With these, a 5-fold cross validation is carried out to determine the best hyperparameter setting.

When imputing categorical variables with prediction models, like random forest, xgboost or generalized linear models, we use the predicted conditional probabilities for each record with a missing value in order to sample a category. Some of the models directly produce these predicted probabilities, like the XGBoost model implemented in R [Chen et al., 2026], while for others this output needs to be specified, see for instance parameter *probability = TRUE* for the implementation of random forest in R package ranger [Wright and Ziegler, 2017].

Additionally, we assess the effect of predictive mean matching (PMM). With PMM, continuous variables are imputed using observed values rather than predicted ones: the k nearest neighbors to the predicted value are considered. If $k > 1$, the final imputed value is computed as either the mean or median of these k neighbors. The function `vimpute()` enables the use of PMM for random forest, xgboost, generalized linear models and robust (generalized) linear models. Within this project we used PMM in combination with random forest and xgboost models.

XGBoost

The method imputes missing values by training a predictive model on the available (complete) observations and then using this model to estimate the missing entries. This is done separately for each variable with missing values, while variables without gaps are left unchanged.

To make these predictions, the model learns patterns in the data through an iterative process that gradually improves its accuracy. It combines many simple decision rules into a stronger overall model, allowing it to capture complex relationships between variables. The approach can handle different types of data, such as numeric and categorical variables, and can be adjusted

through various settings to improve performance.

Once the model is trained, it is used to predict the missing values, which are then inserted back into the dataset.

Transformer Models

This approach imputes missing values using a transformer model that is trained from scratch, rather than relying on pre-trained language models. It first uses only complete observations to learn patterns in the data, converting all values into a standardized text format so they can be processed consistently.

First, the training data is defined as all rows where the variable to be imputed is observed. Next, all values are converted into strings. While this is straightforward for categorical variables, numeric values are standardized to have the same length by fixing the number of digits and decimal places; shorter values are padded with preceding zeros. Finally, the string representations of all variables in a row are concatenated into a single sequence, separated by commas.

Since the transformer model operates on token IDs, all input variables must first be converted into this format. For categorical variables, each category is assigned its own token, while numeric values are represented digit by digit, with additional tokens for signs and decimal points if needed. In essence, this process maps each value to a unique identifier.

Importantly, identical values from different columns are treated separately and receive different token IDs. Missing values in the input variables are also assigned a dedicated token, whereas missing values in the target variable are excluded to avoid learning to reproduce them.

The model follows the standard transformer architecture by first embedding the input token IDs and adding positional encodings to preserve sequence order. It then stacks multiple transformer encoder blocks, each containing a multi-head attention layer and a feed-forward network, to progressively capture complex relationships in the data. The output is processed through pooling, dense, dropout, and normalization layers, with a final dense layer using softmax to produce a conditional probability vector for each possible token in the target variable.

For categorical variables, missing values are imputed by sampling from this probability distribution, while numeric values are predicted digit by digit, using the most likely token at each step and updating the input sequence iteratively. Once all values are predicted, they are converted back from strings to their original factor or numeric format.

5

Results

In this chapter we will see some results from each project but first a brief summary is shown:

Poland (PL-Accommodation): The initial evaluation showed that the **XGB** and **GBR** models achieved the highest and most comparable predictive performance among all tested approaches. Both boosting-based algorithms demonstrated low prediction errors, good stability under different missing-data scenarios, and better overall accuracy compared with single-model approaches such as kNN. Due to their similar performance and strong predictive capabilities, these two models were selected for the final optimization stage.

Luxembourg (LU-Price): The project successfully integrated machine learning into price index theory by implementing a specific centering step in the pipeline. Both **Random Forest** and **XGBoost** consistently achieved the strongest performance, significantly outperforming the linear model and the "Closest Price" baseline across all evaluation metrics (RMSE, MAE, and LRMSE). Methodologically, the results showed that ML-based quality-adjusted indices are less prone to the downward drift observed in standard multilateral indices (GEKS or TPD), proving that ML can produce imputations that are both empirically accurate and conceptually aligned with CPI theory.

Spain (ES-LFS): The study concluded that while **Random Forest** (single imputation) achieved high unit-level predictive accuracy—particularly for the "employed" and "inactive" categories with balanced accuracy above 0.92—it did not provide systematic gains for estimating population aggregates. A key methodological insight was that under the most plausible non-response mechanisms for the Labour Force Survey (MCAR), the current calibration-based reweighting remains more robust. Imputation yielded superior results under the MNAR and MAR assumptions, specifically for the minority "unemployed" class, which is typically the hardest to predict using administrative data. However, the performed analysis indicates that the real non-response behaves as MCAR mechanism.

Portugal (PT-Construction): The project is considered an operational success, with the new ML pipeline reducing processing time from two days to just two hours. **Random Forest** and **XGBoost** were selected as the best-performing models, as they better captured the non-linear relationships and interactions in high-dimensional administrative data compared to regularized linear models (Lasso). The inclusion of growth proxies and enterprise-level administrative features significantly enhanced predictive power, providing a robust solution for cases where historical survey data is missing.

Slovenia (SI-Earnings): The **LightGBM** algorithm was selected for more detailed evaluation due to its operational efficiency, fast processing, and native handling of categorical variables. The LightGBM-based method showed improved predictive performance and, for some of the published statistics and levels of aggregation examined, lower bias. However, a notable finding was its tendency to reduce variability. In addition, the pronounced concentration of values around the minimum wage level was not fully reproduced, with the peak in the imputed distribution shifted slightly towards higher values. Despite this, the imputed values retained finer granularity, while predictive performance remained stable across different missingness rates.

Austria (AT-ITS): Results indicated that **XGBoost** and **Ranger (Random Forest)** form the most robust group for numerical imputation, yielding the lowest Mean Absolute Errors across nearly all simulation scenarios. While Transformer models achieved competitive accuracy, they were deemed unsuitable for production due to extreme runtimes (over 2.5 hours) compared to the five-minute execution of tree-based models. For the categorical classification of CN codes, **kNN** unexpectedly performed best, highlighting that for high-cardinality trade data, proximity-based methods may outperform complex boosting or language models.

5.1. Imputation of foreign tourist in accommodation establishments survey (PL)

5.1.1. General Characteristics of the Results

The predictive performance of five machine learning algorithms, namely Random Forest (RF), Gradient Boosting Regression (GBR), Decision Tree (DT), K-Nearest Neighbors (kNN), and Extreme Gradient Boosting (XGB), was evaluated using monthly data describing the number of international tourists from the five largest source countries: Germany, Russia, Ukraine, the United Kingdom, and the United States. To investigate the robustness of the models under incomplete data conditions, three levels of artificially introduced missing values (10%, 20%, and 30%) were considered.

Model performance was assessed using three complementary statistical measures: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Bias (%). Together, these metrics provide a comprehensive evaluation of predictive performance by measuring the magnitude of prediction errors, sensitivity to large deviations, and the presence of systematic overestimation or underestimation.

The analysis was conducted separately for each country and missing-data scenario. This approach enabled a detailed comparison of model performance under different data conditions and facilitated the identification of the most robust forecasting algorithms.

5.1.2. Overall Performance Comparison

The predictive performance of the evaluated models was compared by jointly considering RMSE, MAE, and Bias values. The most reliable algorithms were those achieving low RMSE and MAE values while maintaining Bias values close to zero, indicating both high prediction accuracy and limited systematic error.

Across all analysed countries and missing-data scenarios, ensemble learning methods consistently outperformed the remaining approaches. Extreme Gradient Boosting (XGB) achieved the lowest prediction errors for Germany, Ukraine, the United Kingdom, and the United States in most missing-data scenarios. For the Russian dataset, Gradient Boosting Regression (GBR) provided the highest predictive accuracy, although XGB produced comparable results.

Random Forest (RF) also demonstrated stable and competitive performance across all analysed datasets. Although its prediction errors were generally slightly higher than those obtained by XGB and GBR, it remained one of the most reliable models, particularly for datasets containing larger proportions of missing values.

Decision Tree (DT) generated higher RMSE and MAE values than the ensemble-based approaches and exhibited greater sensitivity to increasing proportions of missing data. The deterioration in predictive performance became more evident under the 30% missing-data scenario, indicating a lower ability of this algorithm to generalise under incomplete data conditions.

K-Nearest Neighbors (kNN) consistently showed the weakest overall performance. In addition to producing the largest RMSE and MAE values, this model frequently generated substantial negative Bias values, indicating a systematic tendency to underestimate the number of international tourists. This behaviour was observed consistently across all analysed countries.

The influence of missing data on model performance was also investigated. Increasing the proportion of missing values from 10% to 20% did not reduce prediction accuracy. On the contrary, several models achieved slightly lower prediction errors under the 20% missing-data scenario, suggesting that the applied data imputation procedure effectively reduced the influence of irregular observations and noise present in the original datasets.

When the proportion of missing values increased to 30%, a slight deterioration in predictive performance was observed for several models. Nevertheless, the ensemble-based methods

maintained high forecasting accuracy and remained considerably more robust than the single-tree and distance-based approaches.

Overall, the obtained results demonstrate a clear advantage of ensemble learning methods for forecasting international tourist arrivals. Among the evaluated algorithms, XGBoost and Gradient Boosting Regression consistently achieved the highest predictive accuracy while maintaining low systematic errors across different countries and missing-data scenarios. These findings indicate that boosting-based techniques provide the best balance between accuracy, robustness, and stability when predicting tourist arrivals under incomplete data conditions.

Based on these results, XGBoost and Gradient Boosting Regression were selected as the most promising algorithms for further optimisation. The next stage of the study focuses on systematic hyperparameter tuning using a Grid Search procedure to identify the optimal model configuration for each country-specific dataset.

5.1.3. Germany model

5.1.3.1. Prediction Performance Analysis for Tourists from Germany

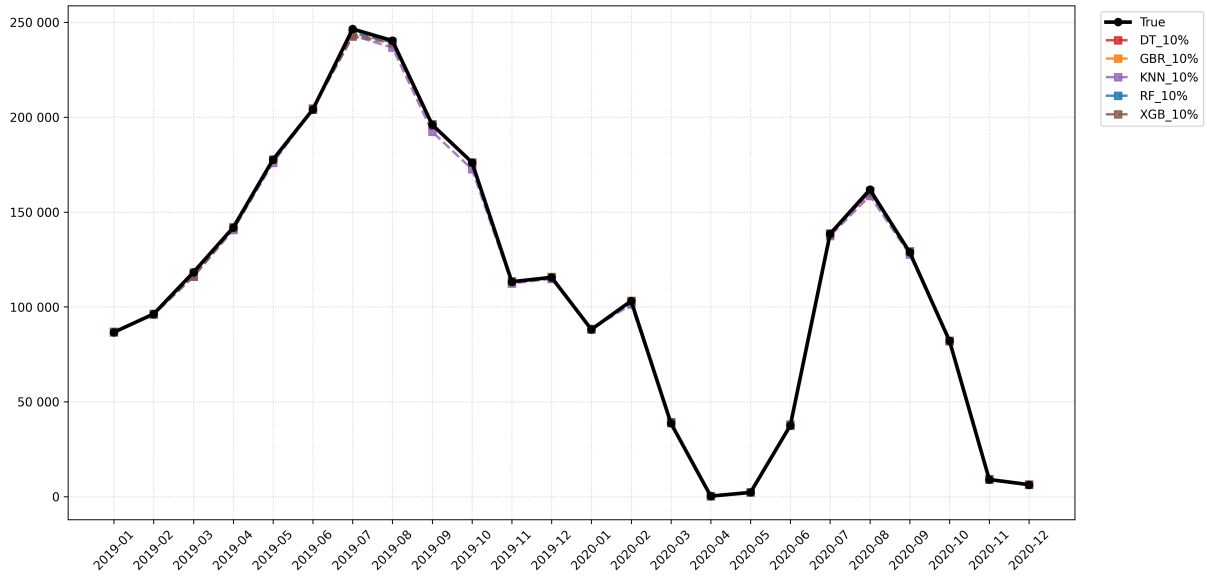


Figure 5.1 Results for Germany with 10% size of gaps

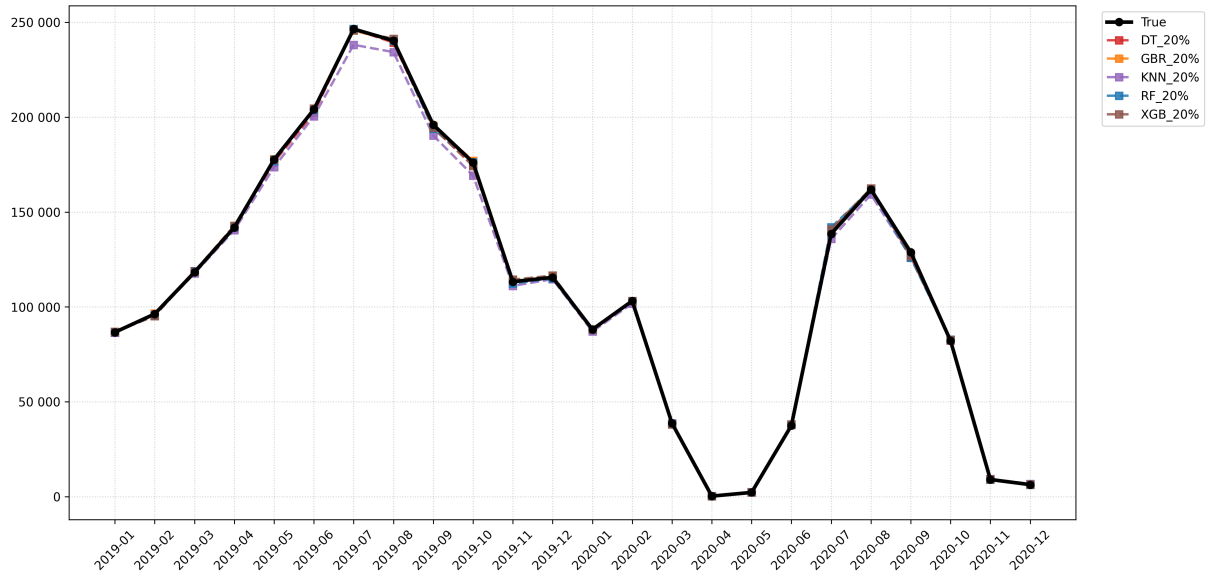


Figure 5.2 Results for Germany with 20% size of gaps

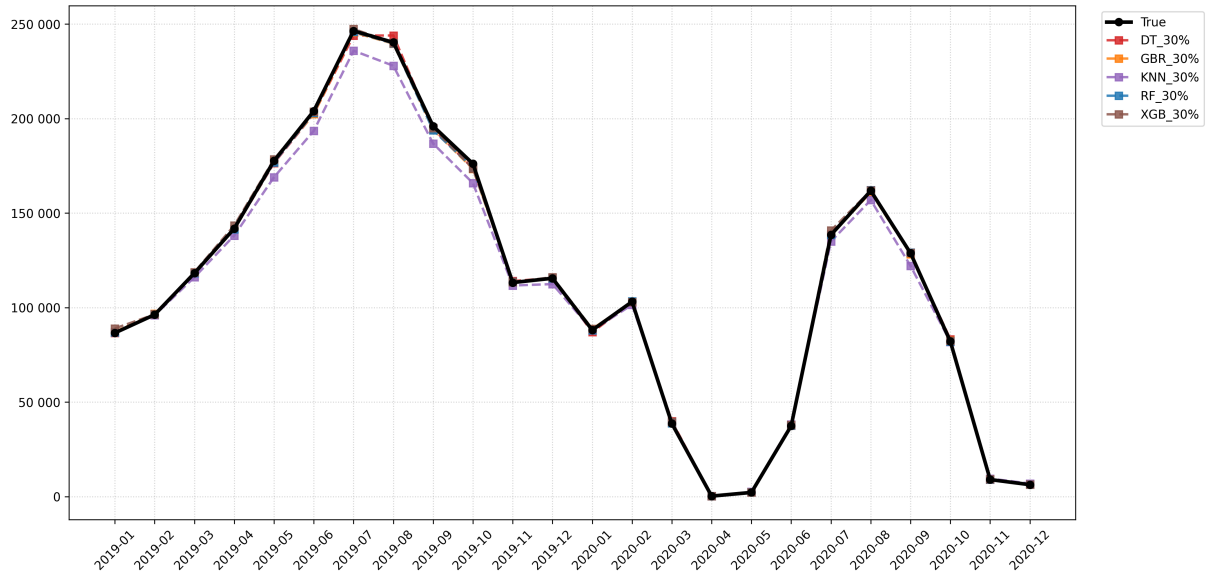


Figure 5.3 Results for Germany with 30% gaps

All evaluated models successfully captured the general pattern of the analysed time series. The results indicate a clear seasonal structure, with an increase in the number of international tourists from the beginning of 2019, reaching maximum values during the summer months (July–August), followed by a gradual decrease in the second half of the year.

The models also correctly reproduced the significant reduction in tourist numbers observed in March and April 2020, which was associated with international travel restrictions introduced during the COVID-19 pandemic. The subsequent partial recovery of tourism activity in the following months was also reflected in the generated predictions.

The highest prediction accuracy was achieved by boosting-based models, namely XGBoost (XGB) and Gradient Boosting Regression (GBR). Predictions generated by these models closely followed the observed values throughout the analysed period, including both high tourism ac-

tivity periods and rapid changes occurring in 2020. This demonstrates their ability to effectively capture seasonal patterns, long-term trends, and sudden variations in the time series.

Random Forest (RF) also provided accurate predictions; however, in several periods its estimates showed slightly larger deviations from observed values compared with XGB and GBR. The largest differences were observed during periods of peak tourist activity and during the sharp decline in tourism caused by the COVID-19 pandemic.

Decision Tree (DT) demonstrated higher prediction variability and lower accuracy when estimating extreme values. K-Nearest Neighbors (kNN) showed a systematic tendency to underestimate tourist numbers, particularly during periods of high tourism intensity. This behaviour was consistent with the higher negative Bias values observed for this model.

The comparison of different missing data scenarios indicates that increasing the proportion of missing values from 10% to 20% did not reduce prediction accuracy. In several cases, XGB and GBR achieved improved agreement with observed values under the 20% missing data scenario. Although a slight deterioration in performance was observed for some models at 30% missing data, boosting-based methods maintained high predictive stability.

5.1.3.2. Analysis of prediction errors for tourists from Germany

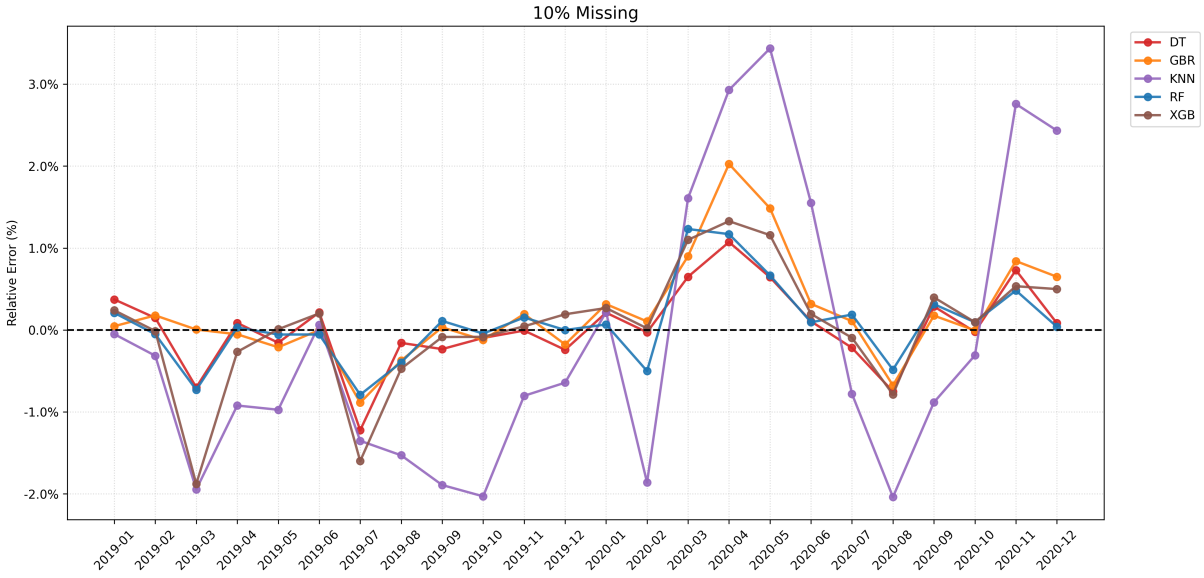


Figure 5.4 Relative Error for Germany with 10% size of gaps

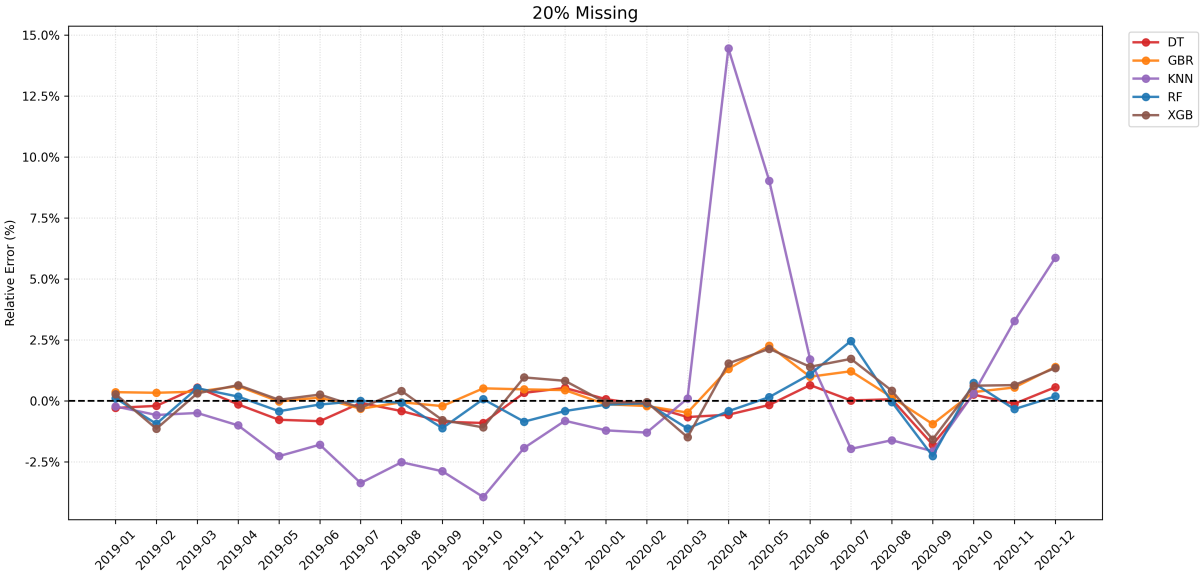


Figure 5.5 Relative Error for Germany with 20% size of gaps

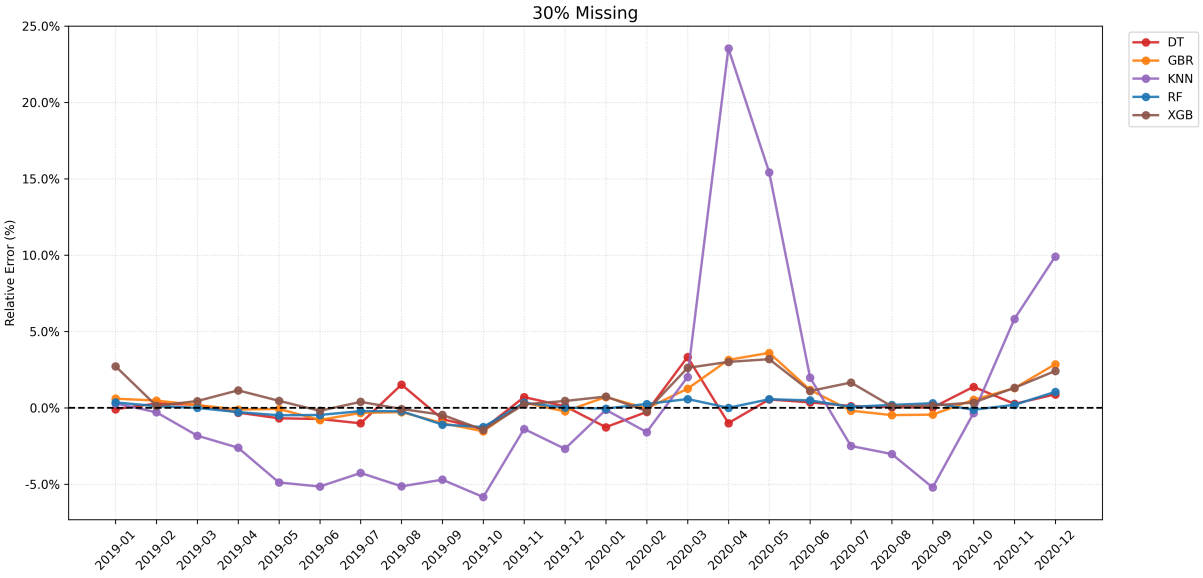


Figure 5.6 Relative Error for Germany with 30% size of gaps

The largest differences between observed and predicted values occurred during periods of rapid changes in the number of international tourists. The most significant deviations were observed between March and June 2020, when tourism activity experienced a sharp decline due to COVID-19-related travel restrictions.

During this period, all evaluated models showed a tendency to overestimate tourist numbers, as they were unable to fully capture the magnitude and suddenness of the observed decrease. The largest prediction errors occurred particularly in May and June 2020.

For the 10% missing data scenario, Random Forest achieved the most accurate predictions during this period, with relative errors ranging approximately between 2% and 4%. In contrast, GBR showed the largest deviations, with prediction errors reaching approximately 10% in May 2020. As the proportion of missing data increased, prediction accuracy generally decreased;

however, the largest deviations remained concentrated mainly between May and July 2020.

Significant differences between predicted and observed values were also identified during the summer season of 2019, when tourist activity reached its highest levels. In July 2019, the observed number of tourists was 789,755, while some models, particularly kNN, underestimated the value by approximately 37,000 tourists, generating a prediction of 752,926 under the 30% missing data scenario.

In the same period, XGB and GBR generated predictions considerably closer to the observed values, with differences generally not exceeding several thousand tourists. Similar behaviour was observed in August and September 2019. kNN consistently underestimated tourist numbers, whereas XGB, GBR, and Random Forest remained closer to the actual seasonal peaks.

These results indicate that ensemble-based methods provide a better representation of seasonal variations and high-demand periods compared with distance-based and single-tree approaches.

The smallest differences between observed and predicted values occurred during periods of relatively stable time series behaviour, particularly at the beginning of 2019 (January–April) and at the end of 2020 (October–December). During these periods, most models generated predictions differing from observed values by only several hundred or several thousand tourists, which represents a relatively small error considering the scale of the analysed data.

Among all evaluated models, XGBoost demonstrated the lowest prediction deviations throughout most of the analysed period. The generated predictions closely followed the observed time series pattern during both seasonal increases and decreases in tourist numbers. GBR achieved similarly strong performance, while Random Forest showed slightly larger deviations.

Decision Tree and especially kNN produced larger differences from observed values. kNN demonstrated a clear tendency to underestimate tourist numbers during periods of high tourism activity and showed increased prediction variability during rapid changes in the time series.

5.1.3.3. Prediction accuracy analysis

Throughout 2019, all evaluated models reproduced the observed number of tourists from the United States with high accuracy. For the Random Forest (RF), Gradient Boosting Regression (GBR), and Extreme Gradient Boosting (XGB) models, relative prediction errors generally remained below 1–2%. For example, in June 2019, the observed number of tourists was 93,107, while the predictions generated by these models differed from the reference value by less than 1%. A similar level of accuracy was maintained during the summer months, when tourist arrivals reached their annual peak.

The largest discrepancies between observed and predicted values occurred during the COVID-19 pandemic, when the time series exhibited abrupt changes.

In April 2020, the observed number of tourists decreased to 1,486. For the dataset with 10% missing values, all models slightly overestimated tourist arrivals, with relative prediction errors ranging from approximately 2% to 8%. For the datasets with 20% and 30% missing values, the predictions remained close to the observed values, and relative errors generally did not exceed 9%.

In May 2020, when 2,631 tourist arrivals were recorded, models trained on the dataset with 10% missing values slightly overestimated the observed values (approximately 2–6%). For the datasets with 20% and 30% missing values, most models underestimated tourist arrivals by up to approximately 14%, with the largest deviations observed for the kNN model under the 30% missing-data scenario.

A similar pattern was observed in June 2020. For the 10% missing-data scenario, the RF, GBR, and XGB models closely reproduced the observed values, with relative errors generally

below 6%. Larger deviations appeared for the 30% missing-data scenario, where the GBR model overestimated tourist arrivals by approximately 13%.

During the recovery period between July and September 2020, the RF, GBR, and XGB models continued to provide highly accurate predictions, with relative errors generally below 3–5%. Higher deviations were mainly observed for the DT and kNN models when 30% of the data were missing.

The final months of 2020 also showed good agreement between predicted and observed values. In November and December, most models maintained relative errors below 5%. Larger deviations were observed for the GBR model, which overestimated tourist arrivals in December by approximately 13%, and for the kNN model, whose prediction errors reached approximately 10–11%.

Overall, the ensemble learning methods (XGB, GBR, and RF) consistently provided the closest agreement with the observed time series across all missing-data scenarios. The largest prediction errors were associated with periods of rapid changes in tourist arrivals, particularly during the pandemic, whereas stable periods of the time series were reproduced with high accuracy by all evaluated models.

5.1.4. Execution time information

Table 5.1 Execution time information

Country	Missing_%	Model	Time_sec
Germany	10	RF	2.97
Germany	10	GBR	16.13
Germany	10	DT	0.30
Germany	10	kNN	0.22
Germany	10	XGB	0.28
Germany	20	RF	2.32
Germany	20	GBR	13.84
Germany	20	DT	0.27
Germany	20	kNN	0.38
Germany	20	XGB	0.29
Germany	30	RF	2.02
Germany	30	GBR	12.04
Germany	30	DT	0.24
Germany	30	kNN	0.49
Germany	30	XGB	0.29
Russia	10	RF	0.85
Russia	10	GBR	4.98
Russia	10	DT	0.11
Russia	10	kNN	0.04
Russia	10	XGB	0.22
Russia	20	RF	1.07
Russia	20	GBR	4.55
Russia	20	DT	0.10
Russia	20	kNN	0.11
Russia	20	XGB	0.22
Russia	30	RF	0.95
Russia	30	GBR	3.93
Russia	30	DT	0.08

Country	Missing_%	Model	Time_sec
Russia	30	kNN	0.15
Russia	30	XGB	0.23
Ukraine	10	RF	2.36
Ukraine	10	GBR	9.53
Ukraine	10	DT	0.20
Ukraine	10	kNN	0.16
Ukraine	10	XGB	0.37
Ukraine	20	RF	2.14
Ukraine	20	GBR	8.24
Ukraine	20	DT	0.16
Ukraine	20	kNN	0.26
Ukraine	20	XGB	0.31
Ukraine	30	RF	1.97
Ukraine	30	GBR	7.37
Ukraine	30	DT	0.15
Ukraine	30	kNN	0.34
Ukraine	30	XGB	0.29
United Kingdom	10	RF	1.79
United Kingdom	10	GBR	6.88
United Kingdom	10	DT	0.13
United Kingdom	10	kNN	0.07
United Kingdom	10	XGB	0.25
United Kingdom	20	RF	1.62
United Kingdom	20	GBR	6.22
United Kingdom	20	DT	0.11
United Kingdom	20	kNN	0.11
United Kingdom	20	XGB	0.28
United Kingdom	30	RF	1.43
United Kingdom	30	GBR	5.43
United Kingdom	30	DT	0.11
United Kingdom	30	kNN	0.14
United Kingdom	30	XGB	0.26
United States	10	RF	1.10
United States	10	GBR	4.47
United States	10	DT	0.08
United States	10	kNN	0.04
United States	10	XGB	0.23
United States	20	RF	0.95
United States	20	GBR	4.02
United States	20	DT	0.08
United States	20	kNN	0.05
United States	20	XGB	0.22
United States	30	RF	0.83
United States	30	GBR	3.56
United States	30	DT	0.07
United States	30	kNN	0.06
United States	30	XGB	0.23

5.2. Imputation of Price Statistics (LU)

Section 3.6.2 describes the ML pipeline which has been implemented and which includes a specific centering step. This step is key to make the link with CPI theory. Indeed, the key conditions that we identified is that the ML model should have no bias: on average, the imputed prices from the ML model in each period should be equal to the observed prices. This condition should hold in each period. We will first review the conceptual assessment below and then provide results of the different imputation methods (ML models).

- **Conceptual assessment**

We denote by $p_{i,t}$ and $q_{i,t}$ the price and quantity sold for item i in period t. The set of items available in period t is denoted by N_t. The number of items available in period t is denoted by n_t. Moreover, we consider a set of price-determining product characteristics (called "features" in the ML field) for product i denoted by $z_{i,k}$ ($k = 1, ..., K$). The K-dimensional vector of product characteristics for product i is written as z_i.

The quality adjustment framework described in this section is based on Chapter 8 of International Monetary Fund et al. [2025]. We assume exact utility maximizing behavior with a linear utility function. In the CPI Theory manual (equation 51, Chapter 8), it is shown that under these assumptions, the following fundamental equation holds:

$$p_{i,t} = P_t h(z_i), \tag{5.1}$$

where P_t is an aggregated price level in period t, and h is a function that measures the quality adjustment factor for item i based on the characteristics z_i of that item.

We now take the logarithm on each side of equation 5.1,define $\rho_t := \ln(P_t)$, $g(z_i) := \ln(h(z_i)$ and add an error term :

$$\ln(p_{i,t}) = \rho_t + g(z_i) + \epsilon_{i,t} \tag{5.2}$$

We now assume that log prices are related to its characteristics in a complex way.

$$\ln(p_{i,t}) = F(z_i, t) + \varepsilon_{i,t} \tag{5.3}$$

We do not restrict ourselves to linear models. In fact, F can be based on any ML method. Typically, we can train the function F on a dataset covering several periods. Pooling data over several time periods usually increases the generalization accuracy of the ML model. It reduces the risk of overfitting to patterns found in small amount of data.

We allow for time-varying relationships between characteristics and prices. We obtain one prediction function per period.

$$f_t(z_i) := F(z_i, t). \tag{5.4}$$

Let $\hat{f}_t$ be the estimated function (or trained model). In order to be in line with equation 5.2, we will require the trained model to satisfy an orthogonality property. As a consequence,

the model can be decomposed into a time component and a time-dependent function that evaluates the product characteristics.

Proposition The function $\hat{f}$ can be decomposed into the form

$$\hat{f}_t(z_i) = \hat{\rho}_t + \hat{g}_t(z_i) \tag{5.5}$$

with

$$\hat{\rho}_t = \frac{1}{n_t} \sum_{i \in N_t} (\ln(p_{i,t}) - \hat{g}_t(z_i)) \qquad \forall t = 1, .., T. \tag{5.6}$$

if and only if

$$\sum_{i \in N_t} \left(\ln(p_{i,t}) - \hat{f}_t(z_i) \right) = 0. \qquad \forall t = 1, .., T \tag{5.7}$$

This proposition has been proved in May and Lamboray [2026].

We conclude from the proposition above that an acceptable model needs to satisfy:

$$\min_f \sum_{t=1,T} \sum_{i \in N_t} (\ln(p_{i,t}) - f_t(z_i))^2 \tag{5.8}$$

subject to the constraint that :

$$\sum_{i \in N_t} (\ln(p_{i,t}) - f_t(z_i)) = 0 \quad \forall t \tag{5.9}$$

Let $\hat{\rho}_t$ and $\hat{g}_t$ be the solutions for this optimization problem. Let us now consider two time periods $t = 1$ and $t = 2$ that we wish to compare. As previously, there are now different ways to calculate quality adjusted average prices in these two periods, using the model from the first or from the second period.

$$\hat{\rho}_1 = \frac{1}{n_1} \sum_{i \in N_1} (\ln(p_{i,1}) - \hat{g}_1(z_i)) \tag{5.10}$$

$$\hat{\rho}_2 = \frac{1}{n_2} \sum_{i \in N_2} (\ln(p_{i,2}) - \hat{g}_2(z_i)) \tag{5.11}$$

$$\hat{\delta}_1 = \frac{1}{n_1} \sum_{i \in N_1} (\ln(p_{i,1}) - \hat{g}_2(z_i)) \tag{5.12}$$

$$\hat{\delta}_2 = \frac{1}{n_2} \sum_{i \in N_2} (\ln(p_{i,2}) - \hat{g}_1(z_i)) \tag{5.13}$$

There are two ways to measure a constant quality price change. The first option is to use the preferences from period $t = 1$, and the second option is to use preference from period $t = 2$. As the two measures are equally acceptable, we take the average of the two as our final measure of constant quality price change between the two periods :

$$\ln(I_{12}^{ML}) = 0.5(\hat{\delta}_2 - \hat{\rho}_1) + 0.5(\hat{\rho}_2 - \hat{\delta}_1) \tag{5.14}$$

Let us now define predicted prices from a trained model:

$$\ln(\hat{p}_{i,t}) := \hat{f}_t(z_i) = \hat{\rho}_t + \hat{g}_t(z_i) \qquad \forall t = 1, .., T, \forall i \in N_t, \tag{5.15}$$

This price index can again be written as an exact imputation index. The imputations directly come from the two ML models. This result is only possible because we require the model to satisfy the orthogonality property.

$$\ln(I_{12}^{ML}) = \frac{1}{2}\left[\frac{1}{n_2}\sum_{i\in N_2}\ln(p_{i,2}) - \frac{1}{n_2}\sum_{i\in N_2}\ln(\hat{p}_{i,1}) + \frac{1}{n_1}\sum_{i\in N_1}\ln(\hat{p}_{i,2}) - \frac{1}{n_1}\sum_{i\in N_1}\ln(p_{i,1})\right] \tag{5.16}$$

In the special case where all items are matched across the two periods ($N1 = N2$), the price index 5.16 reduces to a simple comparison between average prices. This result follows directly from the orthogonality property. Consequently, imputations play no role when the item set remains constant over time.

The price index 5.16 is a bilateral index, as it is constructed from two periods and relies on imputations derived from period-specific prediction functions. However, by construction it satisfies the time-reversal test, since exchanging periods 1 and 2 simply inverts the index. This property makes it suitable for embedding within a multilateral framework such as the GEKS index, allowing consistent comparisons over multiple periods. In this context, the bilateral indices 5.16 can be computed for all pairs of periods and then aggregated using the GEKS formula. The imputed prices used in these bilateral comparisons may be obtained from ML models estimated on pooled data across several periods, which can improve predictive accuracy while remaining fully compatible with the index number framework.

The framework introduced above does not rely on weights. If available, we would recommend to use expenditure weights to measure the importance of each observation. An expenditure weight represents the total amount of purchases for that product. Expenditure weights are not available with data obtained from the web (as in our dataset), but are typically available in transaction dataset. For simplicity, we have ignored the weight issue in our framework. It is possible to extend the framework to the weighted case, although there may be some practical issues how the weights will be taken up in the ML models.

Solving a constrained ML model is not a trivial task. We will heuristically implement the approach by including a specific recentering step in the ML pipeline. We consider any standard ML model $\mathcal{M}$ (which we will call the base learner) and solve the following optimization problem:

$$\min_{\alpha,\beta} \sum_{t=1,..,T}\sum_{i\in N_t}\left(\ln(p_{i,t}) - \alpha\mathcal{M}(t, z_i) - \sum_{s=1,..,T}\beta_s d_s\right)^2, \tag{5.17}$$

where d_s are time-dummies. The independent coefficients α and β_s can always be optimized, so that in the end, the orthogonality constraint will be satisfied. With this strategy,

any base learner can be corrected in order to fit into our framework. If a base learner already satisfies by design the orthogonality property (e.g. linear regressions), this additional optimization problem will be neutral.

- **Empirical assessment**

We apply four models— a linear model, Random Forest, XGBoost, and the Closest Price baseline — to the dataset. Their performance according to several evaluation metrics is shown in Figure 5.7. In these experiments, we fix the number of folds at $k = 5$ and vary n, the horizon at the beginning or end of a price series, between 1 and 3. The results are relatively robust across different choices of n. Random Forest and XGBoost consistently achieve the strongest performance, while the linear model performs the worst. Overall, the linear model outperforms the baseline model (closest price) but remains inferior to both Random Forest and XGBoost.

The resulting price indices, with prices imputed from the respective models, are shown in Figure 5.8. The indices based on Closest Price and XGBoost display broadly similar dynamics and result in the strongest downward movement. In contrast, the linear-model-based index exhibits the smallest decline, whereas the random forest imputations lie between the linear and the XGBoost models.

In addition to the ML based price indices, we also compiled standard multilateral price indices: GEKS, Time Product Dummy and Geary-Khamis. The definition for these three price indices can be found in Eurostat [2022]. These three indices do not require any product characteristics, they only depend on prices by item and period. However, as shown in Figure 5.8, these indices seem to contain some downward drift compared to the ML quality adjusted indices.

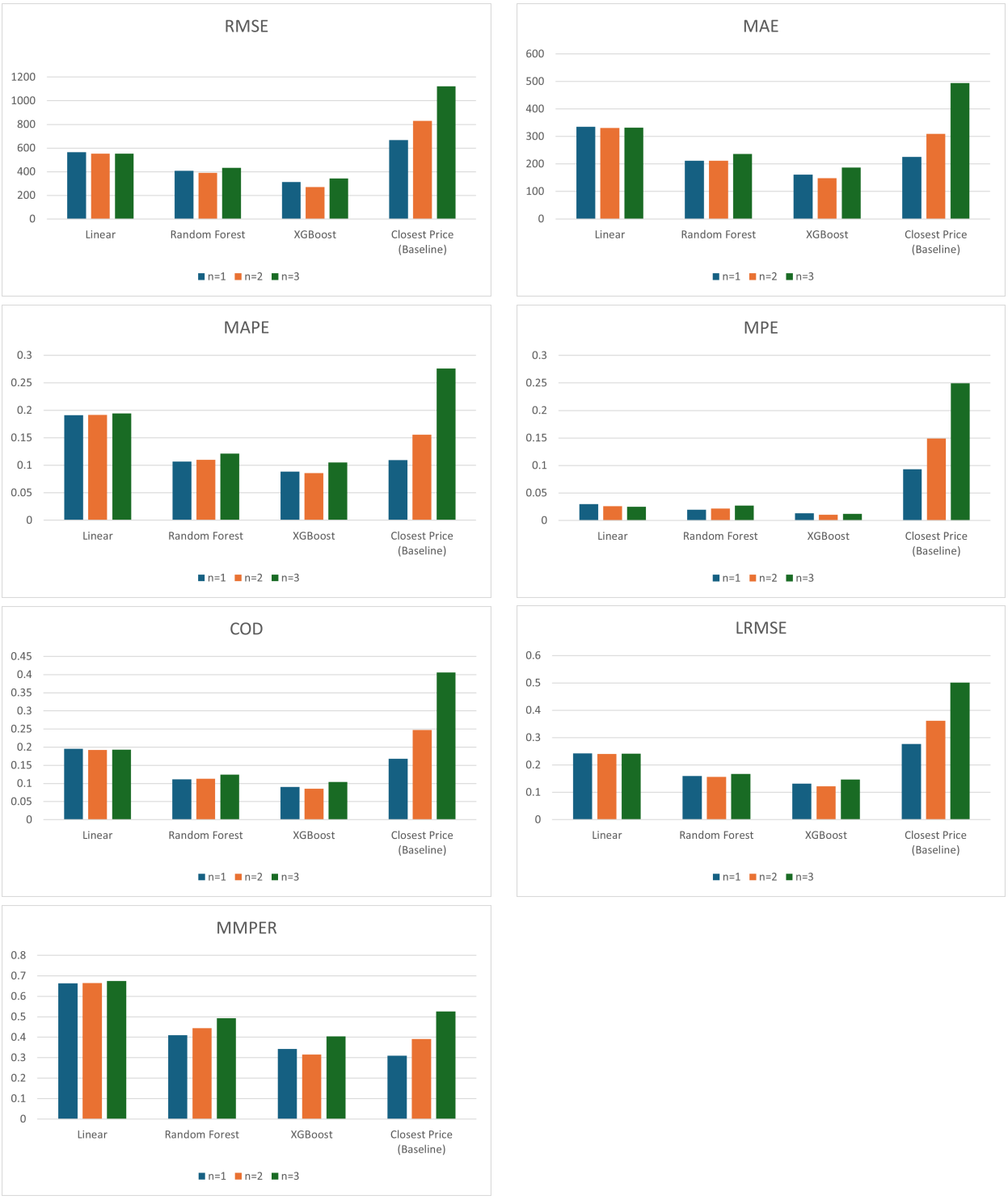


Figure 5.7 Performance measures

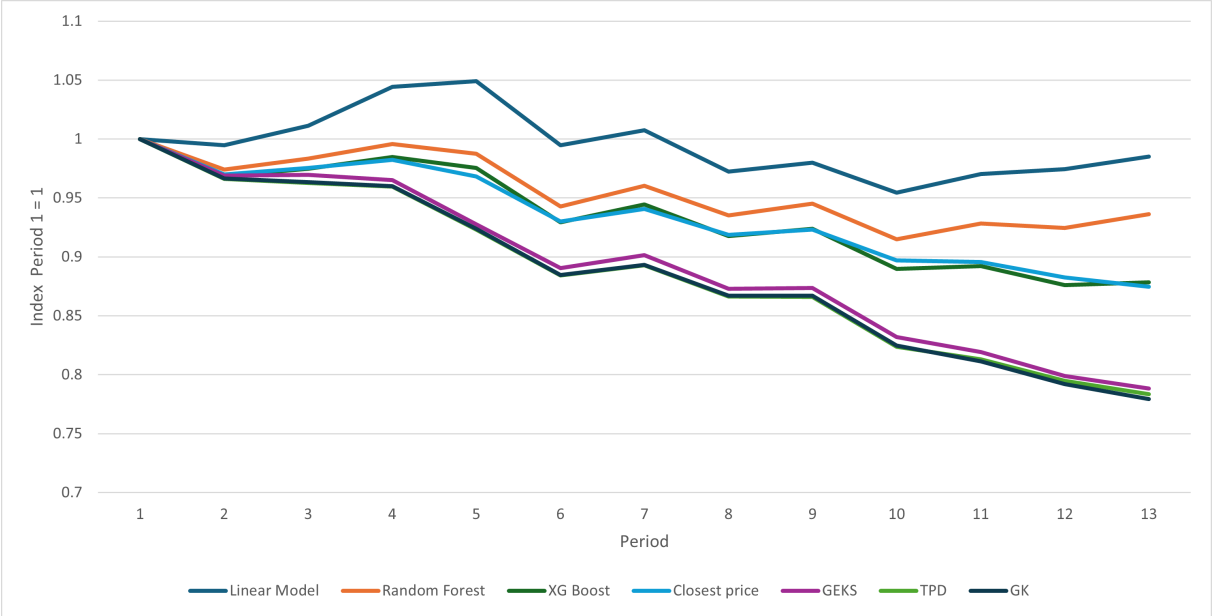


Figure 5.8 Price index

5.2.1. Execution time information

Table 5.2 reports the execution times of the different machine learning approaches. The training phase includes both hyperparameter optimisation and model evaluation using a five-fold cross-validation procedure, while the prediction phase corresponds to the generation of imputations used for price index compilation.

The results show that the computational burden is generally modest on standard consumer hardware. Linear regression and the Closest Price approach require less than 10 seconds to complete the entire workflow, while Random Forest completes in approximately 1.7 minutes. XGBoost is the most computationally demanding method, with a total execution time of around 5.8 minutes, largely due to the hyperparameter optimisation and training stages. In all cases, the computation of the price index itself is negligible, requiring less than 0.1 seconds.

Given these execution times, computational performance is not considered a limiting factor in the present use case. Even the most demanding model can be trained, evaluated and used for prediction within a few minutes on a standard laptop. Consequently, execution time does not constitute a meaningful criterion for model selection. Instead, the choice of model should primarily be driven by statistical performance, such as prediction accuracy, robustness and the resulting quality of the compiled price indices.

Table 5.2 Execution time (seconds) of the different methods (five folds with $n = 1$).

Model	Training-Evaluation	Prediction	Index	Total
Linear Regression	3.41	2.81	0.05	6.27
Random Forest	94.63	6.37	0.05	101.05
XGBoost	347.64	2.95	0.04	350.63
Closest Price	1.38	3.41	0.03	4.82

5.3. Imputation for non-response in Labour Force Survey using synthetic data (ES)

A total of seven imputation methods were evaluated using the `simputation` and `mice` R packages. These include Random Forest, CART, Sequential Hot Deck, K-Nearest Neighbours (with $K = 5$ and $K = 10$), and MICE implementations based on both Random Forest and CART models. The evaluation was carried out by artificially inducing missingness in repeated samples drawn from the synthetic population, considering different missing rates and non-response mechanisms (MCAR, MAR, and MNAR). This Monte Carlo framework allows for a systematic comparison of the robustness of each method across a wide range of scenarios.

Across all simulation settings, the Random Forest single imputation approach consistently achieved the best overall performance in terms of predictive accuracy and stability, and was therefore selected as the preferred imputation method. Using this model, estimates of the employment status distribution were computed and compared with those obtained using the standard calibrated Horvitz–Thompson estimator based on non-response adjusted weights.

With the Random Forest model selected, we then compare the target occupational variable estimates obtained after imputation with those computed using the calibrated and reweighted Horvitz-Thompson estimator. We considere multiple missing percentage values for the target variable, as well as different types of non-response. The calibrated and reweighted estimator generally performed better under MCAR assumption, while imputation yielded better results under the MAR and MNAR assumptions, particularly for estimating unemployment, which is the minority class.

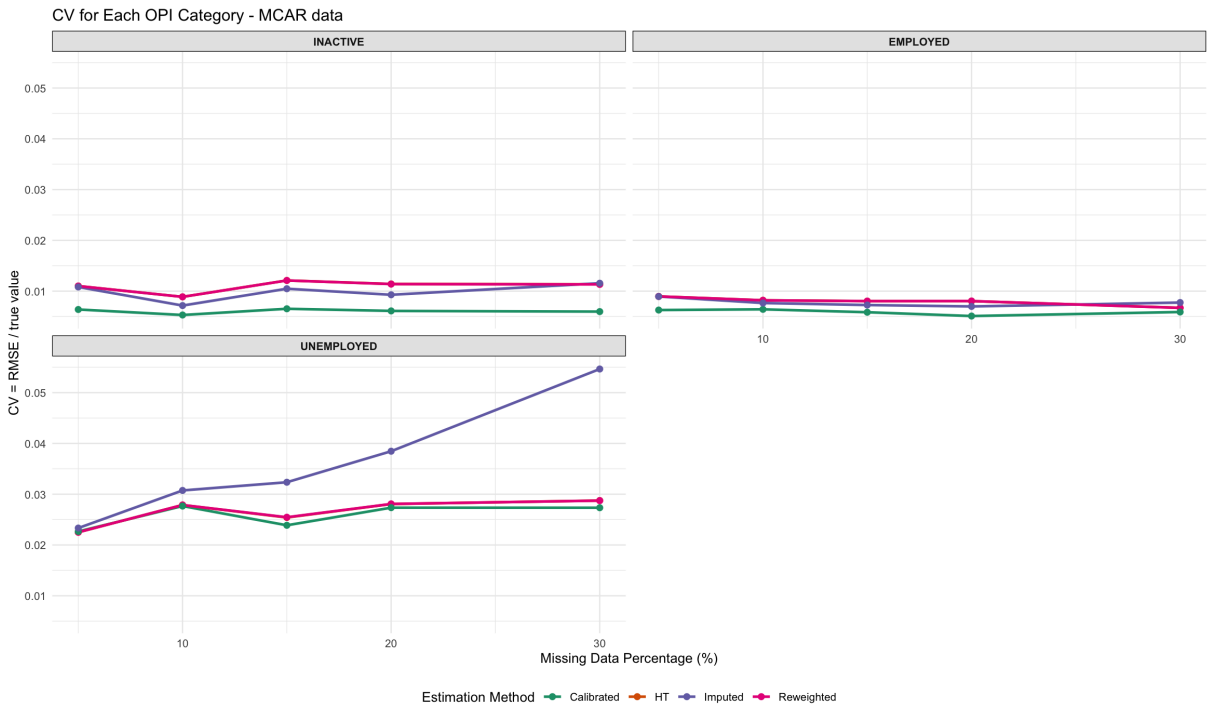


Figure 5.9 Estimator Coefficient of Variation (CV) across employment status categories under MCAR data by missingness rate and estimation strategy

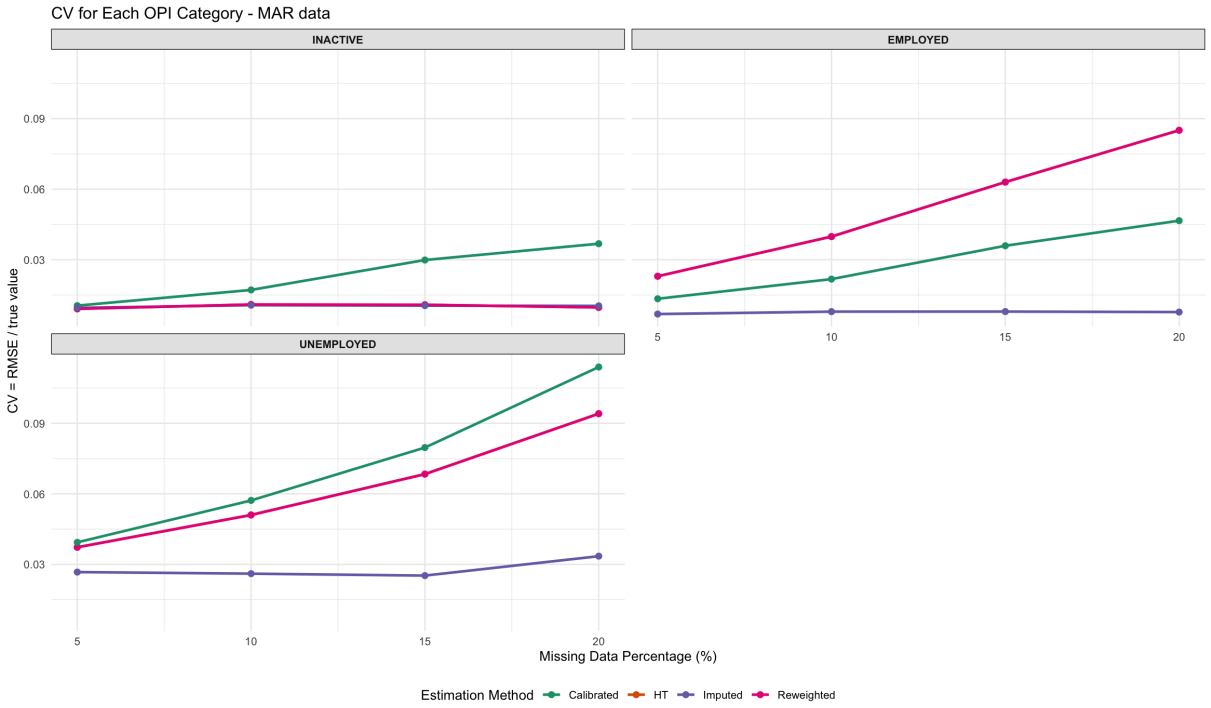


Figure 5.10 Estimator Coefficient of Variation (CV) across employment status categories under MAR data by missingness rate and estimation strategy

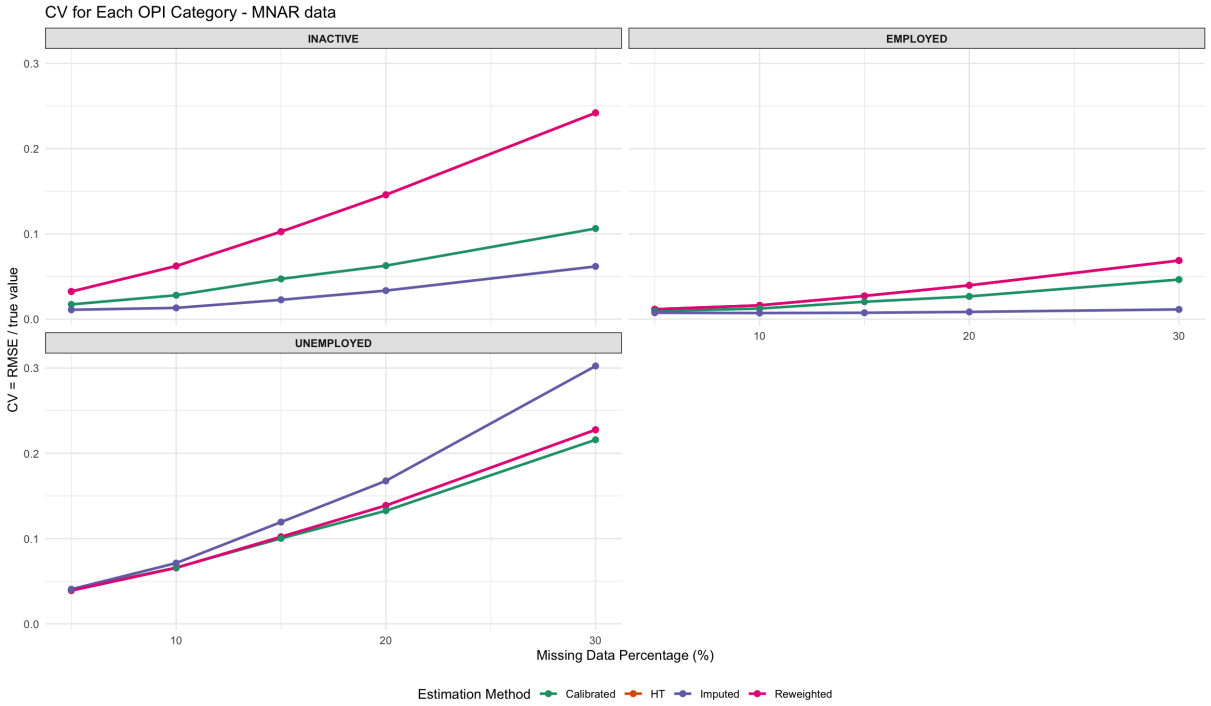


Figure 5.11 Estimator Coefficient of Variation (CV) across employment status categories under MNAR data by missingness rate and estimation strategy

So a final summary could be:

Table 5.3 Recommended estimation strategy by missingness mechanism

Mechanism	Reweighting vs. Imputation	Recommendation
MCAR	Reweighting > Imputation	Reweighting
MAR	Reweighting < Imputation	Imputation
MNAR	Reweighting < Imputation	Imputation

The main results of this project suggest that the choice of estimation strategy has a significant impact on the accuracy of labour market aggregates derived from survey data. While machine learning-based imputation methods can achieve high predictive performance at the unit level, particularly when rich auxiliary information is available, this improvement does not necessarily translate into better performance at the aggregate level.

The results indicate a clear dependence on the underlying missing data mechanism. Under MCAR, the calibrated reweighting approach systematically outperforms imputation in terms of aggregate accuracy. In contrast, under MAR and MNAR scenarios, imputation methods tend to perform better, particularly for capturing the unemployed category, which represents a relatively small and more difficult-to-predict subgroup of the population.

We can investigate how the bias in the estimates varies under different simulated nonresponse mechanisms. As shown in the following figures 5.12, 5.13 and 5.14, the bias remains very small in all cases, becoming slightly more noticeable only at higher nonresponse rates under the MAR and MNAR mechanisms.

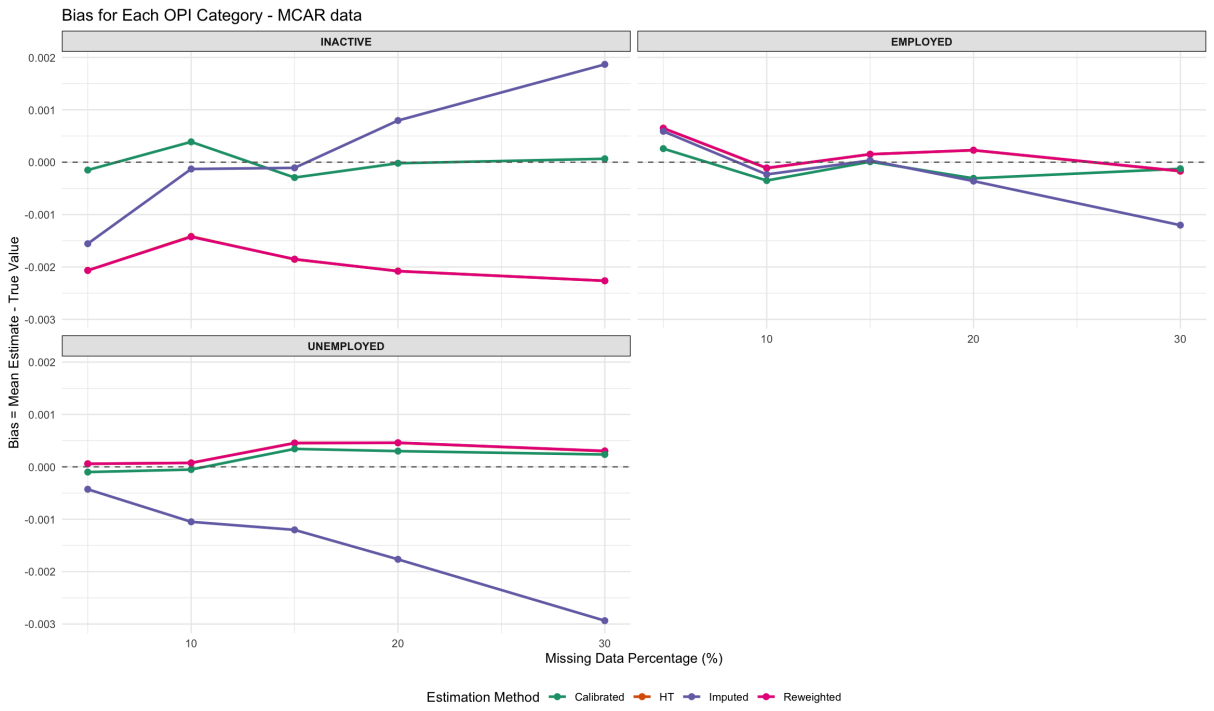


Figure 5.12 Bias across employment status categories under MCAR data by missingness rate and estimation strategy

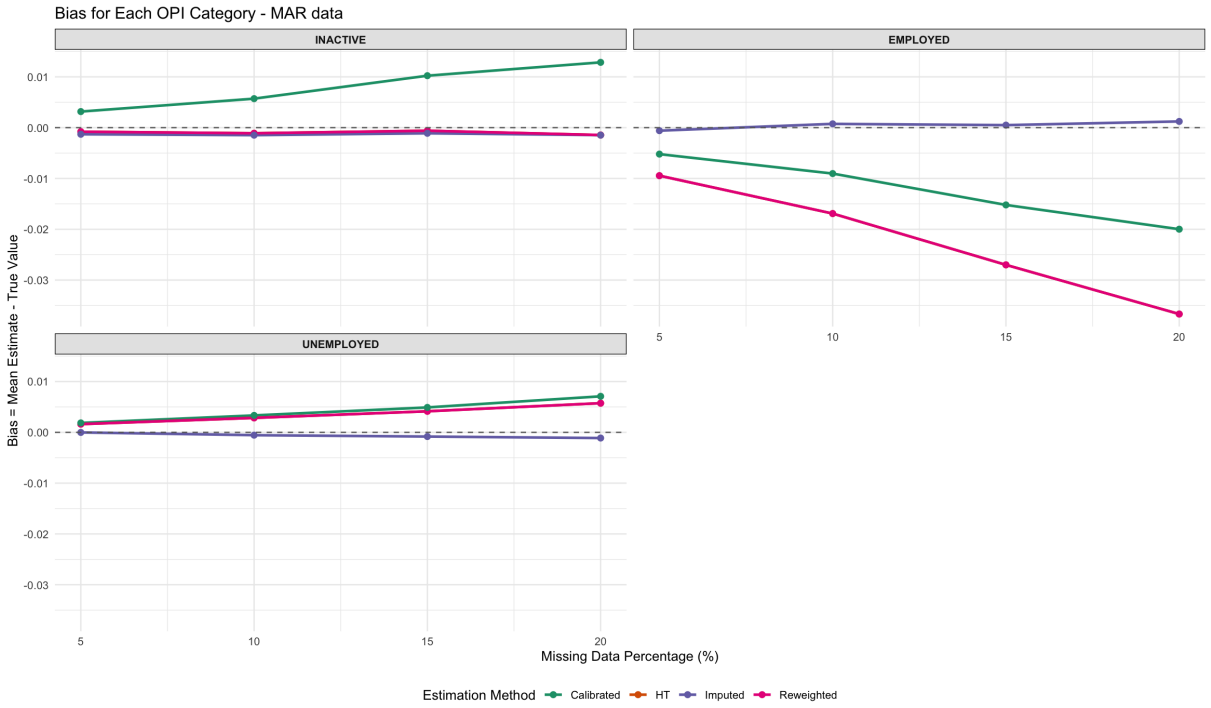


Figure 5.13 Bias across employment status categories under MAR data by missingness rate and estimation strategy

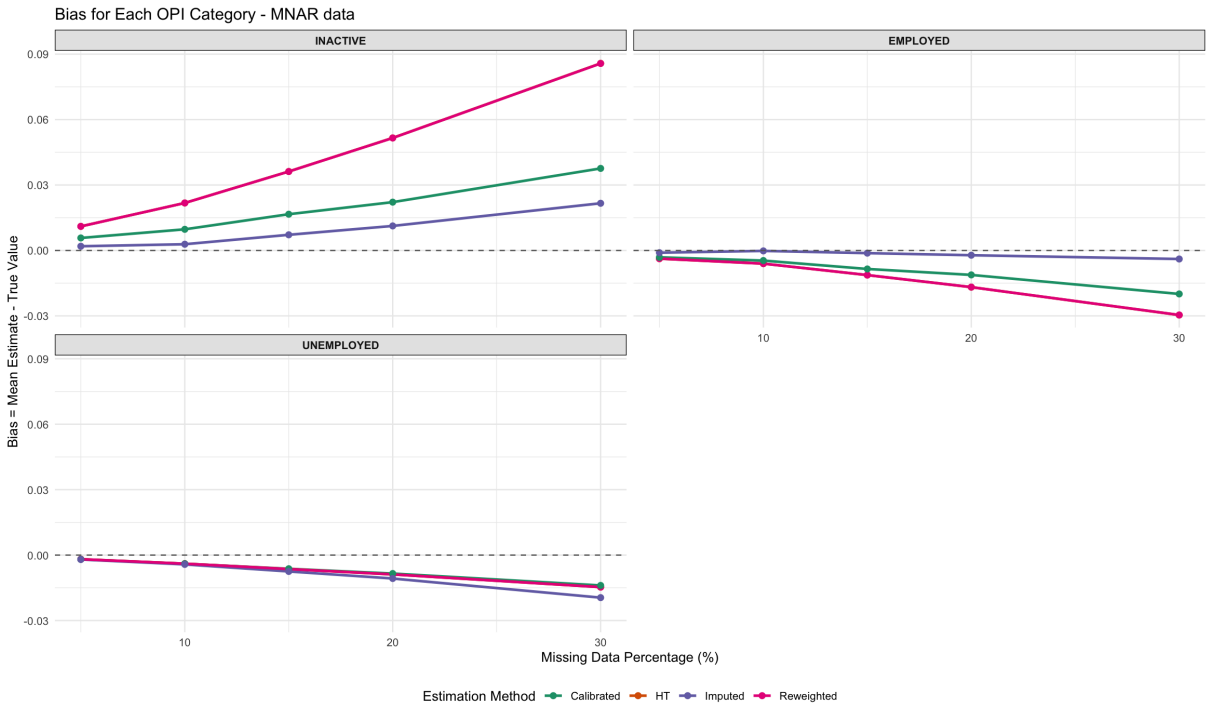


Figure 5.14 Bias across employment status categories under MNAR data by missingness rate and estimation strategy

In relation to the analysis of the distribution before and after the imputation process, some results can be highlighted by comparing the distribution of the target variable before and after imputation. The figure 5.15 shows that the imputation process does not significantly alter the

overall distribution of employment status, indicating that the imputed values are consistent with the observed data.

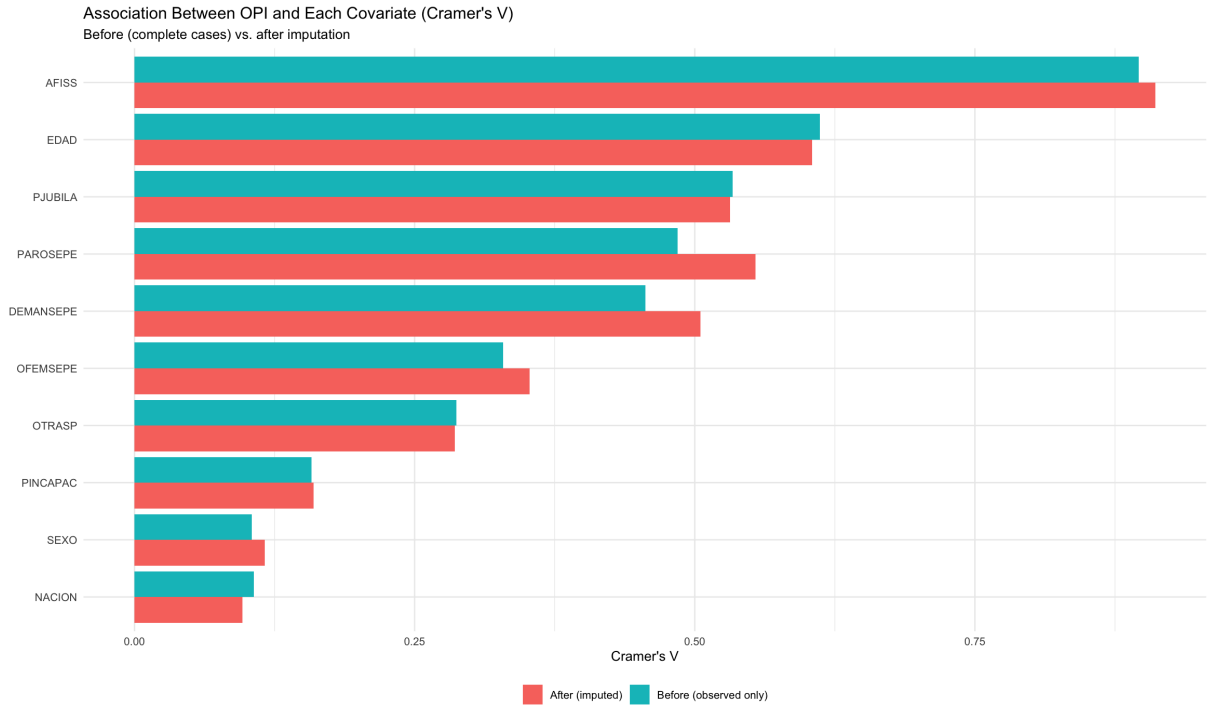


Figure 5.15 Association between OPI and each covariate calculated with the Cramer's V before and after the imputation

A complementary analysis of the non-response mechanism in the Labour Force Survey suggests that the empirical pattern of missingness is more consistent with MCAR mechanisms. In addition, in the real case, nationality and autonomous community emerge as the most relevant predictors of non-response, indicating that missingness is largely driven by observable demographic and geographic factors.

In particular, the simulation study shows that the current approach based on reweighting and calibration remains more robust under the most plausible non-response mechanisms for the Labour Force Survey. Although imputation methods may reduce prediction error for individual employment status values, they can also introduce additional model-based variability and potential bias when used for aggregate estimation.

Overall, the evidence indicates that imputation does not provide systematic gains over the existing methodology for this specific survey context. Therefore, under the assumption that the non-response mechanism remains stable over time, the results do not support replacing the current calibration-based framework with an imputation-based production system. However, this conclusion is conditional on the structure of the available auxiliary information and may not generalise to other surveys with different patterns of non-response or richer predictive signals.

Taken together, these findings suggest that, within the current structure of the Labour Force Survey and its auxiliary information, the existing calibration-based methodology remains the most reliable approach for official production. However, the results also highlight that this conclusion is context-dependent and may differ in surveys where non-response is more strongly

linked to unobserved labour market outcomes or where richer auxiliary data are available.

5.3.1. Execution time information

The computational cost of the imputation procedure is negligible on standard desktop or laptop computers. Similarly, training the Random Forest model requires only modest computational resources. In practice, the overall execution time is driven primarily by the grid-search hyperparameter optimization stage rather than by the imputation or model fitting itself.

5.4. ML Treatment of the Annual Survey on Construction Enterprises Using Admin Data (PT)

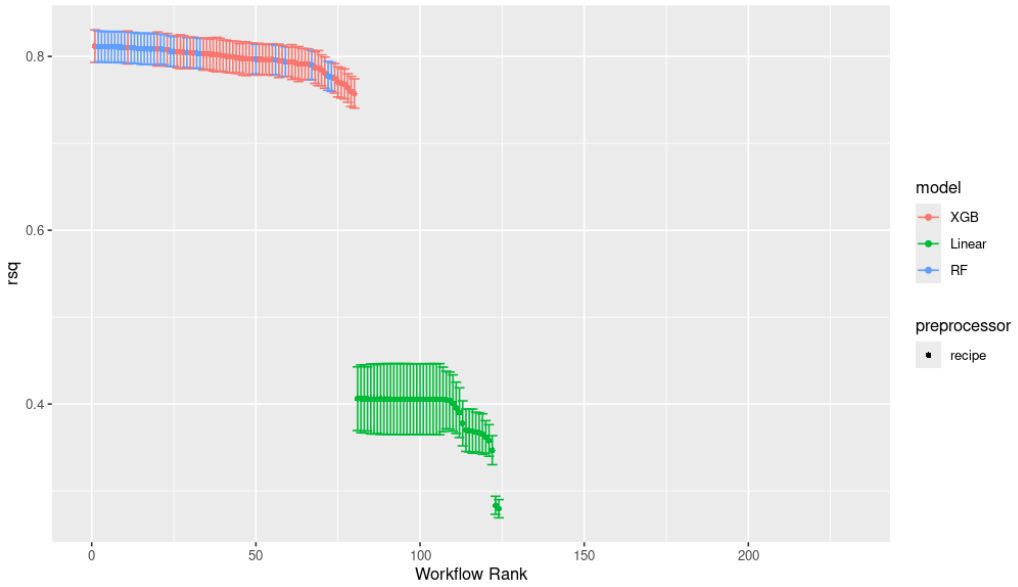


Figure 5.16 Model Comparison

The Random Forest and XGBoost Trees models consistently achieved the highest R^2 values, around 0.80 to 0.83, substantially outperforming the Linear Regression model, which achieved R^2 values mostly between 0.35 and 0.45.

Random Forest appears to provide the best overall performance, with the highest-ranked workflows and relatively stable results across configurations. Boosted Trees achieved comparable performance, although with a slight decline among lower-ranked workflows. In contrast, Linear Regression showed considerably lower predictive accuracy, suggesting that the relationship between predictors and the target variable is not adequately captured by a simple linear model.

The overlap of the error bars for Random Forest and Boosted Trees indicates that some performance differences may not be statistically significant. However, both ensemble-based approaches clearly outperform the linear model, highlighting the importance of modelling non-linear relationships and interactions in the data.

Based on the boxplot, the reference method CLASSIC stands out clearly for its extremely tight error distribution across all three ECAEs (41, 42, 43), with the interquartile range nearly collapsed around zero and minimal whisker extension. This makes it the most precise and unbiased estimator available, but its 67.6% coverage leaves a substantial portion of observations requiring an alternative approach.

CLASSIC0 (missing data replaced by zeros) is the weakest alternative by a considerable margin. Across all ECAEs it shows a strong systematic positive bias, with the median well above zero and an IQR that frequently spans from near zero to over 100%, making it both inaccurate and highly unstable.

CLASSICavg (missing data replaced by strata averages) represents a meaningful improvement over CLASSIC0. Its median sits close to zero in all three ECAEs and the spread, while

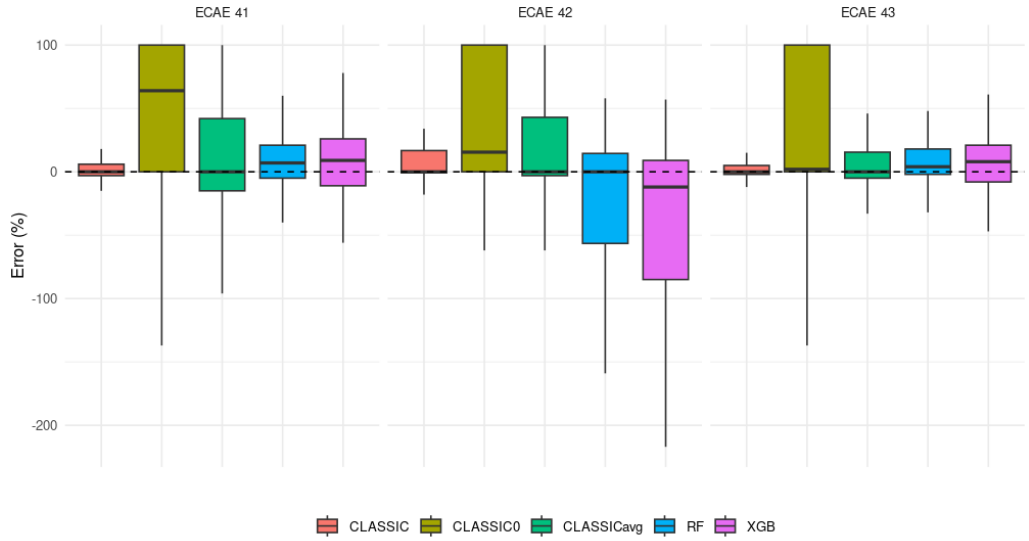


Figure 5.17 Error Distribution by method and ECAE

wider than the reference, remains contained. There is a slight tendency toward positive skew, suggesting occasional overestimation, but overall it behaves as a reasonably stable fallback for the uncovered observations.

RF and XGB perform comparably in ECAEs 41 and 43, both showing low median bias and moderate variance. The critical exception is ECAE 42, where both methods shift notably downward, indicating a systematic negative bias for that subgroup. XGB appears slightly more extreme in this regard. This ECAE-specific deterioration suggests that the distributional characteristics of group 42 interact poorly with both tree-based approaches, and any deployment decision should account for this heterogeneity.

Overall, for the observations not reached by the CLASSIC method, CLASSICavg offers the most consistent and balanced behaviour across ECAEs. RF is a viable second option provided ECAE 42 cases are handled with care. XGB requires the same caution in ECAE 42 and adds little advantage over RF in the remaining groups. CLASSIC0 does not offer a credible alternative given its persistent positive bias and high dispersion throughout.

Table 5.4 Model performance metrics

Model	Coverage (%)	RMSE	MAE	R^2
CLASSIC	67.6	987 026	317 268	0.873
CLASSICavg	100.0	1 078 398	440 033	0.808
CLASSIC0	100.0	1 194 623	459 079	0.777
RF	100.0	956 485	409 916	0.844
XGB	100.0	952 991	453 000	0.850

Coverage reflects the proportion of cases where the method produced a prediction. Metrics for CLASSIC are computed only over cases where a prediction was available.

5.4.1. Execution time information

Table 5.5 Training Infrastructure and Computational Performance

Parameter	**Value**
Infrastructure	
Server	16-core R Server
Training observations	3 854
Variables	15
Tuning Time	
RF	34m 31.8s
XGB	7m 46.6s
Fit Time	
RF	23.802s
XGB	2.910s

5.5. Evaluating Machine Learning Imputation Methods Using Employment Earnings Data (SI)

The quality of the ML method based on the LightGBM algorithm was assessed by comparing the predicted (imputed) regular monthly earnings with the original (true) values across different levels of missingness and by comparing the results with values estimated using traditional imputation methods, namely hot-deck nearest-neighbour donor imputation within strata and mean imputation within strata. The assessment focused on the following indicators: standardized root mean squared error (StdRMSE), relative mean bias at the overall level and within key subpopulations, coefficient of determination (R^2), winsorized variance ratio, and comparisons of distributions and granularity. In addition, the preservation of associations between the modelled target variable, regular earnings per hour, and the predictor variables was examined.

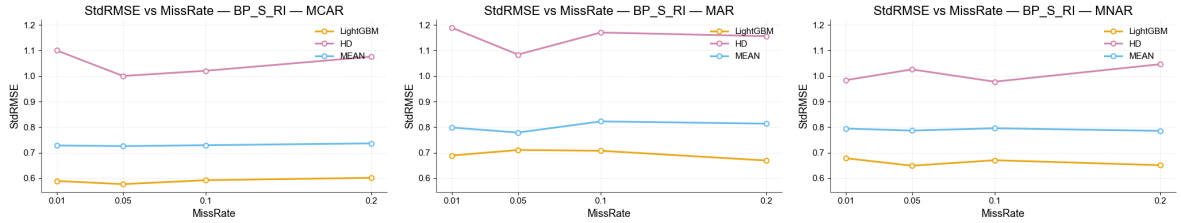


Figure 5.18 Standardized RMSE by Imputation Method, Missingness Mechanism, and Missingness Rate

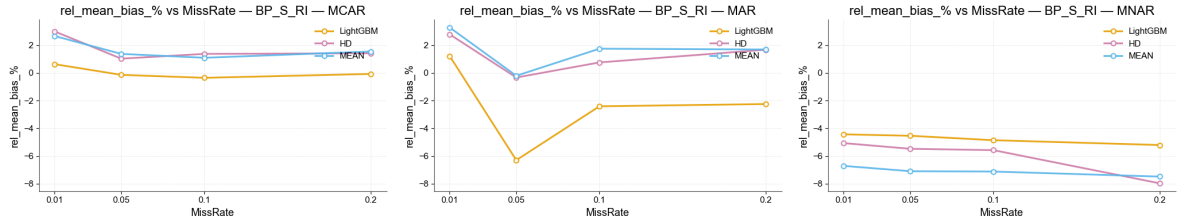


Figure 5.19 Relative Mean Bias by Imputation Method, Missingness Mechanism, and Missingness Rate

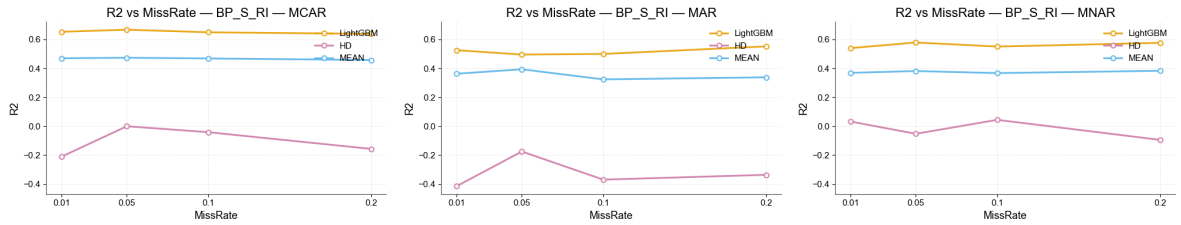


Figure 5.20 Explained Variance by Imputation Method, Missingness Mechanism, and Missingness Rate

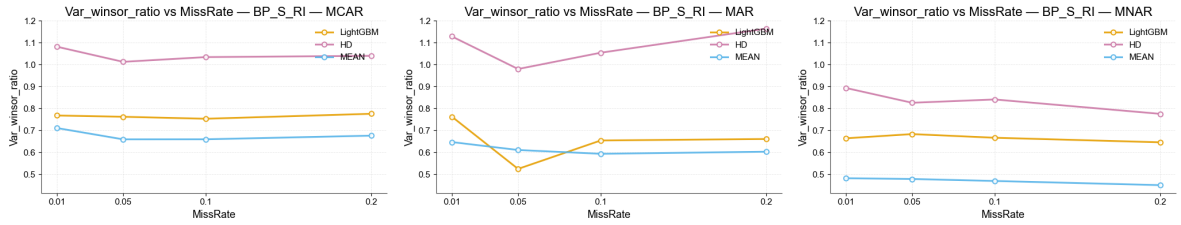


Figure 5.21 Variance Winsor Ratio by Imputation Method, Missingness Mechanism, and Missingness Rate

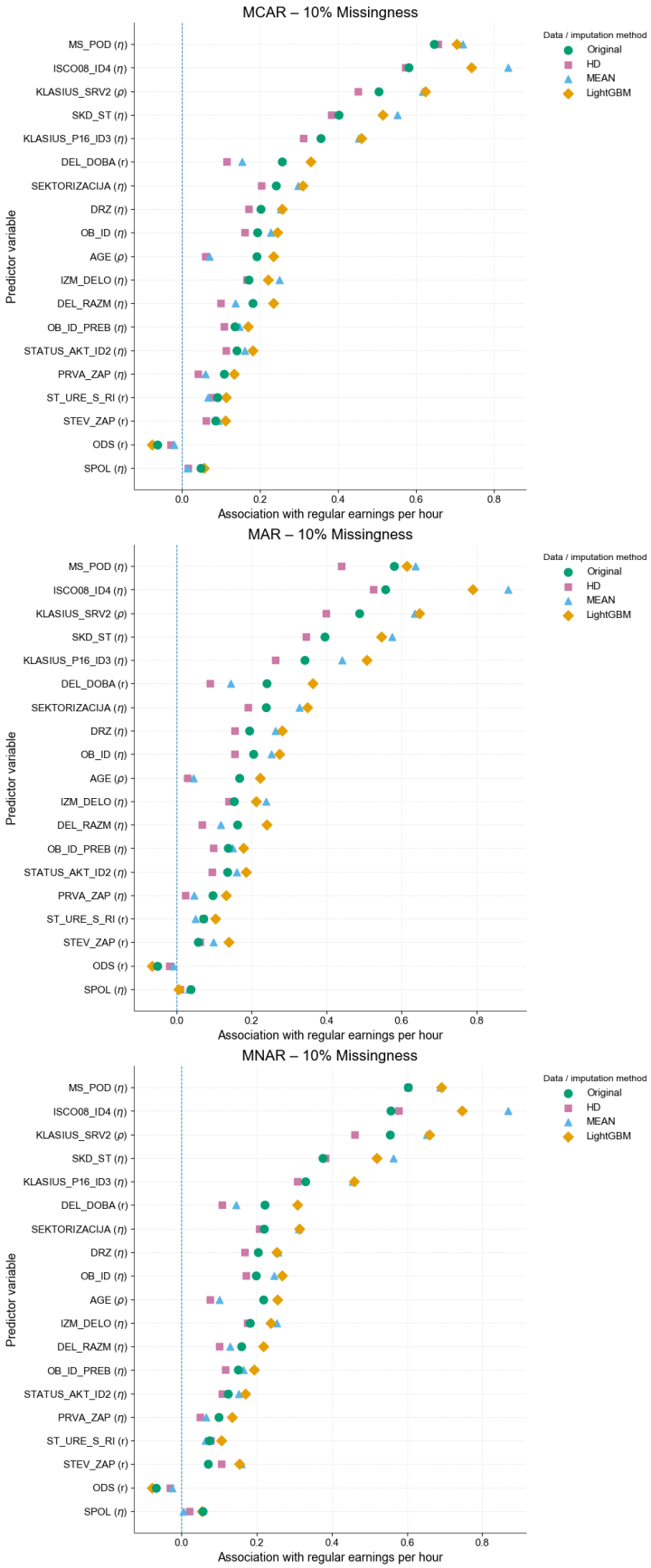


Figure 5.22 Associations with Regular Earnings per Hour by Imputation Method

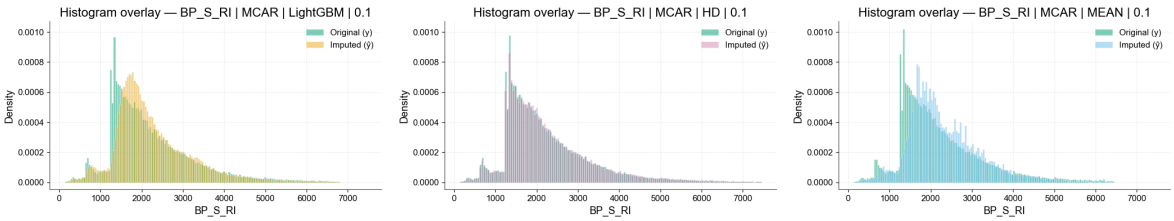


Figure 5.23 Distribution of Original and Imputed Values by Imputation Method

Among the tested algorithms, LightGBM proved to be operationally efficient, relatively fast, and capable of producing reproducible results. The LightGBM-based method maintained stable performance across different levels of missingness and provided better predictive performance, as reflected in lower StdRMSE and higher R^2, compared with traditional imputation methods. Lower bias was also observed for some of the published statistics and levels of aggregation examined. However, the imputed values exhibited lower variability than the original data. The pronounced concentration of values around the minimum wage level was not fully reproduced by LightGBM. The distribution of imputed values showed a lower and smoother peak, with a more gradual increase and decline and a slight shift towards higher values. In contrast, nearest-neighbour hot-deck imputation generally preserved the variability and distributional characteristics of the original data more closely. The associations between regular earnings per hour and the predictor variables were generally preserved after imputation, although the degree of preservation varied across predictors and imputation methods. Hot-deck imputation tended to reproduce the original associations more closely than LightGBM for several predictors, while for others the differences between the methods were small.

Despite the reduced variability, LightGBM produced imputed values with finer granularity than the traditional methods. LightGBM models using a logarithmic transformation of the target variable and a Tweedie objective were also tested to improve the distributional properties of the imputed values; however, these approaches did not result in noticeable improvements. Under the MNAR mechanism, all methods exhibited substantial bias, as the probability of missingness was systematically related to the level of earnings. This highlights a general limitation of imputation approaches in such settings rather than a model-specific issue.

5.5.1. Execution time information

Computational efficiency was explicitly considered when designing the LightGBM workflow. Hyperparameter optimisation was performed on a random sample of 250,000 observations, whereas the selected hyperparameters were subsequently used for final model training on the complete training dataset. The optimisation procedure used early stopping after 100 boosting iterations without improvement and a successive-halving pruning procedure to terminate unpromising hyperparameter configurations before completion. Computational efficiency was further improved by a restricted predefined hyperparameter search space, multithreaded LightGBM training during hyperparameter optimisation, and a maximum of 100 optimisation trials. In addition, the `min_data_in_bin` parameter was evaluated separately in a limited post hoc sweep rather than being included in the full hyperparameter search. Deterministic single-threaded settings were used during the final training stage to ensure reproducibility.

All processing was performed locally on a CPU-based machine with 16 GB of RAM. Data loading and preparation required approximately 3 minutes, hyperparameter optimisation for the 10% missingness scenario required approximately 10 minutes, and final model training

required approximately 3 minutes. The reported processing times are approximate and depend on the hardware configuration, dataset size, missingness scenario, and selected hyperparameters. In particular, the size and complexity of the hyperparameter search space, the number of optimisation trials, the number of cross-validation folds, the early-stopping criterion, the pruning strategy, the optimisation metric, and the resulting hyperparameter configuration can substantially affect computational requirements. The adopted configuration was selected to provide a practical balance between predictive performance, reproducibility, and computational efficiency.

5.6. Imputation of International Trade Statistics (AT)

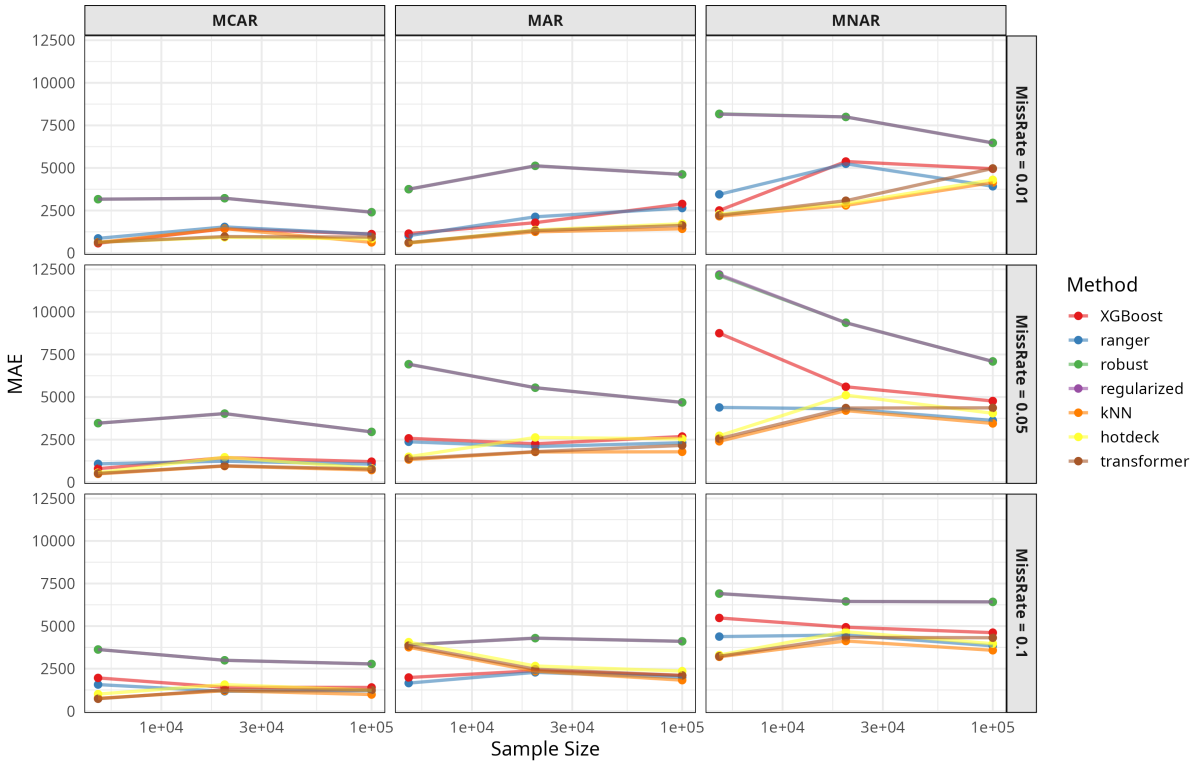


Figure 5.24 Comparison of Imputation Methods based on Mean Absolute Distance (MAE)

Figure 5.24 shows the differences in model performance (mean absolute error) across different sample sizes, missingness mechanisms, and missing rates. Not surprisingly all methods begin to perform worse as the missing mechanism moves from MCAR to MNAR. The group of AI/ML models (XGBoost, transformer and ranger (random forest)) perform yielding the low errors across nearly all scenarios, with the ranger often slightly outperforming or matching XGBoost, particularly at larger sample sizes. In contrast, the linear regression models (robust, regularised) tend to produce higher errors and exhibits greater variability, especially under MAR and MNAR, where it shows pronounced instability at smaller sample sizes. These larger error are most likely the root of poor model specification. In contrast to random forest or XGBoost where interactions between variables are automatically taken into account by the model, the regression models need to have them well and carefully specified. Interesting to see is that the donor base methods, kNN as well as hotdeck both perform well and receive comparable results as the AI/ML models. Overall, the results suggest that the XGBoost and ranger imputations are more robust across varying conditions, while the regularized and robust imputation models are more sensitive to both sample size and the structure of missingness.

Although the transformer imputation yields low MAE values compared to other methods, one major disadvantage with transformer imputation is the run times. This result is shown in Table 5.6, which shows how rapidly runtimes for the transformer imputation increase with growing data set size. While other imputation methods only take around five minutes at most, the transformer imputation can reach runtimes of over 2.5 hours. We find that even though the model performance might be desirable, the high runtimes of this order are not suitable for imputation in production.

Table 5.6 Comparison of time (in seconds) needed to impute variable *weight*.

Sample size	hotdeck	kNN	ranger	regularized	robust	transformer	xgboost
5e+03	14.62	8.85	6.94	6.93	6.98	454.83	23.38
2e+04	26.75	107.29	16.13	21.27	20.06	1864.82	116.62
1e+05	51.33	134.87	398.33	510.63	546.02	10285.68	2407.22

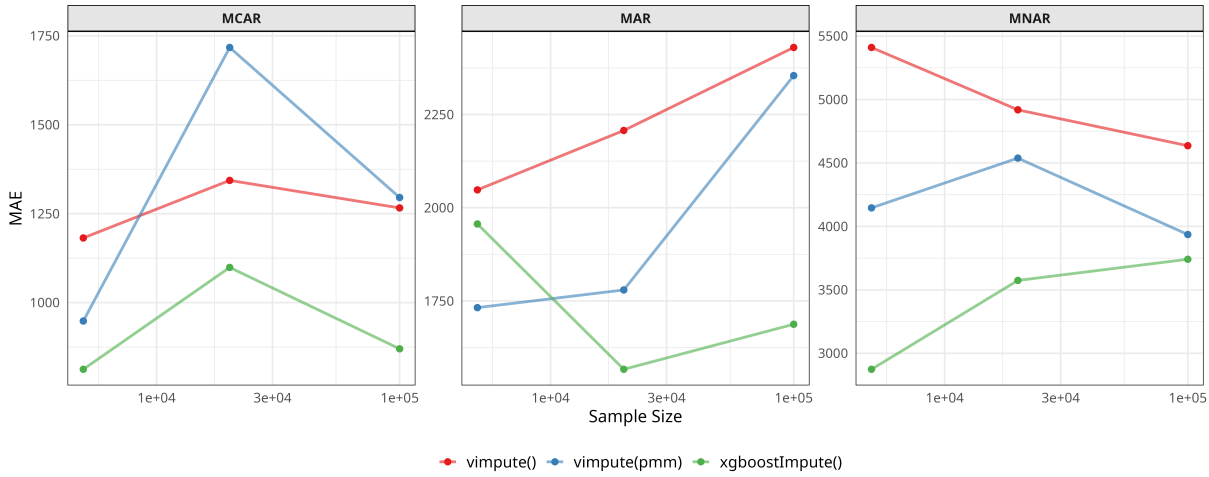


Figure 5.25 Comparison of XGBoost Imputation with different implementations.

Figure 5.25 shows differences between the three XGBoost implementation. The green line shows the imputation using the `xgboostImpute()` function. Red and blue show imputation results where `vimpute`, and `vimpute` with $pmm = 1$ were used, respectively. Under MCAR, the `xgboostImpute` consistently performs best (lowest error) across all sample sizes, followed by the `vimpute` without pmm, while the `vimpute` with pmm performs worst and exhibits a pronounced spike at the medium sample size, indicating instability. Under MAR, the ranking becomes more nuanced: the `xgboostImpute` again achieves the lowest errors overall, particularly at medium sample sizes, while the `vimpute` shows steadily increasing error with larger samples, suggesting limited scalability. The `vimpute` with pmm performs comparably to `vimpute` without using pmm. Overall, the `xgboostImpute` appears most robust to both the missingness mechanism and sample size, whereas the `vimpute` approach is more sensitive to sample size (particularly under MAR).

With imputation, we not only aim to impute missing values as accurately as possible at the unit level, but also to preserve the overall distribution of the data, ensuring that the imputed data closely resembles the true underlying distribution. Figure 5.26 compares the original univariate distribution of variable weights with the distributions obtained after imputation using `xgboostImpute`, `vimpute` with xgboost and `vimpute` with xgboost and predictive mean matching (pmm). The true distribution is highly right-skewed, with a large concentration of observations close to zero and a long tail extending towards higher values. Ideally, an imputation method should reproduce this overall shape without introducing additional patterns that are not present in the original data.

Among the evaluated methods only the `xgboostImpute` shows a strong peak near zero and the long right tail, indicating that the method captures the general structure of the data. However, a noticeable secondary peak appears around values of approximately 6 to 8, which is not present in the true distribution. This suggests that the model tends to assign similar predicted values to groups of observations with comparable characteristics, resulting in an

artificial clustering effect. Despite this distortion, the overall shape remains relatively close to the original distribution.

The `vimpute` imputation also preserves the strong concentration of observations at low values and the general right-skewed nature of the variable. Nevertheless, it introduces a distinct secondary mode around values of 50 to 55. Since this feature is absent in the true distribution, it indicates that the method imputes an excessive number of observations within this range. Consequently, the resulting distribution deviates more substantially from the original distribution than the `xgboostImpute` approach.

The PMM imputation exhibits the largest departure from the true distribution. While the peak near zero is maintained, a pronounced concentration of observations emerges at the upper end of the distribution, around values of 75 to 80. This artificial accumulation suggests that a small number of extreme observed values were repeatedly selected as donors during the imputation process. As a result, the upper tail becomes considerably more pronounced than in the original data, leading to a noticeable distortion of the marginal distribution.

Overall, although some artificial clusters are introduced, all methods preserve the overall skewness of the original distribution.

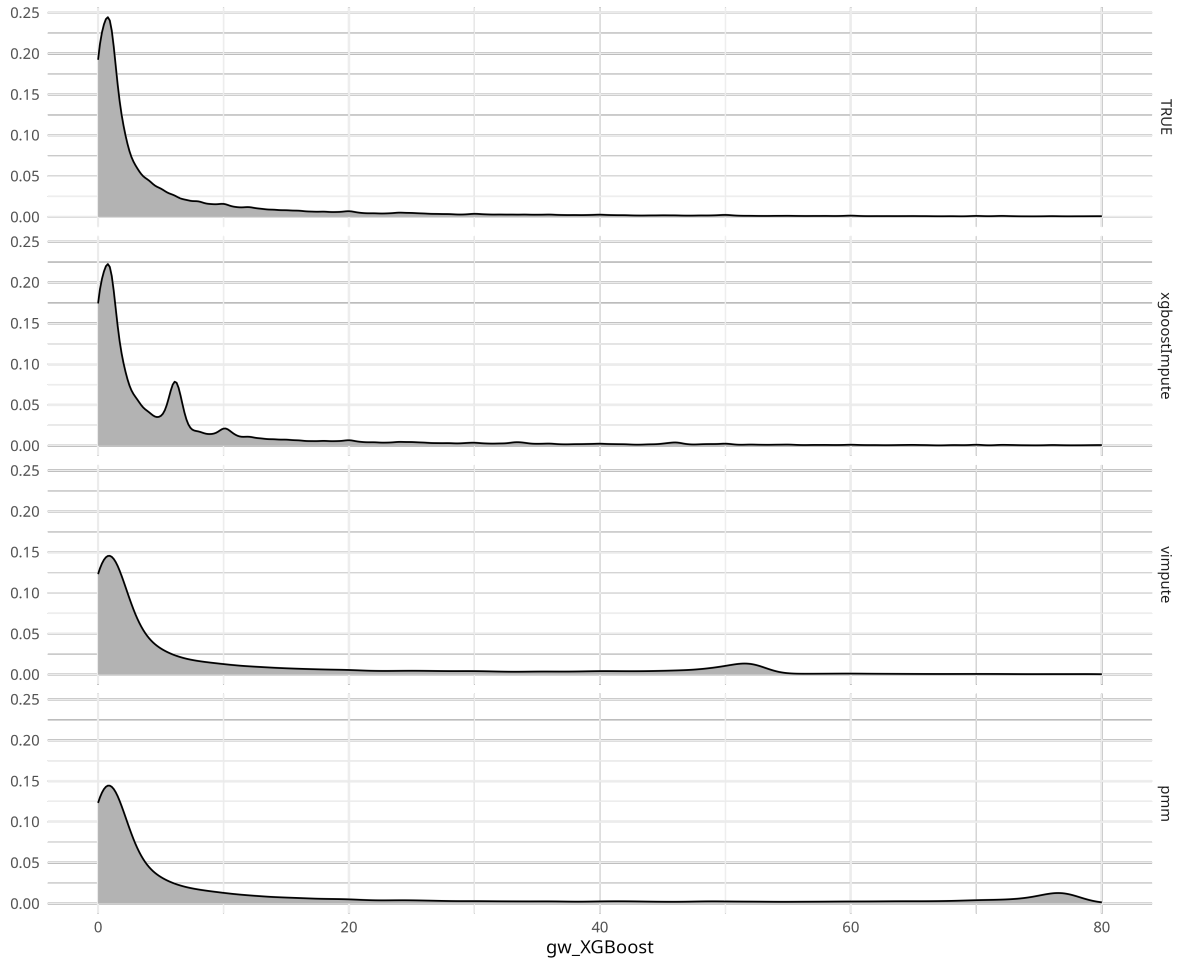


Figure 5.26 Comparison of XGBoost Imputation with different implementations.

In addition to the weight variable, we also conduct tests on the imputation of the CN variable. This is a categorical variable, in detail, a classification system that classifies goods with the EU's specific customs and foreign trade statistics requirements in mind. The original code is up to eight digits long. In our experiments, we only impute the 2-digit CN codes. The results are

summarized in Tables 5.7, 5.8 and 5.9. We observe overall very low performance values across all methods and missingness types. This might be due to the high number of classes combined with a small data set size. Among the compared methods, kNN consistently performs best across all scenarios (MCAR, MAR, and MNAR). It achieves the highest accuracy in each case and also leads in F1 scores and precision–recall balance. XGBoost follows as the second-best method, showing stable performance and relatively strong results, particularly in terms of F1 scores under MNAR conditions. The ranger model and its `vimpute` variant form a middle-performing group. They produce moderate results that are clearly below kNN and XGBoost but still outperform the weaker approaches. In contrast, XGBoost `vimpute` performs worse than standard XGBoost, suggesting that the `vimpute` variant does not provide an advantage for this variable. The transformer model performs by far the worst, with the lowest values across all evaluation metrics and missingness types. This indicates that it is not well suited for imputing the 2-digit CN code in this setting. We consistently observe large difference in macro and micro F1. This indicates that the performance is not the same over all classes, but varies. Regarding the missingness mechanisms, performance is generally highest under MCAR and declines slightly for MAR and MNAR. However, these differences are not dramatic, and the relative ranking of methods remains consistent across all three scenarios.

Table 5.7 Imputation Performance for variable kn2 (MCAR)

	MCAR				
Method	Accuracy	F1_macro	F1_micro	Precision	Recall
transformer	0.0380	0.0801	0.0386	0.0191	0.0159
ranger	0.1059	0.1964	0.1102	0.0625	0.0598
XGBoost	0.1075	0.1730	0.1156	0.0632	0.0557
kNN	0.1340	0.2146	0.1360	0.0924	0.0827
hotdeck	0.1360	0.2172	0.1371	0.0870	0.0765
XGBoost vimpute	0.0900	0.1600	0.0924	0.0547	0.0449
ranger vimpute	0.0960	0.1617	0.0971	0.0616	0.0519

Table 5.8 Imputation Performance for variable kn2 (MAR)

	MAR				
Method	Accuracy	F1_macro	F1_micro	Precision	Recall
transformer	0.0200	0.0907	0.0231	0.0131	0.0107
ranger	0.0675	0.1472	0.0676	0.0511	0.0424
XGBoost	0.0788	0.1546	0.0788	0.0609	0.0494
kNN	0.0960	0.1644	0.0964	0.0723	0.0639
hotdeck	0.0900	0.1559	0.0905	0.0640	0.0554
XGBoost vimpute	0.0640	0.1354	0.0642	0.0431	0.0356
ranger vimpute	0.0685	0.1474	0.0687	0.0504	0.0448

Table 5.9 Imputation Performance for variable kn2 (MNAR)

	MNAR				
Method	Accuracy	F1_macro	F1_micro	Precision	Recall
transformer	0.0160	0.0892	0.0160	0.0137	0.0139
ranger	0.0664	0.1957	0.0671	0.0737	0.0582
XGBoost	0.0750	0.2195	0.0761	0.0864	0.0679
kNN	0.0920	0.2037	0.0930	0.1227	0.0825
hotdeck	0.0790	0.2328	0.0790	0.0980	0.0603
XGBoost vimpute	0.0490	0.1538	0.0500	0.0579	0.0417
ranger vimpute	0.0540	0.1683	0.0560	0.0699	0.0487

5.6.1. Execution time information

Figure 5.27 shows the computational time required for the imputation of the variable CN4, again using the five different methods (hotdeck, kNN, ranger, transformer, and XGBoost) across increasing sample sizes (10,000, 50,000, and 100,000 observations). The left panel presents the full range of runtimes, while the right panel is a zoomed-in version excluding the high runtimes of the transformer method to facilitate comparison among the remaining approaches. Overall, imputation time increased with sample size for all methods, although the magnitude of the increase varied considerably. The transformer-based approach is the most computationally intensive, requiring approximately 8,000 s, 15,500 s, and over 30,000 seconds for sample sizes of 10,000, 50,000, and 100,000, respectively. In contrast, all other methods completed within approximately 4,200 seconds (70 minutes) even for the largest dataset. Among these methods, hotdeck was consistently the fastest, with runtimes remaining close to zero across all sample sizes. Ranger and kNN showed moderate increases in computation time as sample size grew, reaching about 550 and 900 seconds, respectively, at 100,000 observations. XGBoost exhibited the steepest increase among the non-transformer methods, rising from roughly 170 seconds at 10,000 samples to more than 4,000 seconds at 100,000 samples. We attribute the comparably high runtimes for the XGBoost to the high number of classes in the categorical variable CN4. We observe that with increasing number of classes, the XGBoost tends to become much more computationally expensive. These results demonstrate substantial differences in computational efficiency between imputation methods, with transformer models incurring the highest computational cost and hotdeck providing the most time-efficient solution.

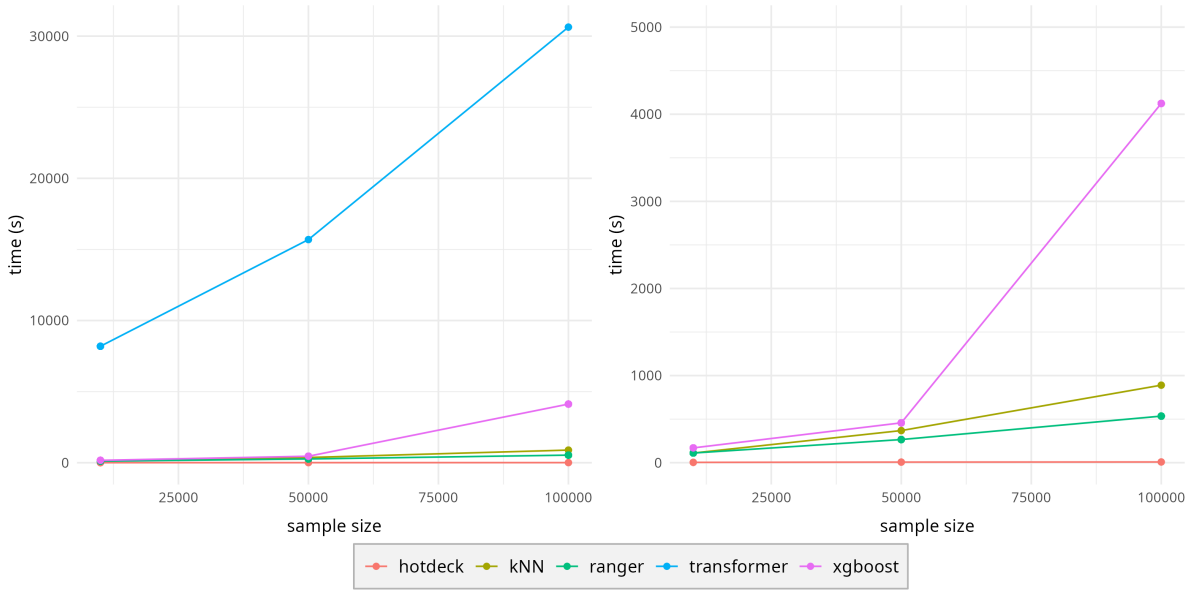


Figure 5.27 Comparison of time needed for Imputation by data size.

6

Conclusions

6.1. Methodological Outcomes

The cross-country experimentation provides a measured perspective on the integration of Machine Learning (ML) into post-collection imputation. While tree-based ensemble methods—specifically **Random Forest, XGBoost, and LightGBM**—demonstrated a superior ability to handle high-cardinality categorical variables and complex non-linear dependencies without extensive manual feature specification, their superiority over traditional methods is not universal and depends heavily on the statistical objective and the underlying missingness mechanism.

Key Findings

The results indicate that the choice of method involves a trade-off between micro-level predictive accuracy and macro-level distributional integrity. The experimentation yielded several critical insights for the modernization of official statistics:

- **Superiority of Tree-Based Ensembles:** Across most projects, Random Forest and Gradient Boosting models consistently outperformed traditional methods, particularly in handling high-dimensional data and complex interactions.
- **Operational Efficiency:** The implementation of ML pipelines demonstrated significant potential for reducing manual intervention. In the Portuguese project, the processing time for the Annual Survey on Construction was reduced from **two days to two hours**.
- **Theoretical Alignment:** The Luxembourg project proved that ML can be successfully integrated into theory-driven frameworks, such as the Consumer Price Index (CPI), by ensuring the models satisfy the orthogonality property.
- **Production Constraints:** While advanced models like Transformers offer high accuracy, their computational cost and runtime (e.g., over 2.5 hours in the Austrian trade project) may limit their suitability for real-time production environments compared to more efficient tree-based models.
- **Predictive Power vs. Distributional Preservation:** In the Slovenian project, although LightGBM offered high operational efficiency, the traditional **NND hot-deck** method remained the most effective approach for preserving the marginal distribution of earnings, avoiding the "variance compression" often observed in ML point estimates.
- **Robustness to Non-response Mechanisms:** The Spanish project revealed that traditional calibration-based reweighting systematically outperformed ML-based imputation under

the scenario **MCAR** (the most plausible in that application), whereas ML-based imputation showed better performance under textbfMAR and textbfMNAR scenarios, particularly for difficult-to-predict minority classes.

- **Algorithm Suitability:** The performance of simpler ML models, such as **kNN** in the Polish project, was found to be relatively poor compared to historical benchmarks, highlighting that the mere application of an ML algorithm does not guarantee improvement unless the model architecture is well-aligned with the specific data dynamics.

Furthermore, the use of synthetic data in the Spanish project proved essential for evaluating design-based errors in a controlled environment [Daas and Puts, 2023].

An additional overarching conclusion emerging from the six use cases is that there is no universally optimal imputation method for official statistics. The performance of both traditional and Machine Learning approaches depends not only on the characteristics of the data, but also on the statistical objective being pursued, whether the focus is micro-level accuracy, preservation of distributions, aggregate estimation, operational efficiency, or explainability. Consequently, model selection should be regarded as a context-specific exercise supported by rigorous validation rather than as a competition to identify a single best-performing algorithm. This reinforces the need for methodological flexibility within National Statistical Institutes, where multiple imputation strategies may coexist and be selected according to the specific production environment and quality requirements of each statistical domain.

In conclusion, while ML techniques offer a powerful addition to the NSI toolkit for addressing complex data structures, they should be viewed as complementary to, rather than a total replacement for, traditional statistical strategies. The "One-Stop-Shop" approach must therefore prioritize a flexible framework where the selection of the imputation method is driven by a rigorous assessment of the data's nature and the final use of the intended statistics.

6.2. Production-Focused Strategy and Lessons Learned

The transition from experimental ML to production-ready official statistics requires balancing predictive accuracy with operational efficiency. Key lessons learned include:

- **Computational Feasibility:** While Transformer-based models offer high accuracy, their runtimes can be prohibitive for large-scale production (e.g., Austria's trade data). Tree-based models offer a better trade-off between speed and performance.

- **Potential Variance Compression:** ML models may produce "conservative" mid-range predictions. While this reduces the risk of implausible outliers, it may also lead to an underestimation of variance (e.g., Slovenia's earnings data).

- **Robustness to Anomalies:** Models trained on historical data require specific adjustments or retraining (avoiding the drift of the models) when faced with structural breaks, such as the COVID-19 pandemic [Steurer et al., 2021].

The project also highlights the importance of establishing standardized evaluation frameworks for AI/ML-based imputation in official statistics. Beyond predictive performance, future methodological developments should promote reproducibility, transparency, uncertainty assessment, and comparability across statistical domains. Such frameworks will be essential to facilitate the sustainable adoption of Machine Learning methods within the European Statistical System while maintaining the trustworthiness and credibility of official statistics.

6.3. Conclusions and Recommendations

The results indicate that ML-based post-collection imputation is a robust and viable alternative to traditional statistical methods for some applications. However, the transition to production requires a "production-focused strategy" that includes continuous model monitoring to detect drift and the development of quality frameworks to ensure transparency.

This deliverable report concludes that while ML significantly improves micro-level accuracy, its impact on macro-level aggregates is highly dependent on the underlying non-response mechanism. Therefore, NSIs are encouraged to adopt a hybrid approach, leveraging ML for its predictive power while maintaining rigorous statistical validation to ensure the continued integrity of official statistics.

6.4. Final Remarks and Future Work

Post-collection imputation using ML is a viable and powerful tool for National Statistical Institutes (NSIs) should be considered. Future work should focus on the development of automated pipelines for model monitoring (detecting model drift) and the refinement of "hybrid" approaches that combine the predictive power of ML with the stability of traditional calibration techniques.

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Appendix A

Appendix: regressors description

A.1. Imputation of foreign tourist in accommodation establishments survey (PL)

Category	Name	Type	Domain	Example	Level
Target	D8R1-number of tourists	Numerical	≥ 0	125	Accommodation establishment
Time	OKRES (month)	Date/Time	YYYYMM	202401	Accommodation establishment
Country	KKR	Categorical	Country codes	Germany	Accommodation establishment
Temporal features	lag_1–lag_12	Numerical	≥ 0	118	Accommodation establishment
	mean_last_3	Numerical	≥ 0	121.3	Accommodation establishment
	mean_last_6	Numerical	≥ 0	126.7	Accommodation establishment
	mean_last_9	Numerical	≥ 0	129.8	Accommodation establishment
	mean_last_12	Numerical	≥ 0	132.5	Accommodation establishment
Spatial statistics	GMN_mean	Numerical	≥ 0	87.5	Municipality
	GMN_sd	Numerical	≥ 0	24.1	Municipality
	GMN_cv	Numerical	≥ 0	0.28	Municipality
	WON_mean	Numerical	≥ 0	104.3	County
	WON_sd	Numerical	≥ 0	31.6	County
	WON_cv	Numerical	≥ 0	0.30	County
	RO_mean	Numerical	≥ 0	119.8	Voivodeship
	RO_sd	Numerical	≥ 0	37.4	Voivodeship
	RO_cv	Numerical	≥ 0	0.31	Voivodeship

Table A.1 Description of features used for predicting the monthly number of foreign tourists in an accommodation establishment. Temporal features were calculated separately for each country of origin (KKR), while spatial statistics were computed at municipality, county and voivodeship levels.

A.2. Imputation of Price Statistics (LU)

Time-independent features are web-scraped characteristics of the different television which are on sale over a 13 months period (June 2021 to June 2022) in a major retailer.

Table A.2 Features of the model to impute non-response in the target of Price Statistics

Category	Name	Type	Domain	Example	Level	Encoding
Target	price	numerical	>0	199	unit	Logarithmic transformation
Time	period	Numerical	>0	1,2, 3,...13	unit	One-hot encoding
Time-independent features	taille_ecran	Numerical	32	unit		
	norme_hd	Categorical	0,1	1	unit	One-hot encoding
	resolution	Categorical	0,1	1	unit	One-hot encoding
	type_ecran	Categorical	0,1,2	1	unit	One-hot encoding
	hdr_high_dynamic_range	Categorical	0,1	1	unit	One-hot encoding
	smart_tv	Categorical	0,1	1	unit	One-hot encoding
	wi_fi	Categorical	0,1	1	unit	One-hot encoding
	bluetooth	Categorical	0,1	1	unit	One-hot encoding
	nombre_ports_hdmi	Numerical	1,2,3,4	1	unit	
	ci	Categorical	0,1	1	unit	One-hot encoding
	consommation_energietique	Numerical	>0	55	unit	
	nombre_haut_parleurs	Numerical	1,2,...,8	1	unit	
	puissance_rms	Numerical		10	unit	
	largeur_sans_pied	Numerical		126	unit	
	hauteur_sans_pied	Numerical		435	unit	
	profondeur_sans_pied	Numerical		126	unit	
	poids_sans_pied	Numerical		126	unit	
	ecran_incurve	Categorical	0,1	1	unit	One-hot encoding
	guide_programme_electronique	Categorical	0,1	1	unit	One-hot encoding
	assistant_vocal	Categorical	0,1	1	unit	One-hot encoding
	sortie_audio_numerique_optique	Categorical	0,1,2	1	unit	One-hot encoding
	nombre_ports_rf	Numerical	0,1,...,4	1	unit	
	rafraichissement_ecran	Categorical	0,1	1	unit	One-hot encoding
	classe_energetique_eu_2020	Categorical	e,f,g	e	unit	Label encoding
	google_chromecast_integre	Categorical	0,1	1	unit	One-hot encoding
	nombre_ports_ethernet_lan	Categorical	0,1	1	unit	One-hot encoding
	canal_retour_audio_arc	Categorical	0,1	1	unit	One-hot encoding
	ai_upscale	Categorical	0,1	1	unit	One-hot encoding
	airplay	Categorical	0,1	1	unit	One-hot encoding
	partage_ecran_screen_mirroring	Categorical	0,1	1	unit	One-hot encoding
	ecran_pivotant	Categorical	0,1	1	unit	One-hot encoding
	multi_view	Categorical	0,1,2	1	unit	One-hot encoding
	hdr_supp	Categorical	0,1	1	unit	One-hot encoding
	hdr	Categorical	0,1	1	unit	One-hot encoding
	hlg	Categorical	0,1	1	unit	One-hot encoding
	dolby	Categorical	0,1	1	unit	One-hot encoding
	crystal_disp	Categorical	0,1	1	unit	One-hot encoding
	ambient_mode	Categorical	0,1	1	unit	One-hot encoding
	norme_hd_4k	Categorical	0,1	1	unit	One-hot encoding
	norme_hd_8k	Categorical	0,1	1	unit	One-hot encoding
	norme_hd_	Categorical	0,1	1	unit	One-hot encoding
	nombre_ports_usb	Numerical	1,2,3	1	unit	
	resol1	Numerical		1366	unit	Logarithmic transformation
	resol2	Numerical		768	unit	Logarithmic transformation

A.3. Imputation for non-response in Labour Force Survey using synthetic data (ES)

Table A.3 Features of the model to impute non-response in the target of the Labour Force Survey

Category	Name	Type	Source	Dimension	Brief description
target	labour_status	char	original_survey	labour_status	Employment status indicator (1 = employed, 2 = unemployed, 3 = inactive).
regressors	autonomous_community	char	sample frame	geographical	Autonomous Community of residence (NUTS 2).
	province	char	sample frame	geographical	Province of residence (NUTS 3).
	age	num	sample frame	demographic	Age in completed years.
	sex	char	sample frame	demographic	Sex of the individual.
	country_of_birth	char	sample frame	demographic	Country of birth for individuals born abroad.
	nationality	char	original_survey	demographic	Nationality of the individual.
	social_security_registration	binary	administrative_register	social_security	Indicator of registration with the Spanish Social Security System.
	sepe_registered_unemployed	binary	administrative_register	labour_market	Indicator of being registered as unemployed with the Spanish Public Employment Service (SEPE).
	sepe_job_seeker_last_3_months	binary	administrative_register	labour_market	Indicator of being registered as a job seeker with SEPE during the last three months.
	receiving_unemployment_benefit	binary	administrative_register	labour_market	Indicator of receiving unemployment benefits or allowances according to SEPE (1 = Yes, 0 = No).
	inss_retirement_pension	binary	administrative_register	pensions	Indicator of receiving a retirement pension from the National Social Security Institute (INSS).
	inss_disability_pension	binary	administrative_register	pensions	Indicator of receiving a disability pension from the National Social Security Institute (INSS).
	inss_allowance	binary	administrative_register	labour_market	Indicator of receiving a social security allowance from the National Social Security Institute (INSS).
	inss_other_pensions	binary	administrative_register	labour_market	Indicator of receiving other pensions from the National Social Security Institute (INSS).

A.4. ML Treatment of the Annual Survey on Construction Enterprises Using Admin Data (PT)

Table A.4 Regressors

Category	Name	Type	Source	Dimension	Brief description
target	QF185001	num	original_survey		Value on building works on reference year
regressors	I187701	char	administrative		Proxy variable on the value of construction works in the administrative data file for the reference year
	I187701_n1	int	administrative		Proxy variable on the value of construction works in the administrative data file year -1
	I187701_n2	int	administrative		Proxy variable on the value of construction works in the administrative data file year -2
	I187701_n3	int	administrative		Proxy variable on the value of construction works in the administrative data file year -3
	QF185001_n1	int	previous survey		Value on building works on year -1
	QF185001_n2	int	previous survey		Value on building works on year -2
	QF185001_n3	int	previous survey		Value on building works on year -3
	VVN	int	administrative data		Enterprise turnover
	NPS	int	administrative data		Enterprise number of employees
	QF186301_n1	int	previous survey		Value on infrastructure works on year -1
	perc_edifn1	int	derived survey		Ratio between buildings and infrastructures values on year -1
	perc_edifn2	int	derived survey		Ratio between buildings and infrastructures values on year -2

A.5. Evaluating Machine Learning Imputation Methods Using Employment Earnings Data (SI)

Table A.5 Evaluating Machine Learning Imputation Methods Using Employment Earnings Data (SI) regressors

Name	Type	Source	Dimension	Formula	Brief description
AGE	num	derived-administrative	socio-demographic	derive_variable (df = data, new_var = 'AGE', var_type = 'numeric', rule = 'LETO - LETO_ROJ - ((MESEC < MESEC_ROJ) | ((MESEC == MESEC_ROJ) & (REF_DAN < DAN_ROJ)))')	Age in years at the end of the reference month
STEV_ZAP	num	derived-statistical register of employment	employer size	aggregate_variable (df = data, new_var = 'STEV_ZAP', group_by_var = 'MS_POD', operation = 'count', condition = None)	Number of persons in paid employment at the legal entity (company)
KLASIUS_SRV2	num	original-statistical register of employment	socio-demographic	National classification; 2-digit code	Educational attainment level
KLASIUS_P16_ID3	char	original-statistical register of employment	socio-demographic	National classification; 3-digit code	Field of education
ISCO08_ID4	char	original-statistical register of employment	work-related	ISCO code; 4-digit code	Occupation performed by the person (ISCO-08)
SKD_ST	char	original-statistical business register	industry classification	NACE code	Statistical NACE code of activity
DRZ	char	original-administrative	socio-demographic	ISO 3166-1 numeric code; 3-digit code	Citizenship at the end of the reference month
SPOL	char	derived-administrative	socio-demographic	Derived at source from the unique identifier of person; 1=male, 2=female	(Biological) sex
MS_POD	char	original-statistical business register	identifier	7-digit code	Employer ID
DEL_RAZM	char	original-administrative	work-related	1=indefinite employment 2=fixed-term employment 3=trainee, indefinite employment 4=trainee, fixed-term employment	Type of employment relationship
IZM_DELO	char	original-administrative	work-related	1=one shift; 2=two shifts; 3=three shifts; 4=four shifts; 5=turnus	Shift work
SEKTORIZACIJA	char	original-statistical business register	institutional / economic sector	Standard classification of institutional sectors (SCIS)	Institutional sector by SNA/ESA
STATUS_AKT_ID2	char	derived-administrative	work-related	Derived at source from codes of the basis for insurance; 11=persons in paid employment by legal persons; 12=persons in paid employment by natural persons	Type of employer
OB_ID	char	original-administrative	work-related	Standard classification of territorial units (municipalities; SKTE-5); 3-digit code	Municipality of workplace

Continued on next page...

Table A.5 Evaluating Machine Learning Imputation Methods Using Employment Earnings Data (SI) regressors *(continued)*

Name	Type	Source	Dimension	Formula	Brief description
OB_ID_PREB	char	original-administrative	socio-demographic	Standard classification of territorial units (municipalities; SKTE-5); 3-digit code	Municipality of residence
PRVA_ZAP	char	derived-administrative	work-related	Derived at source from the date of first employment; 1=first employment; 0=person was already employed in the past	Indicator of first employment in Slovenia
ST_URE_S_RI	num	original-administrative	work-related		Regular work hours
ODS	num	derived-administrative	work-related	derive_variable (data, new_var = 'ODS', var_type = 'numeric', rule = 'BOL_DEL_PC + BOL_DR_PC + DR_ODS_PC + BOL_DEL_SC + BOL_DR_SC + DR_ODS_SC')	Absence from work
DEL_DOBA	num	derived-administrative	work-related	Derived at source	Completed years of work experience

A.6. Imputation of International Trade Statistics (AT)

A.6.1. Description of variables used to impute variable weight (gw)

Name	Type	Source	Brief description
ul	char	original-survey	ISO2C code of the Country of origin, e.g. country from which the imported goods were originally sent.
rc	char	original-survey	ISO2C code of the reporting country, e.g. country which reported the import into Austria
Country_Group	char	derived-survey	Country of origin combined into 6 classes: EU-Western Europe, EU-Southern Europe, EU-Eastern Europe, EU-Northern Europe, EU-Western Asia, Other
kn2	char	derived-survey	2-digit CN code of the product
kn4	char	derived-survey	4-digit CN code of the product
swc	num	original-survey	Value (in €) of the imported goods

A.6.2. Description of variables used to impute variable on the Combined Nomenclature (CN)

Name	Type	Source	Brief description
ul	char	original-survey	ISO2C code of the Country of origin, e.g. country from which the imported goods were originally sent.
rc	char	original-survey	ISO2C code of the reporting country, e.g. country which reported the import into Austria
Country_Group	char	derived-survey	Country of origin combined into 6 classes: EU-Western Europe, EU-Southern Europe, EU-Eastern Europe, EU-Northern Europe, EU-Western Asia, Other
swc	num	original-survey	Value (in €) of the imported goods
gw	num	original-survey	Weight of the reported goods
gw_class	char	derived-survey	Weight of the reported goods combined into 8 classes. Classes were constructed using the 20%, 40%, 50%, 60%, 70%, 80% and 90% percentiles for the weight distribution.
swc_gw	num	derived-survey	Ratio between swc and gw, $\frac{swc}{gw}$.

Appendix B

Appendix: extended results

B.1. Imputation of foreign tourist in accommodation establishments survey (PL)

B.1.1. Russia model

B.1.1.1. Prediction Performance Analysis for Tourists from Russia

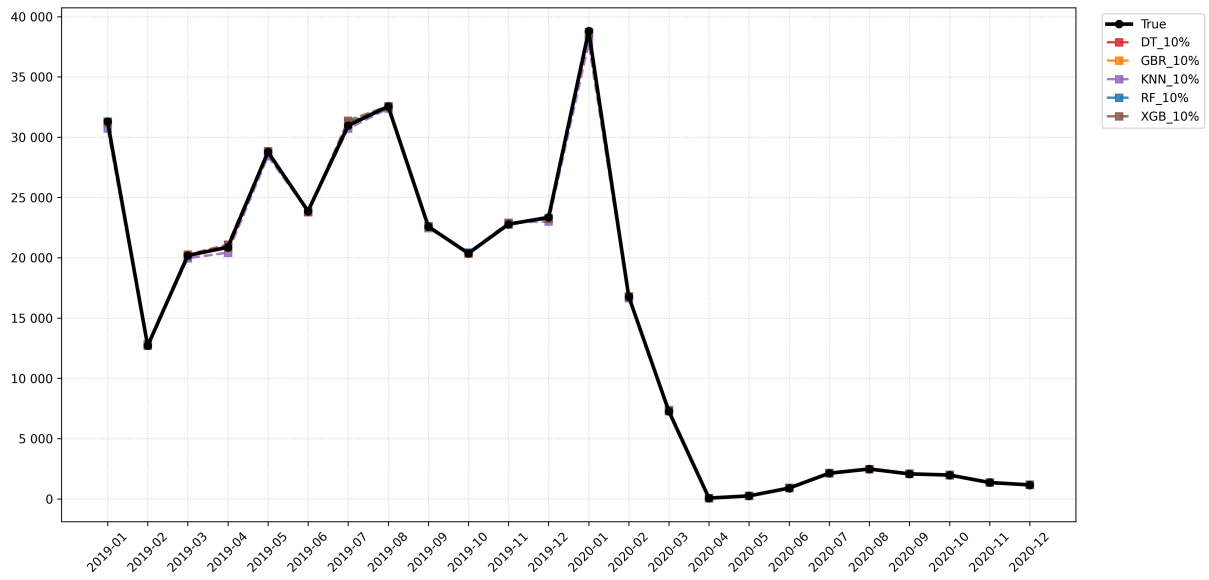


Figure B.1 Results for Russia with 10% size of gaps

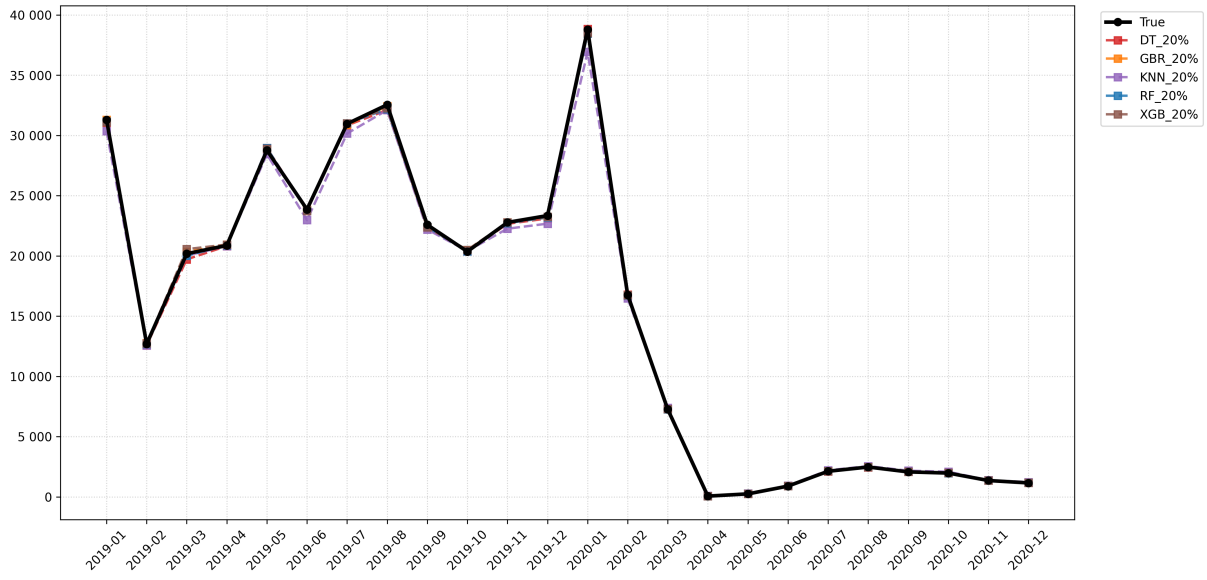


Figure B.2 Results for Russia with 20% size of gaps

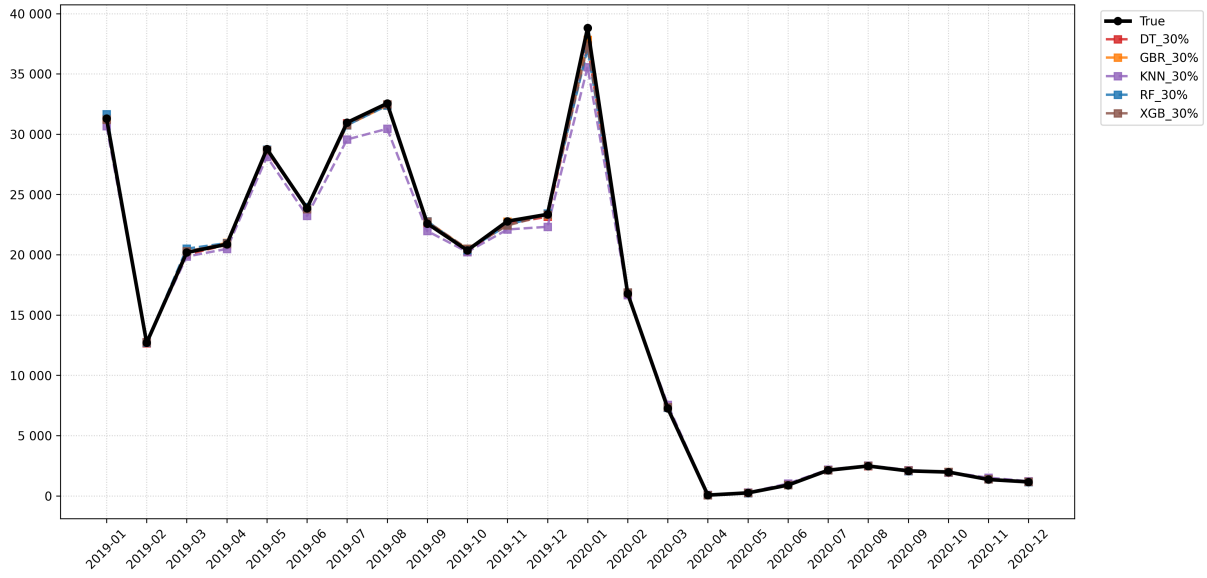


Figure B.3 Results for Russia with 30% gaps

The previous figures present the observed number of tourists from Russia during the period from January 2019 to December 2020, together with predictions generated by the evaluated machine learning models: Random Forest (RF), Gradient Boosting Regression (GBR), Decision Tree (DT), K-Nearest Neighbors (kNN), and Extreme Gradient Boosting (XGB), under three missing data scenarios (10%, 20%, and 30%).

All evaluated models accurately reproduce the overall behaviour of the analysed time series. A clear seasonal pattern is observed, with the highest number of tourists recorded during the summer months of 2019, particularly in July (52,244 tourists) and August (51,000 tourists). This period is followed by a gradual decline in tourist activity, with a significant reduction beginning in March 2020 due to the impact of COVID-19-related travel restrictions.

All models successfully identified the sudden decrease in tourist arrivals and correctly represented the subsequent period of significantly reduced tourism activity.

The closest agreement with observed values was achieved by GBR and Random Forest, whose prediction curves closely followed the actual time series. XGB also demonstrated highly competitive performance, while larger deviations were observed for Decision Tree and particularly kNN. The largest differences for these models occurred during seasonal peaks and periods characterised by rapid changes in tourist activity.

The comparison of different missing data scenarios indicates that increasing the proportion of missing values from 10% to 20% did not negatively affect prediction accuracy. For GBR and XGB, improved agreement with observed values was observed under the 20% missing data scenario, which is also reflected in the calculated error metrics.

At the 30% missing data level, a slight deterioration in prediction accuracy was observed; however, boosting-based models maintained high predictive performance and remained the most stable approaches among the evaluated methods.

Overall, the results confirm that ensemble-based models, particularly GBR and XGB, provide the most accurate reconstruction of tourist flows from Russia and demonstrate strong robustness to different levels of missing data.

B.1.1.2. Analysis of prediction errors for tourists from Russia

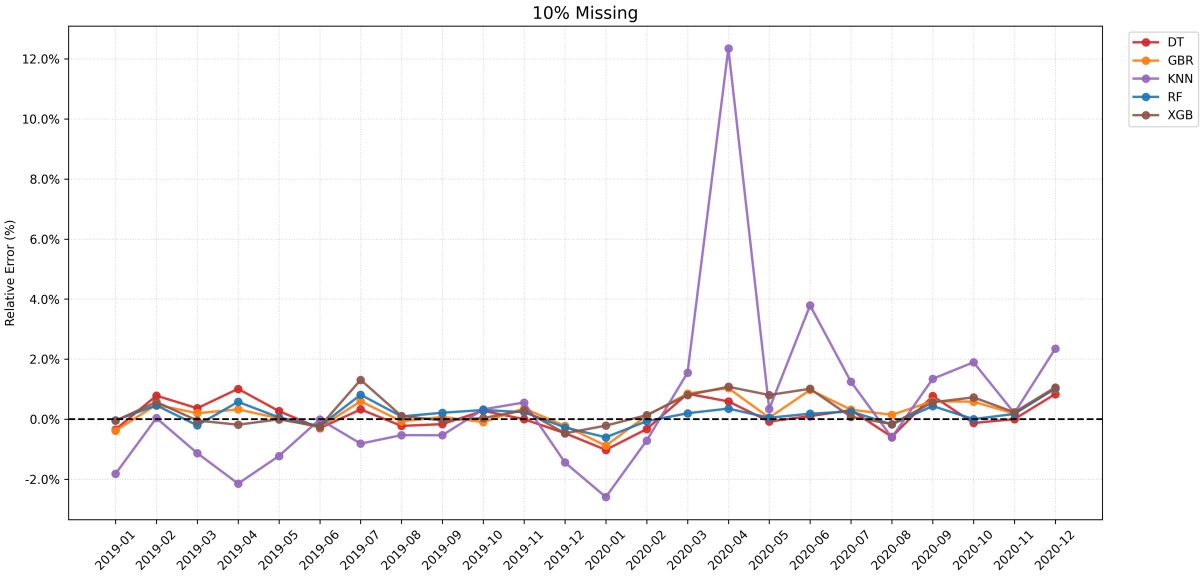


Figure B.4 Relative Error for Russia with 10% size of gaps

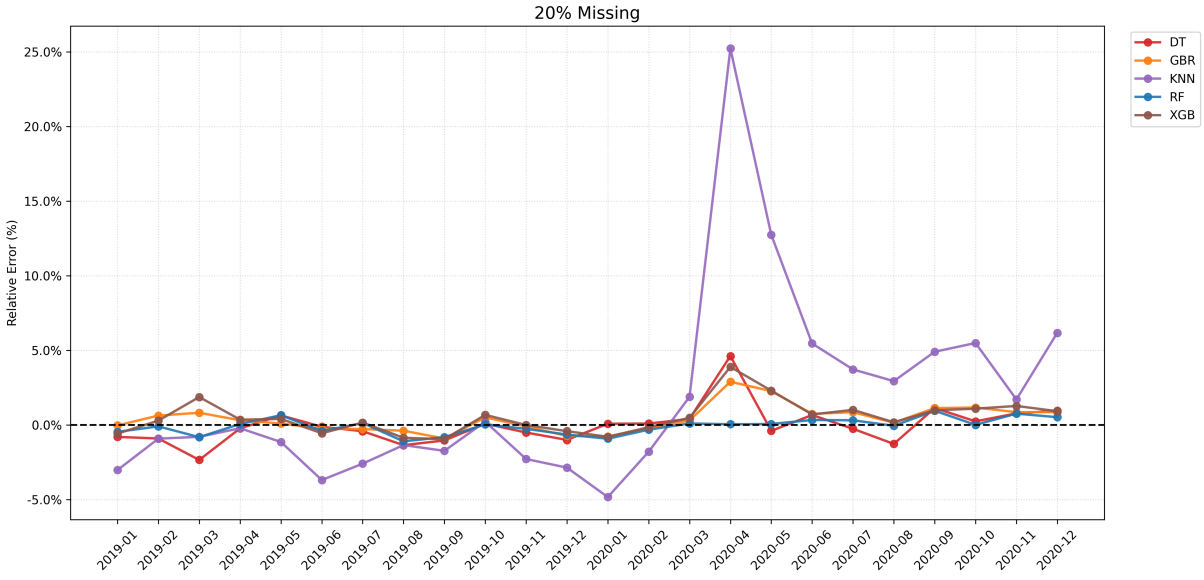


Figure B.5 Relative Error for Russia with 20% size of gaps

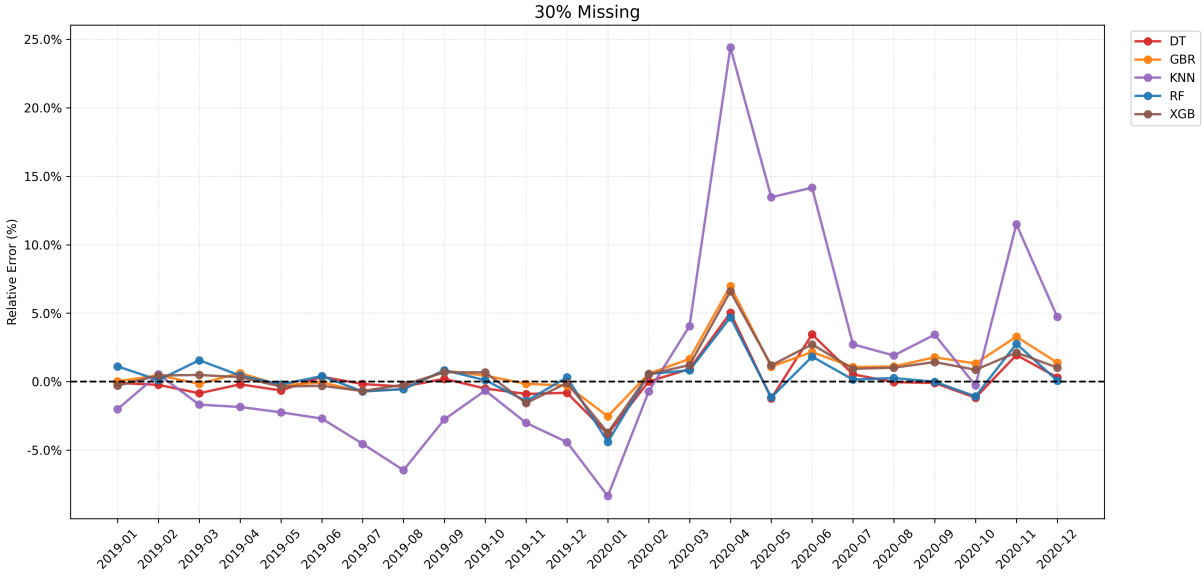


Figure B.6 Relative Error for Russia with 30% size of gaps

The analysis of differences between the actual and predicted values shows that all evaluated models provide highly accurate forecasts for tourists from Russia. In most months, the deviations between observed and predicted values are relatively small and usually range from several dozen to several hundred tourists. The smallest differences are observed during periods with a stable course of the time series, particularly at the beginning of 2019 and at the end of 2020, when the predictions closely follow the actual values.

The largest discrepancies occur during periods of rapid changes in the number of tourists, especially during the COVID-19 pandemic. In April 2020, the actual number of tourists decreased to only 751. For the 10% missing data scenario, all models generated predictions very close to the observed value, ranging approximately between 727 and 740 tourists. This corresponds to a relative error of only approximately 1–3%. However, for the 20% and 30% missing data

scenarios, the models started to underestimate the number of tourists, producing predictions between approximately 630 and 680 tourists, corresponding to relative errors of around 10–17%.

A larger difference can be observed in May 2020, when the actual number of tourists was 1343. For the 10% missing data scenario, RF, GBR, DT, and XGB predicted values between approximately 920 and 1159 tourists, resulting in an underestimation ranging from approximately 14% to 31%. In contrast, for the 20% and 30% missing data scenarios, the predictions were much closer to the observed value, ranging from approximately 1360 to 1372 tourists, with errors not exceeding 2–3%.

Another noticeable deviation occurs in January 2020, when the actual number of tourists reached 68 403. For the 30% missing data scenario, most models underestimated the value, producing predictions between approximately 62 000 and 65 000 tourists. This corresponds to an underestimation of approximately 5–9%. For the 10% and 20% missing data scenarios, the predicted values were considerably closer to the actual observation.

During periods of high tourist activity, such as July and August 2019, the differences between observed and predicted values remained relatively small. Most models achieved errors below 1–3%, indicating a very good ability to reproduce seasonal peaks in tourist arrivals.

The largest relative deviations were observed in October 2020. The actual number of tourists was 5273, while for the 10% missing data scenario most models predicted values close to 3780 tourists, resulting in an underestimation of approximately 28%. For the 20% and 30% missing data scenarios, the predictions were considerably closer to the observed value, with errors reduced to only a few percent.

Among all analysed approaches, kNN demonstrated the largest deviations from the actual values, particularly during seasonal peaks and periods of rapid changes in tourist arrivals. In contrast, GBR, Random Forest, and XGB maintained relatively small differences across most analysed months and provided the best representation of the observed time series behaviour.

Overall, the results indicate that for tourists from Russia all analysed models achieved high forecasting accuracy. The best performance was obtained by Gradient Boosting Regression (GBR) and Extreme Gradient Boosting (XGB), which most accurately reproduced both seasonal variations and sudden changes in tourist arrivals. The largest forecasting errors occurred during periods of rapid changes, especially during the COVID-19-related decline, while during stable periods the prediction errors remained low and generally did not exceed several percent.

B.1.2. Ukraine model

B.1.2.1. Forecasting results for tourists from Ukraine

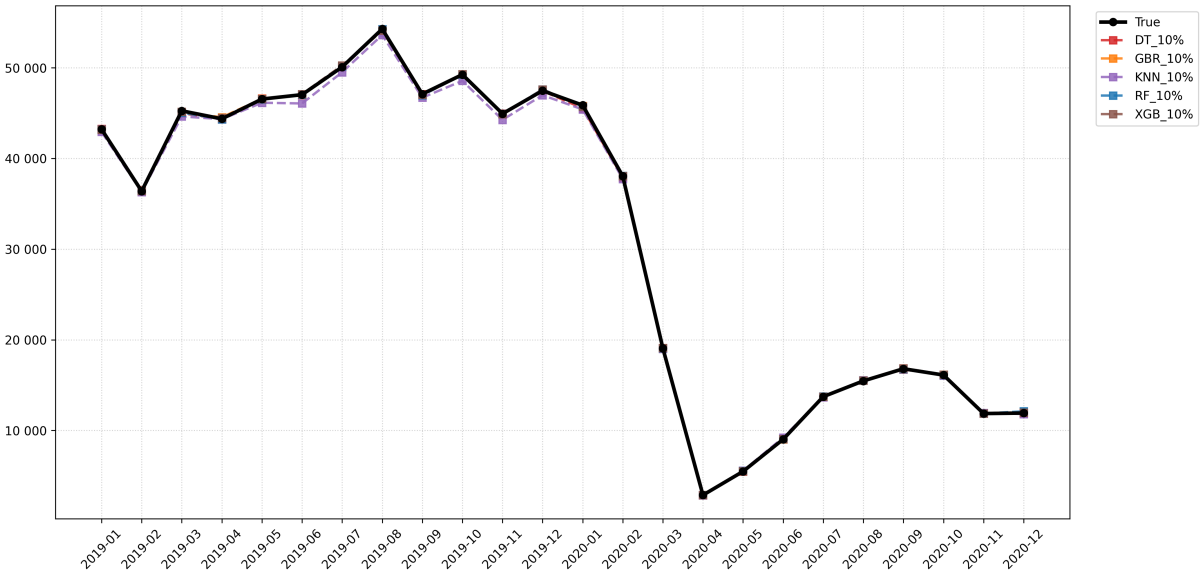


Figure B.7 Results for Ukraine with 10% size of gaps

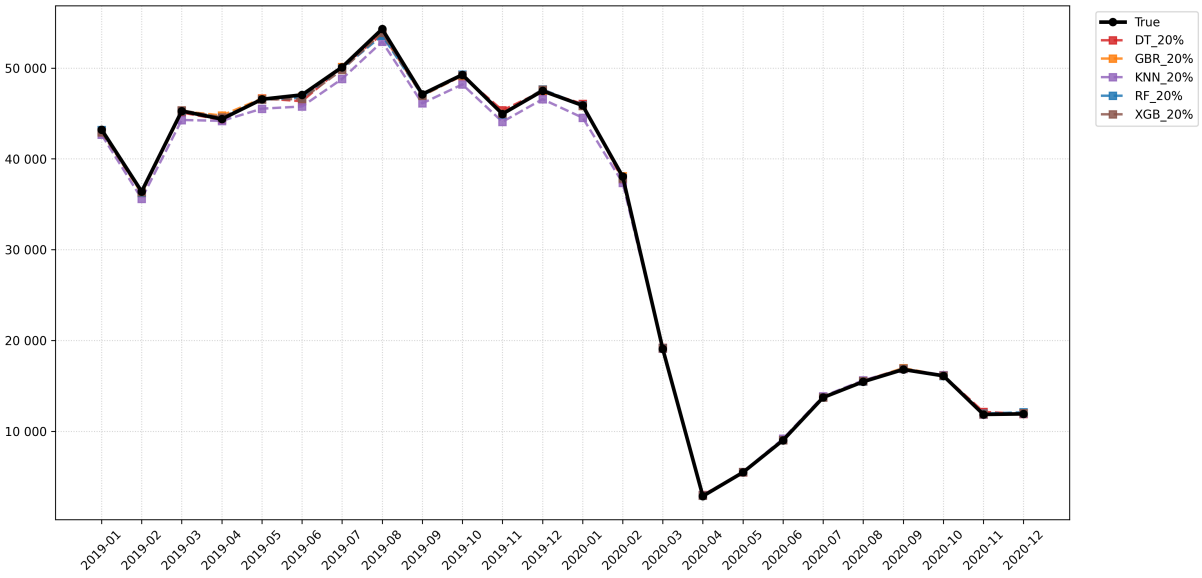


Figure B.8 Results for Ukraine with 20% size of gaps

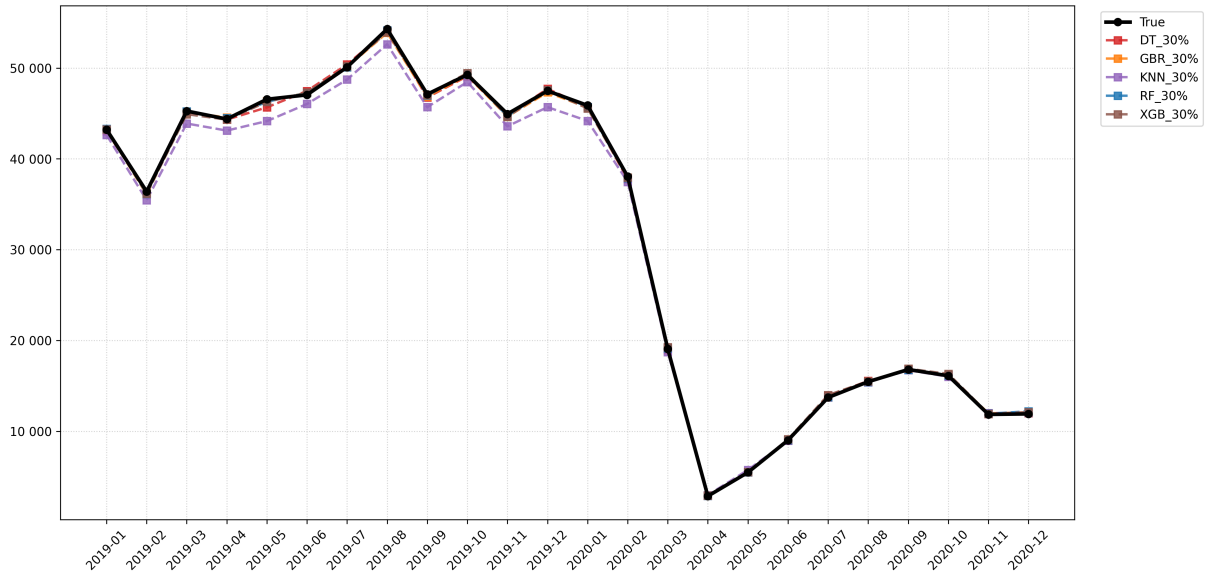


Figure B.9 Results for Ukraine with 30% gaps

Figures B.7, B.8 and B.9 present the actual monthly number of tourists from Ukraine between January 2019 and December 2020 together with the predictions generated by Random Forest (RF), Gradient Boosting Regression (GBR), Decision Tree (DT), K-Nearest Neighbors (kNN), and Extreme Gradient Boosting (XGB) under three missing data scenarios (10%, 20%, and 30%).

B.1.2.2. Analysis of prediction errors for tourists from Ukraine

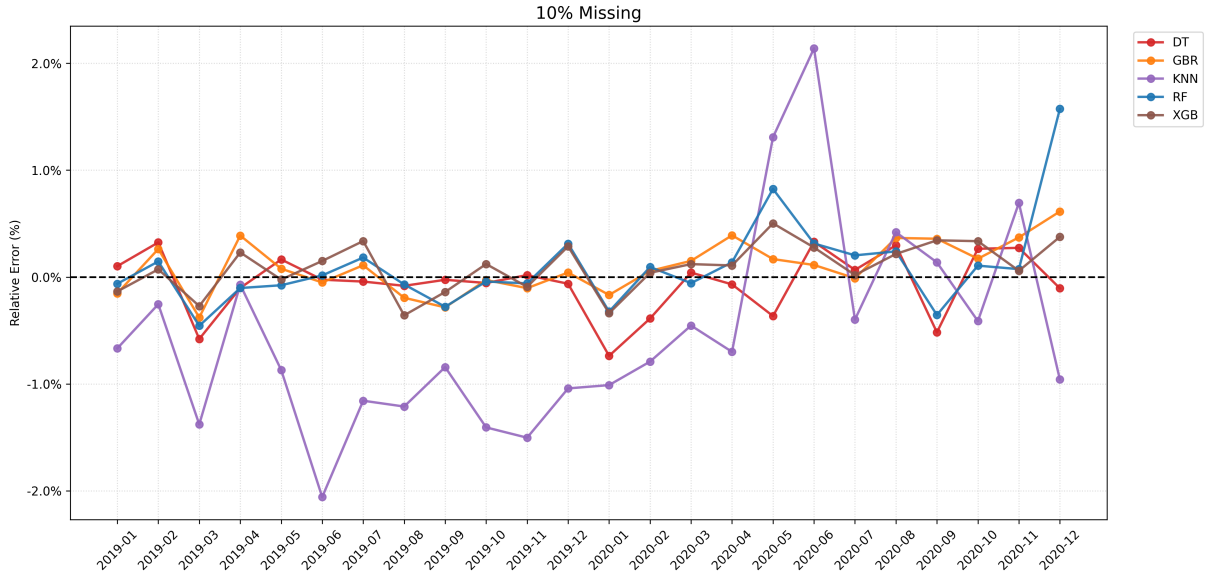


Figure B.10 Relative Error for Ukraine with 10% size of gaps

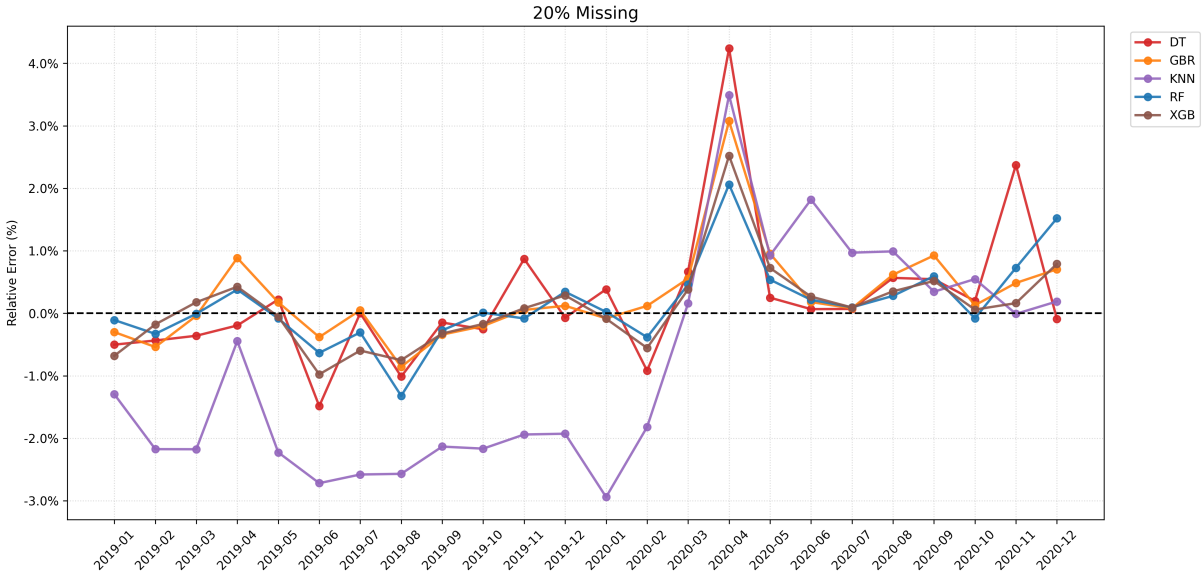


Figure B.11 Relative Error for Ukraine with 20% size of gaps

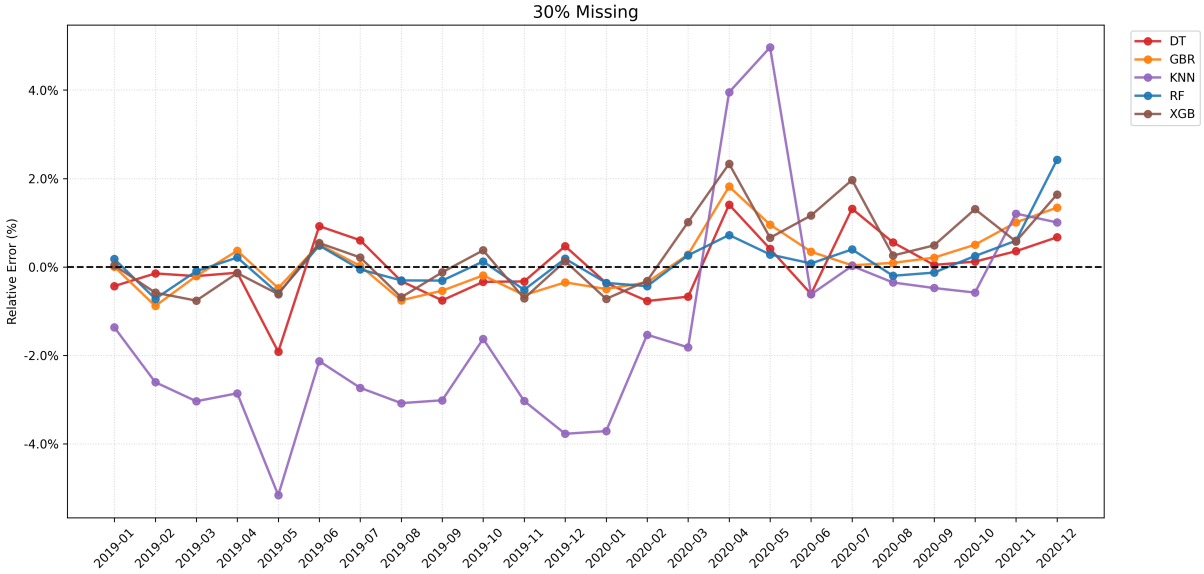


Figure B.12 Relative Error for Ukraine with 30% size of gaps

The comparison between the observed and predicted values indicates that all evaluated models achieved high forecasting accuracy throughout most of the analysed period. During 2019, the differences between the actual and predicted numbers of tourists remained relatively small. For Random Forest (RF), Gradient Boosting Regression (GBR), and Extreme Gradient Boosting (XGB), the relative prediction errors generally did not exceed 1–3%, indicating a close agreement with the observed data.

The largest discrepancies occurred during periods characterised by rapid changes in tourist arrivals. The first such period was April 2020, when the actual number of tourists decreased to 55 614. All models underestimated the observed value. For the 10% missing data scenario, the deviations were approximately 2–4%, whereas for the 20% and 30% missing data scenarios the underestimation increased to approximately 10–15%.

Noticeable differences were also observed in June 2020. With an actual value of 61 722 tourists, the models trained using the 10% missing data scenario predicted between approximately 55 000 and 59 000 tourists, corresponding to an underestimation of approximately 5–11%. Under the 30% missing data scenario, several models slightly overestimated the observed value, producing forecasts exceeding 65 000 tourists, corresponding to an overestimation of approximately 5–6%.

Another period with larger prediction errors occurred in September 2020, when the observed number of tourists reached 99 763. For the 10% missing data scenario, RF, GBR, and XGB underestimated the observed value by approximately 1–2%. Under the 30% missing data scenario, the deviations became more pronounced. For example, XGB predicted approximately 95 340 tourists, corresponding to an underestimation of approximately 4.4%, while several other models produced errors exceeding 10%.

Larger deviations were also observed in December 2020. The actual number of tourists was 76 255, whereas the predicted values ranged between approximately 71 000 and 81 000 tourists, resulting in deviations of up to approximately 7% in both directions.

During periods characterised by relatively stable tourist arrivals, particularly throughout the first half of 2019, all evaluated models generated predictions that closely followed the observed values, with only minor percentage differences.

The largest relative errors were consistently observed for Decision Tree and K-Nearest Neighbors, which were less effective in capturing rapid changes in tourist arrivals and more frequently produced both underestimation and overestimation. In contrast, Random Forest, Gradient Boosting Regression, and especially Extreme Gradient Boosting maintained relatively small prediction errors across most analysed months, providing the most accurate representation of the observed time series.

B.1.3. United Kingdom model

B.1.3.1. Forecasting results for tourists from the United Kingdom

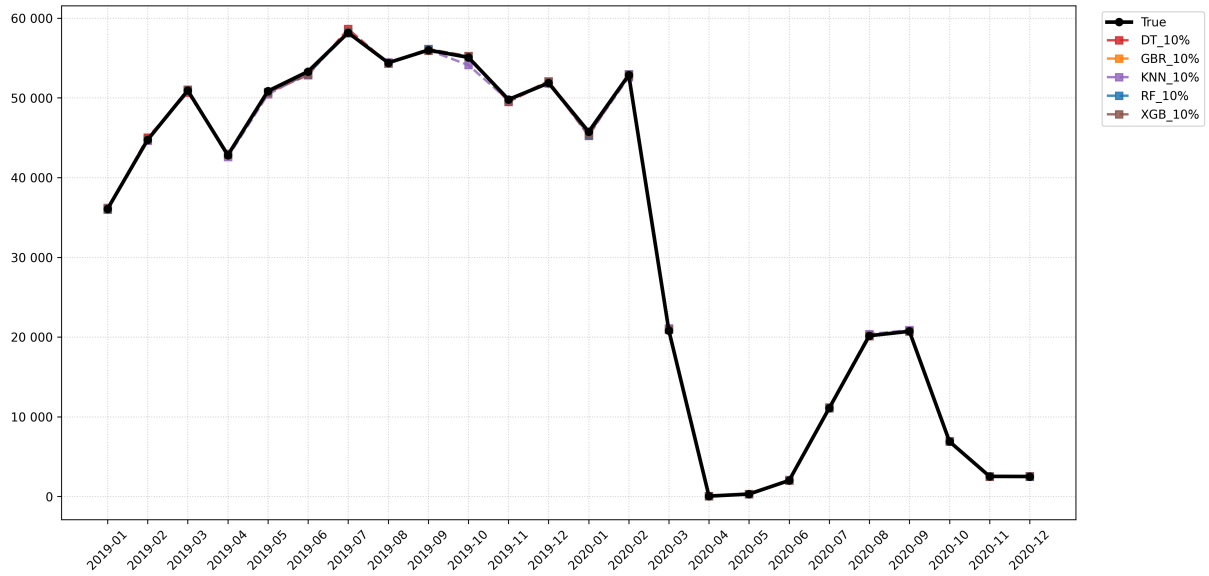


Figure B.13 Results for United Kingdom with 10% size of gaps

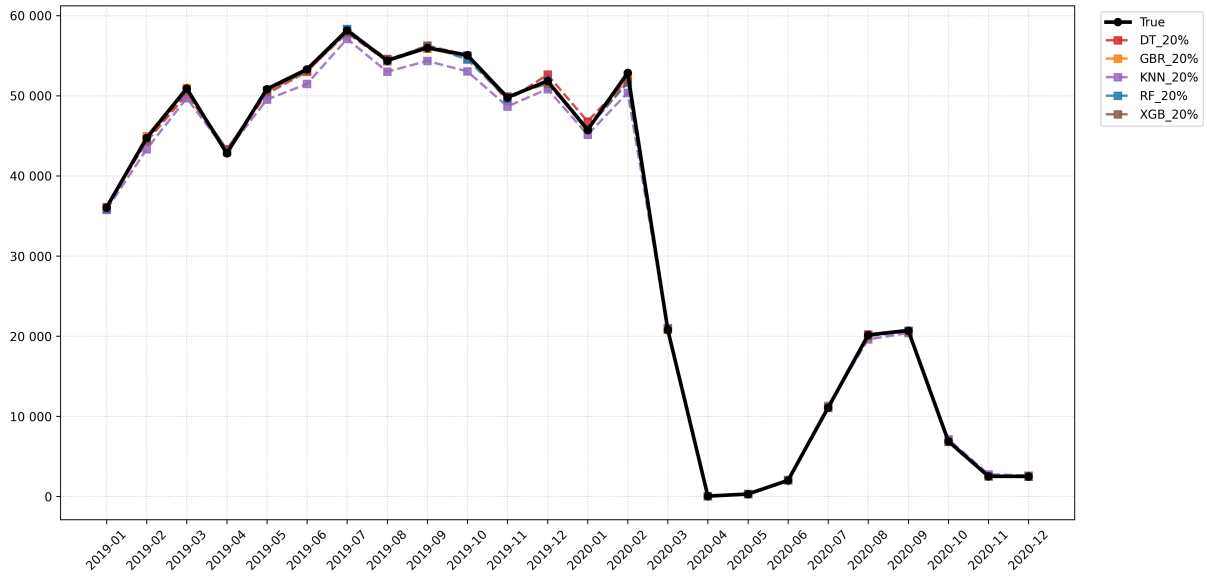


Figure B.14 Results for United Kingdom with 20% size of gaps

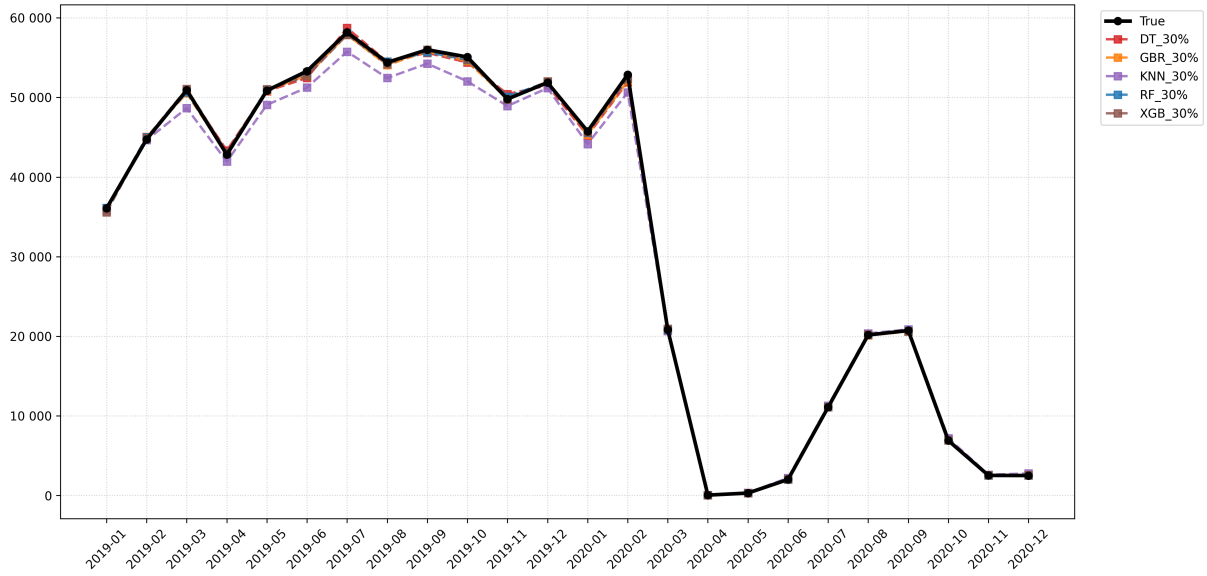


Figure B.15 Results for United Kingdom with 30% gaps

Figures B.13, B.14 and B.15 present the actual monthly number of tourists from the United Kingdom between January 2019 and December 2020 together with the predictions generated by Random Forest (RF), Gradient Boosting Regression (GBR), Decision Tree (DT), K-Nearest Neighbors (kNN), and Extreme Gradient Boosting (XGB) under three missing data scenarios (10%, 20%, and 30%).

All evaluated models accurately reproduce the overall behaviour of the analysed time series. Throughout 2019, tourist arrivals remained relatively stable, reaching seasonal peaks in July (136 794 tourists) and August (129 153 tourists). At the beginning of 2020, tourist arrivals remained at a high level before declining sharply from March onwards due to the COVID-19 pandemic and the associated travel restrictions. The lowest value was recorded in April 2020, when only 234 tourists were observed. A gradual recovery followed during the subsequent

months, reaching a local maximum of 48 133 tourists in September 2020, after which tourist arrivals declined again.

Random Forest (RF), Gradient Boosting Regression (GBR), and Extreme Gradient Boosting (XGB) closely follow the observed values throughout the analysed period, accurately capturing both the seasonal patterns and the rapid changes associated with the pandemic. Slightly larger deviations are observed for Decision Tree (DT) and K-Nearest Neighbors (kNN), particularly during periods characterised by rapid fluctuations in tourist arrivals.

The comparison of the three missing data scenarios shows that all models maintained high forecasting accuracy for the 10% missing data scenario. Under the 20% missing data scenario, slightly larger deviations were observed during the pandemic period, while the 30% missing data scenario resulted in a further reduction in forecasting accuracy, particularly during the recovery phase in the second half of 2020. Despite this deterioration, the boosting-based models continued to provide the most accurate and stable predictions across all missing data levels.

B.1.3.2. Analysis of prediction errors for tourists from the United Kingdom

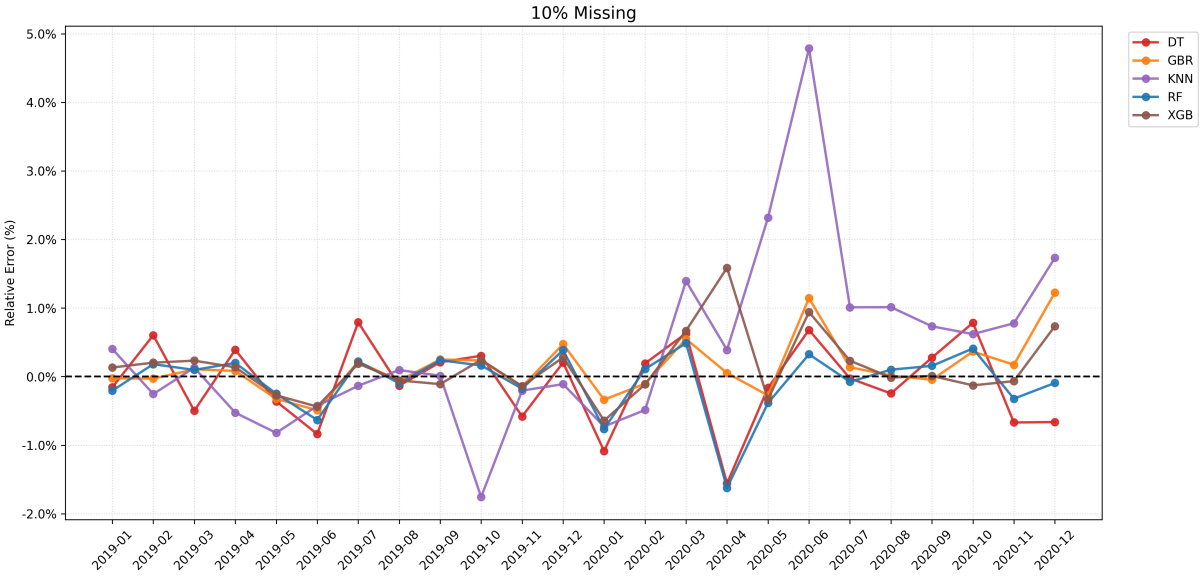


Figure B.16 Relative Error for United Kingdom with 10% size of gaps

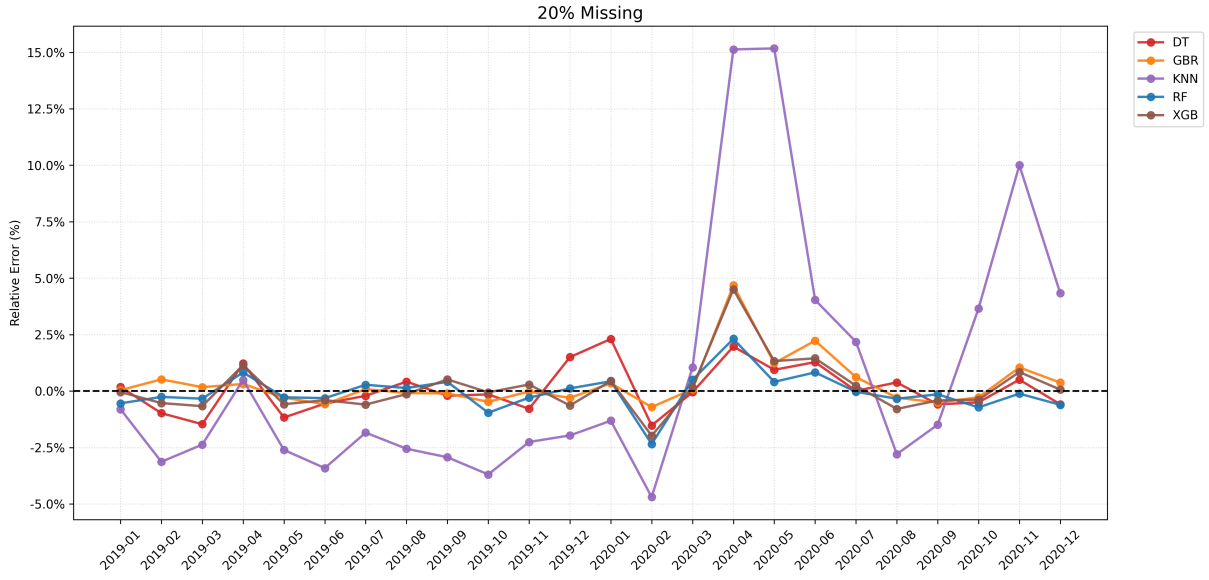


Figure B.17 Relative Error for United Kingdom with 20% size of gaps

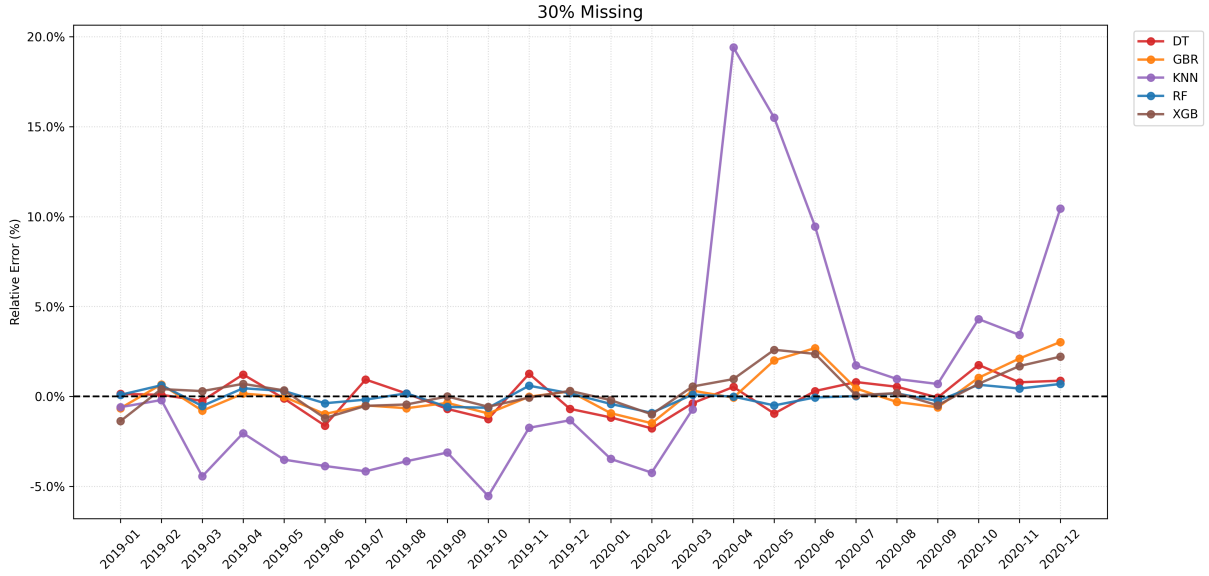


Figure B.18 Relative Error for United Kingdom with 30% size of gaps

The comparison of observed and predicted values indicates that all evaluated models achieved high forecasting accuracy throughout most of the analysed period. During 2019, the differences between the actual and predicted numbers of tourists remained minimal. For Random Forest (RF), Gradient Boosting Regression (GBR), and Extreme Gradient Boosting (XGB), the relative prediction errors generally remained below 1%, demonstrating an excellent agreement with the observed data.

The largest discrepancies occurred during periods characterised by rapid changes in tourist arrivals. The first such period was April 2020, when the observed number of tourists dropped to only 234. Under the 10% missing data scenario, all models generated predictions very close to the observed value, ranging from approximately 228 to 231 tourists, corresponding to relative errors of approximately 1–3%. In contrast, for the 20% and 30% missing data scenarios, the models

consistently underestimated the observed value, producing predictions between approximately 180 and 196 tourists, resulting in relative errors of approximately 20–30%.

Noticeable differences were also observed in May 2020, when the actual number of tourists reached 795. For the 10% missing data scenario, most models predicted between 744 and 771 tourists, corresponding to an underestimation of approximately 3–6%. Under the 20% missing data scenario, the predictions remained very close to the observed value, with forecasts ranging from approximately 800 to 810 tourists. For the 30% missing data scenario, larger deviations reappeared. For example, GBR predicted approximately 738 tourists, corresponding to an underestimation of approximately 7%, while XGB predicted approximately 724 tourists, representing an underestimation of approximately 9%.

Another period with increased prediction errors occurred in June 2020. With an observed value of 4 068 tourists, the models trained under the 10% missing data scenario generated forecasts with relative errors below 2%. Under the 20% missing data scenario, several models slightly overestimated the observed value by approximately 5–9%, while similar deviations were observed for the 30% missing data scenario.

During the recovery phase, particularly in August and September 2020, the predictions generated by RF, GBR, and XGB remained very close to the observed values. Relative errors generally did not exceed 2–3%, indicating that these models effectively captured the recovery of tourist arrivals.

Larger deviations were again observed towards the end of 2020. In November, the actual number of tourists was 5 855, whereas predictions under the 30% missing data scenario ranged from approximately 5 584 to 6 148 tourists, corresponding to relative errors of approximately 5–7%. A similar pattern was observed in December 2020, when the observed value was 5 218 tourists and the predicted values ranged from approximately 4 918 to 5 860 tourists, resulting in deviations of up to approximately 12%.

The largest prediction errors were consistently associated with Decision Tree (DT) and K-Nearest Neighbors (kNN), particularly under the 30% missing data scenario. These models exhibited greater variability and were less effective in reproducing rapid changes in tourist arrivals. In contrast, Random Forest, Gradient Boosting Regression, and especially Extreme Gradient Boosting maintained relatively small prediction errors throughout most of the analysed period, providing the most accurate representation of the observed time series.

B.1.4. United States model

B.1.4.1. Forecasting results for tourists from the United States

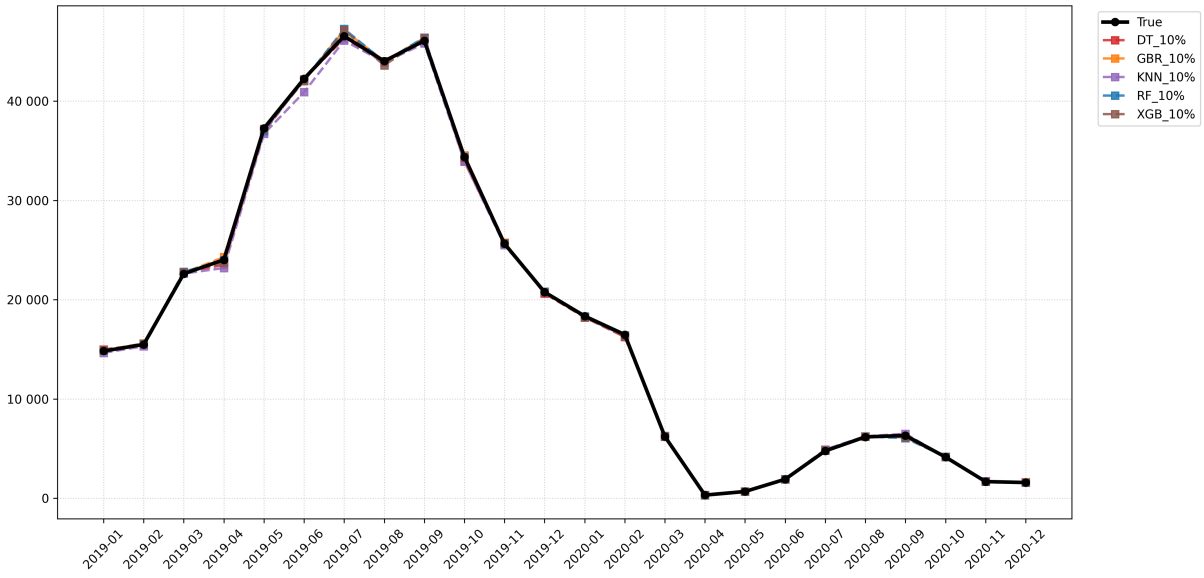


Figure B.19 Results for United States with 10% size of gaps

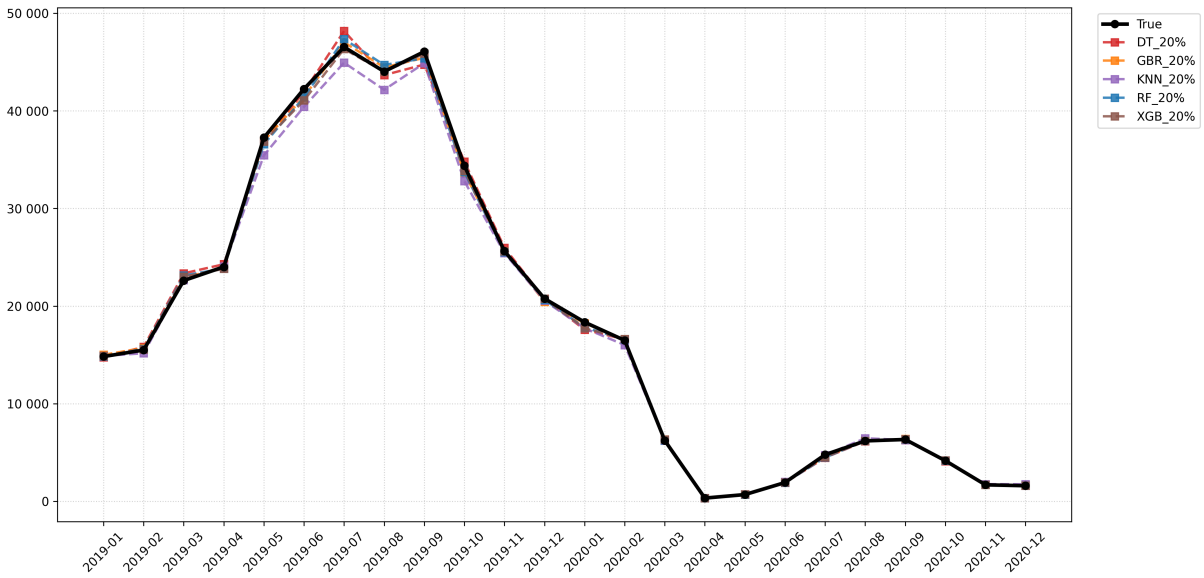


Figure B.20 Results for United States with 20% size of gaps

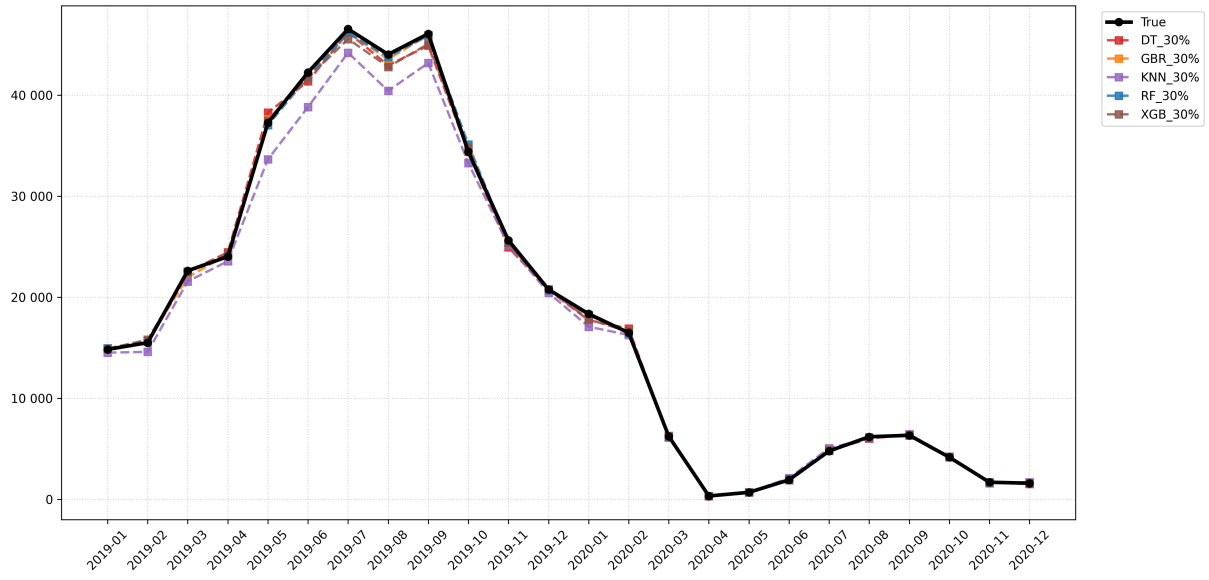


Figure B.21 Results for United States with 30% gaps

Figures B.19, B.20 and B.21 presents the actual monthly number of tourists from the United States visiting Poland between January 2019 and December 2020 together with the predictions generated by Random Forest (RF), Gradient Boosting Regression (GBR), Decision Tree (DT), K-Nearest Neighbors (kNN), and Extreme Gradient Boosting (XGB) under three missing data scenarios (10%, 20%, and 30%).

The analysed time series exhibits a clear seasonal pattern throughout 2019. The lowest numbers of tourist arrivals were recorded during the winter months (approximately 33 000–43 000 tourists), followed by a steady increase during spring and a peak in the summer season, reaching 113 664 tourists in July and 113 210 in August. Tourist arrivals gradually declined during the remaining months of the year.

At the beginning of 2020, tourist arrivals remained at a level comparable to that observed in the previous year. However, a sharp decline occurred from March onwards as a consequence of the COVID-19 pandemic and the associated travel restrictions. The lowest value was recorded in April 2020, when only 1 486 tourists visited Poland. A gradual recovery began in May and continued throughout the second half of the year, reaching a local maximum of 19 618 tourists in August 2020. Nevertheless, tourist arrivals remained substantially below the pre-pandemic levels.

Random Forest (RF), Gradient Boosting Regression (GBR), and Extreme Gradient Boosting (XGB) accurately reproduced both the seasonal fluctuations and the abrupt changes associated with the pandemic. The prediction curves generated by these models closely followed the observed values throughout the analysed period. Slightly larger deviations were observed for Decision Tree (DT) and K-Nearest Neighbors (kNN), particularly under higher levels of missing data.

The results also demonstrate the influence of the percentage of missing data on forecasting performance. For the 10% missing data scenario, all evaluated models produced predictions that closely matched the observed values. As the proportion of missing data increased to 20% and 30%, the variability of the predictions became more pronounced, particularly during periods of rapid changes in tourist arrivals caused by the COVID-19 pandemic. Nevertheless, the boosting-based models maintained high forecasting accuracy across all analysed scenarios.

B.1.4.2. Analysis of prediction errors for tourists from the United States

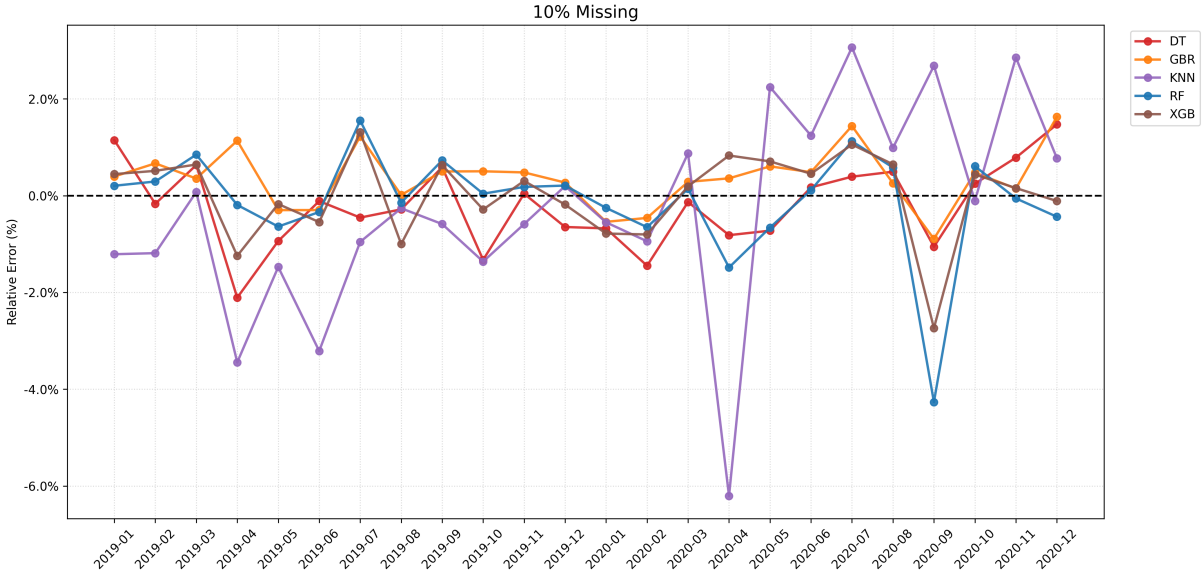


Figure B.22 Relative Error for United States with 10% size of gaps

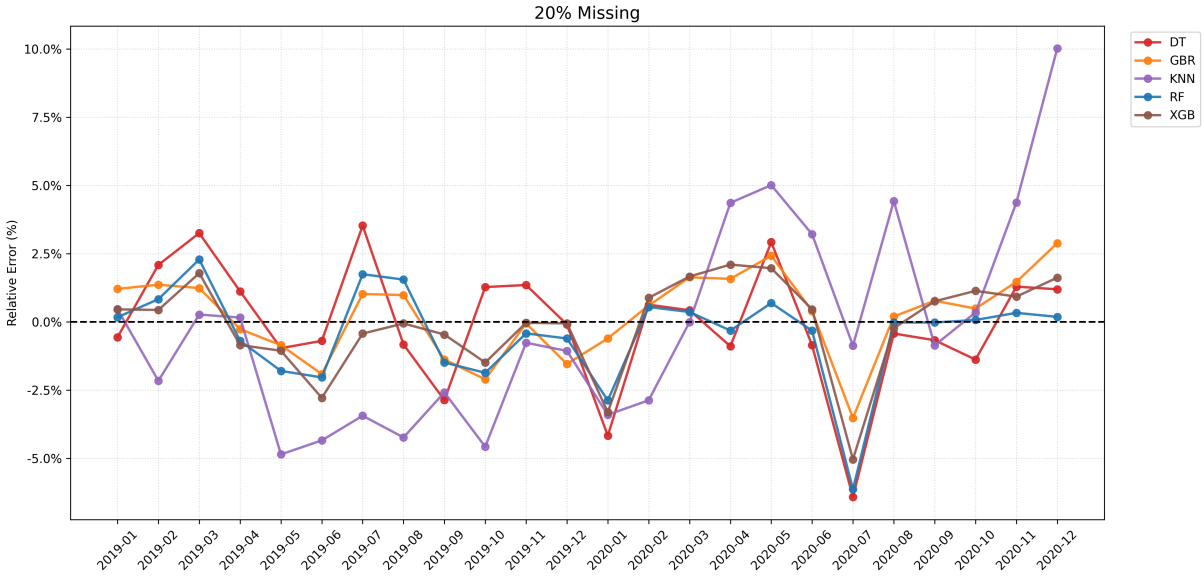


Figure B.23 Relative Error for United States with 20% size of gaps

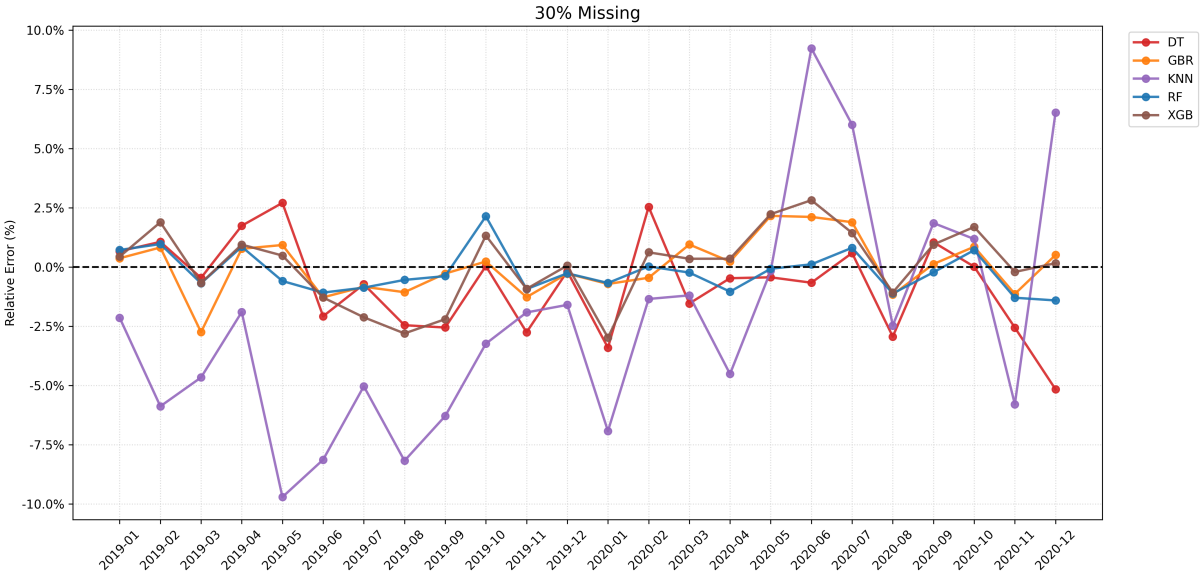


Figure B.24 Relative Error for United States with 30% size of gaps

The previous figures present the relative prediction errors obtained for all evaluated machine learning models under three missing-data scenarios (10%, 20%, and 30%).

During 2019, the relative prediction errors remained low for all ensemble-based models. For the RF, GBR, and XGB models, the absolute relative error generally did not exceed 5%, while in most months it remained below 2%. The largest deviations during this period occurred in July and August, corresponding to the seasonal peak in tourist arrivals. Even under these conditions, the ensemble methods maintained relatively small prediction errors, whereas the DT and particularly the kNN model exhibited larger deviations.

A noticeable increase in prediction errors was observed during the first months of the COVID-19 pandemic. The largest relative errors occurred in April and May 2020, when tourist arrivals decreased abruptly. For the 10% missing-data scenario, most models slightly overestimated the observed values in April, with relative errors ranging from approximately 6% to 8%. In contrast, for the 20% and 30% missing-data scenarios, the prediction errors were generally smaller, indicating that the imputation procedure remained robust despite the increased proportion of missing observations.

In May 2020, the relative errors increased for all models under the 20% and 30% missing-data scenarios. Most models underestimated the number of tourist arrivals, with prediction errors reaching approximately 10–13%. The largest deviation was observed for the kNN model, whose relative error exceeded 18% for the 30% missing-data scenario.

During the recovery period between June and September 2020, the ensemble models (RF, GBR, and XGB) rapidly regained prediction accuracy. Relative errors were typically below 5%, demonstrating their ability to capture the recovery trend after the initial pandemic shock. The DT and kNN models remained less stable and produced noticeably larger deviations, particularly for the 30% missing-data scenario.

The final months of 2020 were characterised by relatively small prediction errors for the RF and XGB models, generally remaining below 5%. Larger deviations were observed for the GBR

model in December (approximately 6.5%) and for the kNN model under higher missing-data levels. Nevertheless, the ensemble methods consistently outperformed the remaining approaches throughout the analysed period.

Overall, the analysis of relative prediction errors confirms the conclusions obtained from the RMSE, MAE, and Bias metrics. The XGB, GBR, and RF models demonstrated the highest prediction accuracy and robustness to missing data, while the kNN model produced the largest and most variable relative errors, particularly during periods of rapid changes in tourist arrivals.

In the following, detail results from the models can be seen.

Table B.1 Comparison of different machine learning models for imputation of number of foreign tourist by country of origin

Country	Missing_%	Model	RMSE	MAE	Bias_%	Time_sec
Germany	10	RF	25,204	5,3555	-1,19	8,73
		GBR	22,119	5,7697	-1,1	24,56
		DT	33,5702	8,1804	-1,75	0,45
		kNN	43,0702	20,5178	-9,52	13,49
		XGB	49,663	6,5777	-2,46	0,46
	20	RF	58,3235	6,551	-0,69	7,93
		GBR	29,8779	5,9767	0,68	28,75
		DT	35,4014	8,0082	-1,53	0,4
		kNN	48,6302	20,3752	-8,69	22
		XGB	47,3522	6,5913	0,27	0,36
	30	RF	29,3022	5,6003	-0,65	8,04
		GBR	25,0635	5,851	-0,72	19,29
		DT	46,7117	9,1792	-0,35	0,36
		kNN	47,5369	20,8338	-10,48	33,11
		XGB	44,1418	6,0382	1,03	0,35
Russia	10	RF	10,8738	1,9952	0,74	2,72
		GBR	7,9384	1,9012	-0,05	7,07
		DT	9,5746	2,7599	-0,42	0,15
		kNN	12,5674	6,3807	-8,76	2,08
		XGB	8,4456	1,8836	1,1	0,34
	20	RF	8,1051	1,9176	-1,68	3,15
		GBR	5,387	1,7495	-0,58	7,48
		DT	11,9982	2,9961	-2,33	0,16
		kNN	12,1226	6,4479	-8,89	4,03
		XGB	6,9465	1,8794	-0,39	0,23
	30	RF	19,4773	2,2797	-1,05	2,21
		GBR	8,4309	1,8474	-0,66	6,26
		DT	15,0459	2,9816	-2,16	0,11
		kNN	17,5652	6,7859	-10,14	4,96
		XGB	14,3529	1,7708	-1,39	0,29
Ukraine	10	RF	6,4044	1,803	-0,07	5,07
		GBR	5,6232	1,9052	0,1	14,14
		DT	7,0698	2,6671	-0,8	0,27

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Table B.1 – Continued from previous page

Country	Missing_%	Model	RMSE	MAE	Bias_%	Time_sec
		kNN	13,3301	6,7662	-8,15	7,68
		XGB	5,6886	1,6621	0,26	0,34
		RF	7,5524	1,8237	-0,48	5,15
		GBR	5,9943	1,8931	0,02	12,27
	20	DT	12,731	2,9825	-0,66	0,23
		kNN	12,4859	6,5923	-8,19	13,56
		XGB	7,7902	1,6483	-0,72	0,38
		RF	7,3823	1,9985	-0,2	3,85
		GBR	5,7897	1,8816	-0,54	10,42
	30	DT	10,1339	2,9698	-0,61	0,21
		kNN	14,7139	6,8007	-8,29	17,35
		XGB	8,178	1,9583	-0,1	0,34
		RF	11,8812	3,1249	-0,04	4,04
		GBR	8,6052	3,2714	0,25	10,29
	10	DT	18,4852	4,8652	-0,33	0,2
		kNN	23,9758	11,3632	-2,18	2,91
		XGB	8,8097	2,928	0,02	0,36
		RF	20,3413	4,0031	-1,03	4,34
		GBR	16,6199	3,9214	-0,43	10,73
United Kingdom	20	DT	23,9512	5,8863	-0,62	0,18
		kNN	31,0904	12,6145	-10,42	5,36
		XGB	17,1752	3,5715	-1,21	0,27
		RF	16,4697	3,8187	-0,31	3,16
		GBR	13,5997	3,8691	-1,3	7,76
	30	DT	21,1918	5,5592	-0,77	0,17
		kNN	31,6782	12,9724	-8,68	6,42
		XGB	14,8089	3,5606	-0,61	0,27
		RF	24,6917	4,5466	1,51	2,55
		GBR	16,9242	3,824	3,31	6,42
	10	DT	24,3016	5,1299	-3,88	0,12
		kNN	29,427	10,1794	-10,23	1,05
		XGB	20,8635	3,9038	-0,5	0,24
		RF	32,0019	4,6591	-1,96	2,29
		GBR	22,4181	4,0993	-1,32	5,67
United States	20	DT	43,6669	5,8284	0,31	0,1
		kNN	29,166	10,3578	-12,02	2,53
		XGB	23,2955	3,8978	-2,9	0,2
		RF	25,4745	4,4173	-0,73	2,18
		GBR	17,625	3,7715	-1,37	5,4
	30	DT	29,9423	5,6756	-2,42	0,11
		kNN	38,2947	11,0509	-16,64	3,75
		XGB	22,3405	4,1822	-2,43	0,31

The following figures B.25, B.26, B.27, B.28 and B.29 illustrate the prediction results obtained using both the XGBoost (XGB) and Gradient Boosting Regressor (GBR) models for each analyzed country. Each figure compares the observed and predicted numbers of tourists. The best-

performing model for a given country, as determined by the evaluation metrics, is highlighted in green, whereas the lower-performing model is highlighted in red.

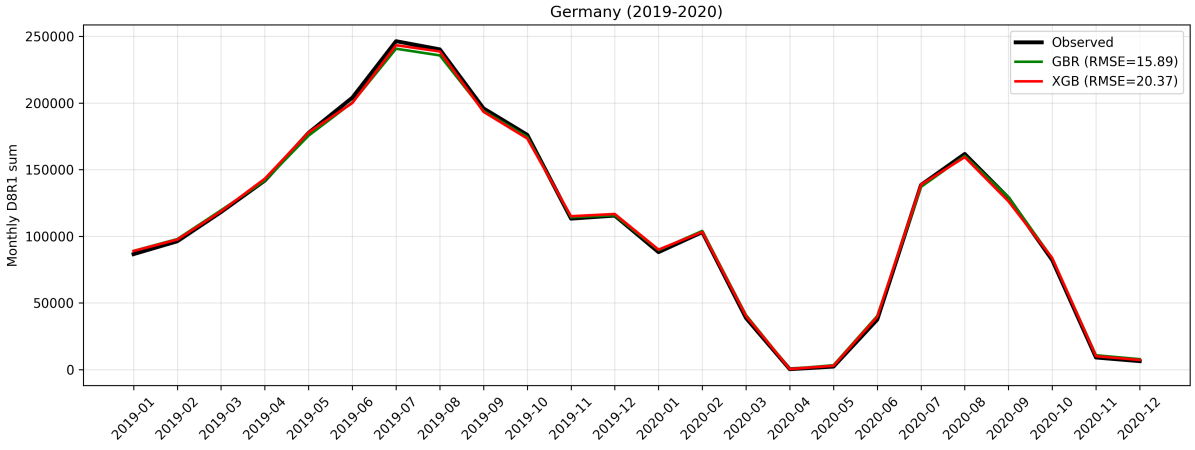


Figure B.25 .



Figure B.26 .



Figure B.27 .

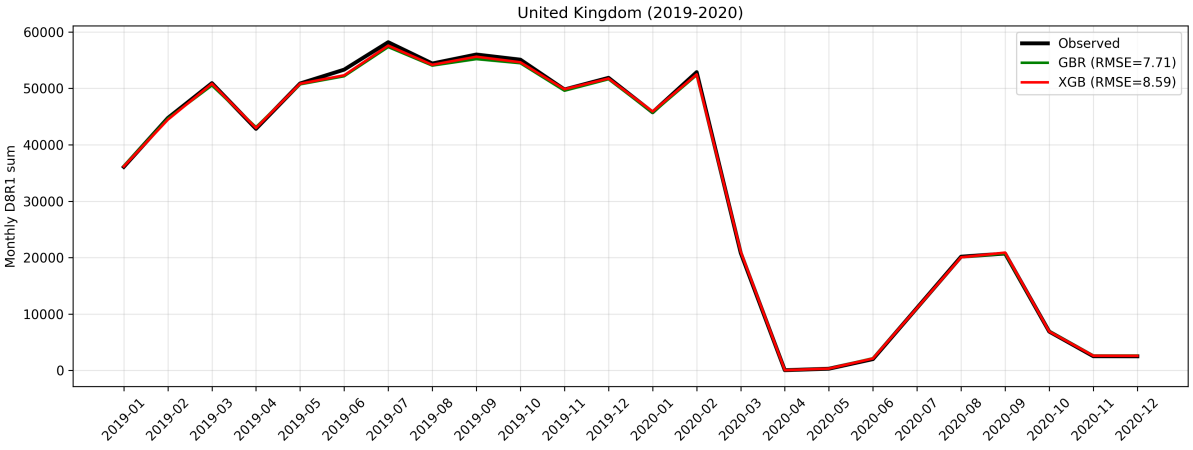


Figure B.28 .

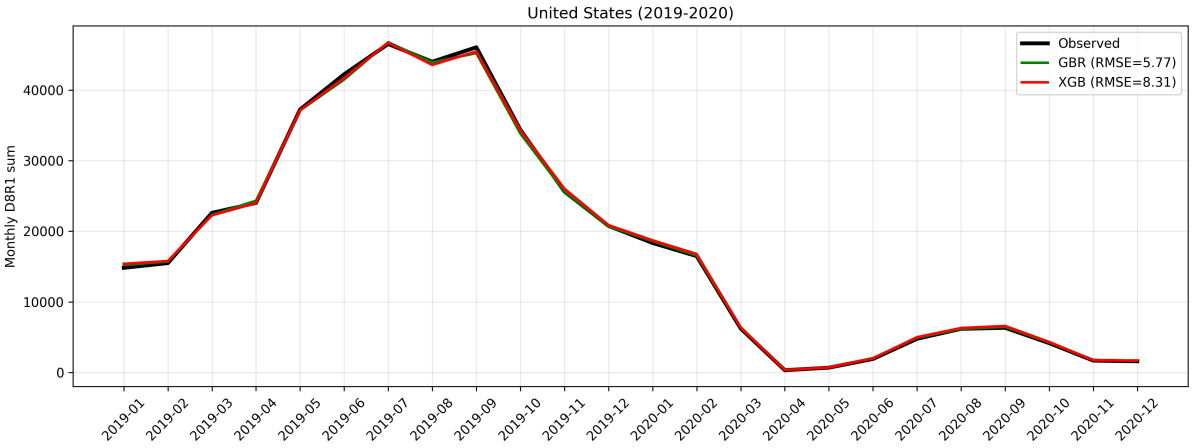


Figure B.29 .